Phenix Cryo-EM Tools

21 documentation pages
Generated from local Phenix documentation

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Docking a model into a map

cryo-em-dock-model.html

Why

Obtaining the structure of a macromolecule generally requires building an atomic model that fits the cryo-EM map. A docking procedure is used to do this fitting if the model, or a part of the model, is known but has not been placed into the map.

For example, your cryo-EM map might show a molecular complex assembled from components available from other experiments, such as X-ray crystallography. Then, you can dock these components' models into your cryo-EM map to obtain a model for the entire complex.

How

The program phenix.dock_in_map automatically docks one or several models into a map, using a convolution-based shape search to find a part of the map that is similar to the model. Initially, it performs a low-resolution search that focuses on the overall shape of the molecule, which is performed without rotation to optimize runtime. Alternatively, you can perform an initial fast search by matching the moments of inertia of the model and map, but this option requires an accurate map that looks a lot like the model. Next, if the placement is satisfactory, a full search can be run on multiple processors using real-space rigid-body refinement with the full resolution of the map.

In a standard run of phenix.dock_in_map, you input the model (PDB file) that you want to place, a map (CCP4, MRC, or other map format file), the nominal resolution of the map, and the number of processors to use.The program outputs a search PDB file transformed to superpose on the target map file

Getting help from the Phenix chatbot

You can use the free Phenix chatbot to get help on any Phenix program. Just ask it any question about using Phenix.

How to use phenix.dock_in_map: Click here

Common issues

- None

Related programs

- cryo_fit: This tool performs flexible fitting of a model into a cryo-EM map.

Map improvement (cryo-EM)

cryo-em-map-manipulations.html

Why

Cryo-EM maps can be improved by methods such as sharpening/blurring and density modification.

Sharpening can reveal the high-resolution details concealed in the cryo-EM map. Often, cryo-EM maps can appear smooth and can lack a high level of detail (contrast) because high-resolution amplitudes of corresponding Fourier map coefficients decay from causes such as radiation damage, sample movement, sample heterogeneity, and errors in the reconstruction procedure. We note that sharpening (and blurring) modify the amplitudes of Fourier coefficients and leave the phases unchanged.

Density modification can be used to improve a cryo-EM map by adjusting the Fourier coefficients to better agree with both the original map and the expected features. This improvement relies on two ideas: (1) the Fourier coefficients representing the map are uncorrelated, and (2) some features in a map region are known in advance.

How

The program phenix.map_sharpening performs map sharpening by optimizing the detail and connectivity of a cryo-EM map. It tests several resolution-dependent functions and outputs the one with maximal clarity, based on adjusted surface area (default) or kurtosis. The program works best on maps with 4.5 Å or better resolution. At lower resolutions, it may be unable to distinguish variations in the quality of the map as the sharpening is varied. Typically, optimal sharpening is determined by examining a box cut out of the map, and then applying this to the entire map; however, other options are available.

The program phenix.resolve_cryo_em performs density modification of a cryo-EM map using two unmasked half-maps, the FSC-based resolution, and a sequence file specifying the contents of the map. You can also supply a model, which will be randomized, refined, and used in model-based density modification. You normally access the program's functionality by running the ResolveCryoEM tool in the Phenix GUI.

How to use phenix.map_sharpening: Click here

How to use phenix.resolve_cryo_em: Click here

Common issues

- phenix.map_sharpening doesn't work for map_box-produced maps: map_sharpening should not be used on maps produced with the exact-unique option of map_box. These maps are closely masked around the density of a single molecular, so much of the map is set to zero. Therefore, the map is missing noise information and auto-sharpen doesn't work properly.
- phenix.resolve_cryo_em doesn't work well: Poor results could be due to using masked half-maps, having non-uniform solvent noise, or getting interference from having very prominent density away from the macromolecule. The program can also crash if you have limited memory and are using multiple processors. See phenix.resolve_cryo_em for ways to address these issues.

Related programs

- phenix.map_symmetry: This tool finds internal symmetries in a map, allowing the map to be reduced to the repeating unit.
- phenix.map_box: This tool cuts out a box from a larger map. It is used to extract the unique part of a map.

- phenix.combine_focused_maps: This tool creates a weighted composition map from a set of locally focused maps and associated models, where each part of each map is weighted by its correlation with the corresponding model. The maps do not have to be superimposed or even have the same gridding or size.

Assessing Map Quality (cryo-EM)

cryo-em-map-quality.html

Why

The first step after data processing is analysis to assess data quality and detect any pathologies that might thwart structure solution. The quality of the reconstruction, and therefore the interpretability of a cryo-EM map, can deteriorate from many causes including structural heterogeneity, radiation damage, and beam-induced sample movement.

Data quality is typically expressed by the resolution, but this term has a different meaning for cryo-EM than crystallographic experiments. For a cryo-EM map, the overall resolution (dFSC) is usually defined as the maximum spatial resolution at which the information content of the map is reliable.

The cryo-EM resolution is typically obtained by analyzing the Fourier shell correlation (FSC) for two cryo-EM half-maps binned in resolution shells. If the macromolecule is structurally heterogeneous, a single resolution value is adequate. 'Local resolutions' can be also assigned to different map regions.

How

The phenix.mtriage program estimates the resolution of cryo-EM maps using several approaches, and then provides a summary of the map statistics. The tool requires only a full cryo-EM map (in CCP4, MRC, or other related format). However, you can obtain additional information by inputting two half-maps and/or an atomic model (in PDB or mmCIF format).

How to use the phenix.mtriage: Click here

Common issues

Program stops: All input maps are expected to be on the same grid. Also, box information (e.g., unit cell of CRYST1 record in PDB files) must match across all of the inputs.

Related programs and Documentation

- phenix.show_map_info: This program lists the properties of a map, including the origin, grid points, unit cell, and map size.
- phenix.map_box: This program cuts out a box around selected atoms from a larger map (MTZ, CCP4, or X-plor format).
- phenix.map_symmetry: Some molecules (e.g., viruses) have high internal symmetry, so you can reduce the map to the repeating unit. This program finds such symmetries, and phenix.map_box can extract the unique part of the map (using the extract-unique option).
- phenix.combine_focused_maps: This program creates a weighted composite map from a set of locally focused maps and associated models, where each part of each map is weighted by its correlation with the corresponding model.

Map Symmetry (cryo-EM)

cryo-em-map-symmetry.html

Why

Some molecules, such as viruses, have high internal symmetry so you can reduce the cryo-EM map to the repeating unit. This can decrease the required computational resources and enable better averaging for improved signal to noise.

How

The phenix.map_symmetry program uses the density at a set of randomly-chosen points in a cryo-EM map to identify symmetries, which are selected from a set of pre-defined symmetry types (e.g., point-group symmetry and helical symmetry). The tool then creates a file with the symmetry operations.

By default, phenix.map_symmetry assumes the principle axes of symmetry in a map are along x, y, and z and that the helical symmetry is along z. The user can specify a specific symmetry type or look for all elements in the database of 45 common symmetry elements and point groups.

How to use the phenix.map_symmetry: Click here

Common issues

None.

Related programs and Documentation

- phenix.map_box: This program cuts out a box around selected atoms from a larger map. It can be used to extract the unique part of the map (using the extract-unique option).
- phenix.combine_focused_maps: This program creates a weighted composite map from a set of locally focused maps and associated models, where each part of each map is weighted by its correlation with the corresponding model.

Automated model building (cryo-EM)

cryo-em-model-building.html

Why

Determining the structure of a macromolecule typically requires building an atomic model that fits the cryo-EM map. If the structure of the molecule or its components are unknown, the model has to be built from scratch. This task is challenging because molecules are typically very large and map interpretation is difficult at low resolution. Therefore, manual interpretation of cryo-EM maps is time-consuming and error-prone. For many cases, however, automated methods can be used to build models for both proteins and nucleic acids.

How

The program phenix.map_to_model interprets a cryo-EM map and automatically builds an atomic model. If desired, the map is sharpened using phenix.map_sharpening. The unique parts of the structure are then identified by taking the reconstruction symmetry into account. The procedure identifies which parts of the map correspond to protein or RNA/DNA. To save runtime, you can use phenix.map_sharpening to sharpen the map and phenix.map_box to cut out the unique part before you run phenix.map_to_model.

The phenix.map_to_model tool requires you to input a CCP4-style (MRC, etc) cryo-EM map or MTZ map coefficients and a sequence file. For optimal model-building results, the resolution of the map should be 4.5 Å or better. The program outputs a PDB file with a model that is superimposed on the original map.

How to use the phenix.map_to_model GUI: Click here

Common issues

- One or more sub-processes crash: If you have a very large structure, your computer may not have enough memory to run phenix.map_to_model. The easiest solution is to run the tool on just the unique part of the structure. You can also cut out boxes and work on each section individually. Or you can cut back on the resolution to save memory.
- Want to re-run without all the steps: Partially-completed jobs can occur when your queueing system crashes during a run or one or more sub-processes crash. If you have a partially-completed or completed run, you can carry on from where you left off by using the keyword carry_on=True. You can also use this keyword to build up your model in pieces or to run many jobs in parallel.

Related programs

- phenix.map_sharpening: This program performs map sharpening by optimizing the detail and connectivity of a cryo-EM map.
- phenix.map_box: This program cuts out a box around selected atoms from a larger map (MTZ, CCP4, or X-plor format).
- phenix.fix_insertions_deletions: This program fixes register errors in the model by comparing the density of side-chain positions to a sequence file.
- phenix.trace_and_build: This tool rapidly builds protein models by tracing the path of a polypeptide chain, finding the CB positions, and building the atomic model. Normally, this is done by running phenix.map_to_model, but you can also run phenix.trace_and_build directly.

Real-space refinement (cryo-EM)

cryo-em-real-space-refinement.html

Why

Once an atomic model has been built using a cryo-EM map, you need to optimize it to best fit the experimental data while also preserving good agreement with prior chemical knowledge. This process alternates between automated refinement, validation, and either manual or automated model corrections.

A target function guides the refinement by linking the model parameters to the experimental data and by scoring the model-versus-data fit. For cryo-EM data, refinement of the model is done in real space and the target function is formulated in terms of a three-dimensional map. Because there are generally too many model parameters, refinement requires additional restraints that modify the target function by creating relationships between independent parameters.

How

The program used to refine atomic models in real space is phenix.real_space_refine. The algorithm uses a simplified refinement target function that speeds up calculations, so optimal data-restraint weights can be identified with little runtime cost. The procedure is robust, working at resolutions from 1 to 6 Å. The input and output models can be in PDB or mmCIF formats. The input map can be in CCP4 or MTZ format.

This program makes use of restraints on covalent geometry, secondary structure, Ramachandran plots, rotamer outliers, and internal molecular symmetry. The default mode performs gradient-driven minimization of the entire model. However, optimization can also be performed using simulated annealing, morphing, rigid-body refinement, and/or systematic side-chain improvement — at the potential expense of longer runtimes.

How to use phenix.real_space_refine: Click here

Common issues

- Poorly refined model: If you use secondary structure restraints, an incorrectly refined model is likely if there are errors in the secondary structure annotation (e.g., incorrect SHEET and HELIX records in the PDB file header). Secondary structure annotation strongly depends on the quality of the input model.

Cryo-EM Structure Deposition

cryo-em-structure-deposition.html

Why

Once a protein structure is solved, it is usually deposited in the World Wide Protein Data Bank (wwPDB). This is necessary for publication as most journals require a PDB ID to be included in any manuscript that describes a new crystallographic structure. The wwPDB is an invaluable resource that contains over 190,000 structures (as of July 2021).

Procedure

The model files (PDB and mmCIF) generated by phenix.real_space_refine contain information that is required when depositing a structure to the wwPDB. However, the wwPDB currently only accepts model files in mmCIF for deposition.

To generate a model file suitable to deposition, a two stage process is currently recommended:

- By default, phenix.real_space_refine will output model files in mmCIF and PDB format. If you turned off the option for mmCIF output, run a final cycle of phenix.real_space_refine that writes mmCIF files for model and data. This can be set in the Output section of the GUI.
- You can then process the model file (mmCIF) and a sequence file with the mmtbx.prepare_pdb_deposition program to create a mmCIF file with the sequence. This program requires the full sequence for the macromolecule to be provided. In the GUI, this program is in the "PDB Deposition" section of tools.

How to use mmtbx.prepare_pdb_deposition: Click here

Related programs

- phenix.validation_cryoem: This program can be used to generate the table of statistics.
- phenix.get_pdb_validation_report: This program submits your model and data in CIF format to the PDB OneDep server to get a validation report in PDF an XML formats.

References

Announcing mandatory submission of PDBx/mmCIF format files for crystallographic depositions to the Protein Data Bank (PDB). P.D. Adams, P.V. Afonine, K. Baskaran, H.M. Berman, J. Berrisford, G. Bricogne, D.G. Brown, S.K. Burley, M. Chen, Z. Feng, C. Flensburg, A. Gutmanas, J.C. Hoch, Y. Ikegawa, Y. Kengaku, E. Krissinel, G. Kurisu, Y. Liang, D. Liebschner, L. Mak, J.L. Markley, N.W. Moriarty, G.N. Murshudov, M. Noble, E. Peisach, I. Persikova, B.K. Poon, O.V. Sobolev, E.L. Ulrich, S. Velankar, C. Vonrhein, J. Westboork, M. Wojdyr, M. Yokochi, and J.Y. Young. Acta Cryst. D75, 451-454 (2019).

- Announcing mandatory submission of PDBx/mmCIF format files for crystallographic depositions to the Protein Data Bank (PDB). P.D. Adams, P.V. Afonine, K. Baskaran, H.M. Berman, J. Berrisford, G. Bricogne, D.G. Brown, S.K. Burley, M. Chen, Z. Feng, C. Flensburg, A. Gutmanas, J.C. Hoch, Y. Ikegawa, Y. Kengaku, E. Krissinel, G. Kurisu, Y. Liang, D. Liebschner, L. Mak, J.L. Markley, N.W. Moriarty, G.N. Murshudov, M. Noble, E. Peisach, I. Persikova, B.K. Poon, O.V. Sobolev, E.L. Ulrich, S. Velankar, C. Vonrhein, J. Westboork, M. Wojdyr, M. Yokochi, and J.Y. Young. Acta Cryst. D75, 451-454 (2019).

Validation (cryo-EM)

cryo-em-validation.html

Why

Comprehensive Validation (cryo-EM) is a tool that can be used to evaluate the plausibility of a model, including its agreement with cryo-EM data. Note that for crystallographic (X-ray or neutron) data you should use instead Comprehensive Validation (X-ray).

Many steps are involved to get from the initial idea about a molecule to the final atomic model and its interpretation. Errors can occur in each of these steps, so validation procedures are necessary to ensure that the atomic models are reasonable and fit the experimental data.

In particular, validation addresses data, model, and model-versus-data quality:

- Data. The quality of experimental data, such as the resolution or amount of noise, defines the quality of the derived models. Therefore, it is vital to understand the data quality so that you use the correct model-building and refinement procedures appropriate for the data at hand. (See the Assessing Map Quality (cryo-EM) overview for more details.)
- Model. The underlying principles behind model validation are the same for any experimental method: a good model should make chemical sense and be consistent with empirical statistics for high-quality prior structures. Thus, atomic models derived from cryo-EM reconstructions should conform to prior knowledge about covalent geometry, non-bonded interactions, and known distributions of side-chain and main-chain conformations in both proteins and nucleic acids. Generally, the goal is to have as few outliers as is feasible, where outliers can be explained by their environment and/or experimental data.
- Model-versus-data fit. Model-verses-data validation criteria require the model to describe its own data well. A perfectly good model from a stereochemistry perspective may not fit the 3D reconstruction sufficiently or at all. In reciprocal space, model-versus-data agreement is assessed by curves of the Fourier shell correlation (FSC) as a function of resolution. In real space, the agreement is measured by correlation coefficients (CCs).

How

Comprehensive validation (cryo-EM) is a one-stop program in Phenix that validates model, data, and model-versus-data fit. The minimal input is a map (in MRC, CCP4, or related format) and an atomic model (in PDB or mmCIF format). However, providing a map, model, and two half-maps is desirable.

For geometric validation, Phenix uses MolProbity, which is fully integrated into the Comprehensive validation (cryo-EM) package. The graphics programs Coot and PyMOL are also fully integrated, enabling communication of the validation results.

How to use the validation tools in the Phenix GUI: Click here

Common issues

- Outliers versus errors: In most structures, there will be Ramachandran and side-chain rotamer outliers. If the data (the map) and surrounding chemistry supports the outlier conformation, then it is most likely an outlier from prior expectations. In the absence of supporting density and/or conflicting local chemistry, it is probably an error that needs correction. For example, an atomic model derived from low-resolution data is expected to have no geometrical outliers because they are unlikely to be supported by the experimental data.

Related programs

- phenix.cablam_validation: The C-Alpha Based Low-Resolution Annotation Method (CaBLAM) validates protein backbone conformations in models determined at low resolution (2.5-4 Å), where it is difficult to determine peptide orientations because carbonyl O atoms cannot be discerned in the density maps. The program uses protein C-alpha geometry to evaluate the main-chain geometry and identify areas of probable secondary structure.
- phenix.mtriage: This program computes various cryo-EM map statistics, including resolution estimates, Fourier shell correlation curves, and more.

Cryo-EM structure solution with Phenix

cryo-em_index.html

Click on boxes in the figure to read an overview of each step. Also see the AlphaFold models in Phenix overview and video tutorials for using AlphaFold models in Phenix. You can use the Phenix chatbot to get help on using any Phenix program. Just ask it any question about using Phenix. GUI documentation for most-used programs: Predict a structure with AlphaFold using PredictModel Automated structure determination using AlphaFold models with PredictAndBuild Assess the quality of your map with Mtriage Optimize a map with MapSharpening Density modify a map with RESOLVE cryo-EM Identify symmetry in a map with Map-Symmetry Extract a box with map and model with Map-Box Dock a model into a map with Dock-in-map Flexibly fit a model to a map with CryoFit Build a model with Map-to-model Refine your model against a map with phenix.real_space_refine Generate geometry restraints Build ordered water with Douse Model heterogenicity by refinement against multiple maps phenix.varref Complete list of programs for each structure solution step Other map tools: Calculate a local resolution map Combine best parts of focused maps Convert map to structure factors Segment a map Other model building tools: Rapid model-building Guess sequences from map Compare CA/P in two models Adjust a CA/CB model Fix register errors

GUI documentation for most-used programs:

- Predict a structure with AlphaFold using PredictModel
- Automated structure determination using AlphaFold models with PredictAndBuild
- Assess the quality of your map with Mtriage
- Optimize a map with MapSharpening
- Density modify a map with RESOLVE cryo-EM
- Identify symmetry in a map with Map-Symmetry
- Extract a box with map and model with Map-Box
- Dock a model into a map with Dock-in-map
- Flexibly fit a model to a map with CryoFit
- Build a model with Map-to-model
- Refine your model against a map with phenix.real_space_refine
- Generate geometry restraints
- Build ordered water with Douse
- Model heterogenicity by refinement against multiple maps phenix.varref
- Complete list of programs for each structure solution step

Other map tools: Calculate a local resolution map Combine best parts of focused maps Convert map to structure factors Segment a map

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Auto-sharpen (deprecated, use MapSharpening)

auto_sharpen.html

Author(s)

- auto_sharpen: Tom Terwilliger

Purpose

The routine auto_sharpen will automatically identify optimal sharpening/blurring/map adjustment for the input map and will write out an optimized version of the map.

Video Tutorial

The tutorial video is available on the Phenix YouTube channel and covers the following topics:

- basic overview
- how to run phenix.map_to_model via the GUI

GUI

A Graphical User Interface is available.

Usage

How auto_sharpen works:

Auto-sharpen adjusts the resolution dependence of the map to maximize the clarity of the map. You can choose to use map kurtosis or the adjusted surface area of the map which is the default for this purpose.

The auto-sharpen tool works best at resolutions of about 4.5 A or better. At lower resolutions it may be unable to distinguish variations in the quality of the map as the sharpening is varied.

Kurtosis is a standard statistical measure that reflects the peakiness of the map.

The adjusted surface area is a combination of the surface area of contours in the map at a particular threshold and of the number of distinct regions enclosed by the top 30% (default) of those contours. The threshold is chosen by default to be one where the volume enclosed by the contours is 20% of the non-solvent volume in the map. The weighting between the surface area (to be maximized) and number of regions enclosed (to be minimized) is chosen empirically (default region_weight=20).

Several resolution-dependent functions are tested, and the one that gives the best adjusted surface area (or kurtosis) is chosen. In each case the map is transformed to obtain Fourier coefficients. The amplitudes of these coefficients are then adjusted, keeping the phases constant. The available functions for modifying the amplitudes are:

```
No sharpening (map is left as is)

Sharpening b-factor applied over entire resolution range (b_sharpen
applied to achieve an effective isotropic overall b-value of b_iso).

Sharpening b-factor applied up to resolution specified with the
resolution=xxx keyword, then blurred beyond this resolution (with
```

```
transition specified by the keyword k_sharpen, b_iso_to_d_cut). If
the sharpening b_sharpen is negative (blurring the map),
the blurring is applied over the entire resolution range.

Resolution-dependent sharpening factor with three parameters.
First the resolution-dependence of the map is removed by normalizing the
amplitudes. Then a scale factor S is to the data, where
log10(S) is determined by coefficients b[0],b[1],b[2] and a resolution
d_cut (typically d_cut is the nominal resolution of the map).
The value of log10(S) varies smoothly from 0 at resolution=infinity, to b[0]
at d_cut/2, to b[1] at d_cut, and to b[1]+b[2] at the highest resolution
in the map. The value of b[1] is limited to being no larger than b[0] and the
value of b[1]+b[2] is limited to be no larger than b[1].
```

You can also choose to specify the sharpening/blurring parameters for your map and they will simply be applied to the map. For example you can apply a sharpening B-value (b_sharpen) to sharpen the map, or you can specify a target overall B-value (b_iso) to obtain after sharpening.

Box of density in sharpening

Normally phenix.auto_sharpen will determine the optimal sharpening by examining the density in a box cut out of your map, then apply this to the entire map.

Local sharpening

You can choose to apply autosharpening locally if you want. In this case the auto-sharpening parameters are determined in many boxes cut out of the map, and corresponding sharpened maps are calculated. The map that is produced is a weighted map where the density at a particular point comes most from the sharpened map based on a box near that point.

Half-map-based sharpening

You can identify the sharpening parameters using two half-maps if you want. The resolution-dependent correlation of density in the two half-maps is used to identify the optimal resolution-dependent weighting of the map. This approach requires a target resolution which is used to set the overall fall-off with resolution for an ideal map. That fall-off for an ideal map is then multiplied by an estimated resolution-dependent correlation of density in the map with the true map (the estimation comes from the half-map correlations).

Model-based sharpening

You can identify instead the sharpening parameters using your map and a model. This approach requires a guess of the RMSD between the model and the true model. The resolution-dependent correlation of model and map density is used as in the half-map approach above to identify the weighting of Fourier coefficients.

Using crystallographic maps

You can use phenix.auto_sharpen with a crystallographic map (represented as map coefficients).

Shifting the map to the origin

Most crystallographic maps have the origin at the corner of the map (grid point [0,0,0]), while most cryo-EM maps have the orgin in the middle of the map. An output map with the origin shifted to the corner of the map is optionally written out.

Output files from auto_sharpen

sharpened_map.ccp4: Sharpened map.

shifted_sharpened_map.ccp4: Sharpened map, shifted to place the origin on grid point (0,0,0) and sharpened

sharpened_map_coeffs.mtz: Sharpened map, shifted to place the origin on grid point (0,0,0) and sharpened, represented as map coefficients.

Examples

Standard run of auto_sharpen:

Running auto_sharpen is easy. From the command-line you can type:

```
phenix.auto_sharpen my_map.map resolution=2.6
```

where my_map.map is a CCP4, mrc or other related map format, and you specify the nominal resolution of the map.

Possible Problems

Specific limitations and problems:

Maps produced with the extract-unique option of map_box should not be sharpened with auto-sharpen. These maps are closely masked around the density of a single molecule and are set to zero in much of the map, so the information about noise in the map that normally is available is missing and auto-sharpen does not work properly.

Literature

Automated map sharpening by maximization of detail and connectivity. T.C. Terwilliger, P.V. Afonine, Sobolev, OV, and P.D. Adams. Acta Cryst. D74, 545-559 (2018).

- Automated map sharpening by maximization of detail and connectivity. T.C. Terwilliger, P.V. Afonine, Sobolev, OV, and P.D. Adams. Acta Cryst. D74, 545-559 (2018).

Additional information

List of all available keywords

job_title = None Job title in PHENIX GUI, not used on command lineinput_filesmap_file = None File with CCP4-style maphalf_map_file = None Half map (two should be supplied) for FSC calculation. Must have grid identical to map_fileexternal_map_file = None External map to be used to scale map_file (power vs resolution will be matched)map_coeffs_file = None Optional file with map coefficientsmap_coeffs_labels = None Optional label specifying which columns of of map coefficients to usepdb_file = None If a model is supplied, the map will be adjusted to maximize map-model correlation. This can be used to improve a map in regions where no model is yet built.ncs_file = None File with NCS information (typically point-group NCS with the center specified). Typically in PDB format. Can also be a .ncs_spec file from phenix. Created automatically if symmetry is specified.seq_file = None Sequence file (unique chains only, 1-letter code, chains separated by blank line or greater-than sign.) Can have chains that are DNA/RNA/protein and all can be present in one file.input_weight_map_pickle_file = None Weight map pickle fileoutput_filesshifted_map_file =

shifted_map.ccp4 Input map file shifted to new origin.sharpened_map_file = None Sharpened input map file. In the same location as input map.shifted_sharpened_map_file = None Input map file shifted to place origin at 0,0,0 and sharpened.sharpened_map_coeffs_file = None Sharpened input map (shifted to new origin if original origin was not 0,0,0), written out as map coefficientsoutput_weight_map_pickle_file = weight_map_pickle_file.pkl Output weight map pickle fileoutput_directory = None Directory where output files are to be written applied.crystal_infois_crystal = None Defines whether this is a crystal (or cryo-EM). Default is True if use_sg_symmetry=True and False otherwise.resolution = None Optional nominal resolution of the map.solvent_content = None Optional solvent fraction of the cell.solvent_content_iterations = 3 Iterations of solvent fraction estimation. Used for ID of solvent content in boxed maps.molecular_mass = None Molecular mass of molecule in Da. Used as alternative method of specifying solvent content.ncs_copies = None You can specify ncs copies and seq file to define solvent contentwang_radius = None Wang radius for solvent identification. Default is 1.5* resolutionbuffer_radius = None Buffer radius for mask smoothing. Default is resolutionpseudo_likelihood = None Use pseudo-likelihood method for half-map sharpening. (In development)map_modificationb_iso = None Target B-value for map (sharpening will be applied to yield this value of b_iso). If sharpening method is not supplied, default is to use b_iso_to_d_cut sharpening.b_sharpen = None Sharpen with this b-value. Contrast with b_iso that yields a targeted value of b_isob_blur_hires = 200 Blur high_resolution data (higher than d_cut) with this b-value. Contrast with b_sharpen applied to data up to d_cut. Note on defaults: If None and b_sharpen is positive (sharpening) then high-resolution data is left as is (not sharpened). If None and b_sharpen is negative (blurring) high-resolution data is also blurred.resolution_dependent_b = None If set, apply resolution_dependent_b (b0 b1 b2). Log10(amplitudes) will start at 1, change to b0 at half of resolution specified, changing linearly, change to b1/2 at resolution specified, and change to b1/2+b2 at d_min_ratio*resolutionnormalize_amplitudes_in_resdep = False Normalize amplitudes in resolution-dependent sharpeningd_min_ratio = 0.833 Sharpening will be applied using d_min equal to d_min_ratio times resolution. Default is 0.833scale_max = 100000 Scale amplitudes from inverse FFT to yield maximum of this valueinput_d_cut = None High-resolution limit for sharpeningrmsd = None RMSD of model to true model (if supplied). Used to estimate expected fall-of with resolution of correct part of model-based map. If None, assumed to be resolution times rmsd_resolution_factor.rmsd_resolution_factor ...

- job_title = None Job title in PHENIX GUI, not used on command line
- input_filesmap_file = None File with CCP4-style maphalf_map_file = None Half map (two should be supplied) for FSC calculation. Must have grid identical to map_fileexternal_map_file = None External map to be used to scale map_file (power vs resolution will be matched)map_coeffs_file = None Optional file with map coefficientsmap_coeffs_labels = None Optional label specifying which columns of of map coefficients to usepdb_file = None If a model is supplied, the map will be adjusted to maximize map-model correlation. This can be used to improve a map in regions where no model is yet built.ncs_file = None File with NCS information (typically point-group NCS with the center specified). Typically in PDB format. Can also be a .ncs_spec file from phenix. Created automatically if symmetry is specified.seq_file = None Sequence file (unique chains only, 1-letter code, chains separated by blank line or greater-than sign.) Can have chains that are DNA/RNA/protein and all can be present in one file.input_weight_map_pickle_file = None Weight map pickle file
- map_file = None File with CCP4-style map
- half_map_file = None Half map (two should be supplied) for FSC calculation. Must have grid identical to map_file
- external_map_file = None External map to be used to scale map_file (power vs resolution will be matched)
- map_coeffs_file = None Optional file with map coefficients

- map_coeffs_labels = None Optional label specifying which columns of of map coefficients to use
- pdb_file = None If a model is supplied, the map will be adjusted to maximize map-model correlation. This can be used to improve a map in regions where no model is yet built.
- ncs_file = None File with NCS information (typically point-group NCS with the center specified). Typically in PDB format. Can also be a .ncs_spec file from phenix. Created automatically if symmetry is specified.
- seq_file = None Sequence file (unique chains only, 1-letter code, chains separated by blank line or greater-than sign.) Can have chains that are DNA/RNA/protein and all can be present in one file.
- input_weight_map_pickle_file = None Weight map pickle file
- output_filesshifted_map_file = shifted_map.ccp4 Input map file shifted to new origin.sharpened_map_file = None Sharpened input map file. In the same location as input map.shifted_sharpened_map_file = None Input map file shifted to place origin at 0,0,0 and sharpened.sharpened_map_coeffs_file = None Sharpened input map (shifted to new origin if original origin was not 0,0,0), written out as map coefficientsoutput_weight_map_pickle_file = weight_map_pickle_file.pkl Output weight map pickle fileoutput_directory = None Directory where output files are to be written applied.
- shifted_map_file = shifted_map.ccp4 Input map file shifted to new origin.
- sharpened_map_file = None Sharpened input map file. In the same location as input map.
- shifted_sharpened_map_file = None Input map file shifted to place origin at 0,0,0 and sharpened.
- sharpened_map_coeffs_file = None Sharpened input map (shifted to new origin if original origin was not 0,0,0), written out as map coefficients
- output_weight_map_pickle_file = weight_map_pickle_file.pkl Output weight map pickle file
- output_directory = None Directory where output files are to be written applied.
- crystal_infois_crystal = None Defines whether this is a crystal (or cryo-EM). Default is True if use_sg_symmetry=True and False otherwise.resolution = None Optional nominal resolution of the map.solvent_content = None Optional solvent fraction of the cell.solvent_content_iterations = 3 Iterations of solvent fraction estimation. Used for ID of solvent content in boxed maps.molecular_mass = None Molecular mass of molecule in Da. Used as alternative method of specifying solvent content.ncs_copies = None You can specify ncs copies and seq file to define solvent contentwang_radius = None Wang radius for solvent identification. Default is 1.5* resolutionbuffer_radius = None Buffer radius for mask smoothing. Default is resolutionpseudo_likelihood = None Use pseudo-likelihood method for half-map sharpening. (In development)
- is_crystal = None Defines whether this is a crystal (or cryo-EM). Default is True if use_sg_symmetry=True and False otherwise.
- resolution = None Optional nominal resolution of the map.
- solvent_content = None Optional solvent fraction of the cell.
- solvent_content_iterations = 3 Iterations of solvent fraction estimation. Used for ID of solvent content in boxed maps.
- molecular_mass = None Molecular mass of molecule in Da. Used as alternative method of specifying solvent content.
- ncs_copies = None You can specify ncs copies and seq file to define solvent content
- wang_radius = None Wang radius for solvent identification. Default is 1.5* resolution

- buffer_radius = None Buffer radius for mask smoothing. Default is resolution
- pseudo_likelihood = None Use pseudo-likelihood method for half-map sharpening. (In development)
- map_modificationb_iso = None Target B-value for map (sharpening will be applied to yield this value of b_iso). If sharpening method is not supplied, default is to use b_iso_to_d_cut sharpening.b_sharpen = None Sharpen with this b-value. Contrast with b_iso that yields a targeted value of b_isob_blur_hires = 200 Blur high_resolution data (higher than d_cut) with this b-value. Contrast with b_sharpen applied to data up to d_cut. Note on defaults: If None and b_sharpen is positive (sharpening) then high-resolution data is left as is (not sharpened). If None and b_sharpen is negative (blurring) high-resolution data is also blurred.resolution_dependent_b = None If set, apply resolution_dependent_b (b0 b1 b2). Log10(amplitudes) will start at 1, change to b0 at half of resolution specified, changing linearly, change to b1/2 at resolution specified, and change to b1/2+b2 at d_min_ratio*resolutionnormalize_amplitudes_in_resdep = False Normalize amplitudes in resolution-dependent sharpeningd_min_ratio = 0.833 Sharpening will be applied using d_min equal to d_min_ratio times resolution. Default is 0.833scale_max = 100000 Scale amplitudes from inverse FFT to yield maximum of this valueinput_d_cut = None High-resolution limit for sharpeningrmsd = None RMSD of model to true model (if supplied). Used to estimate expected fall-of with resolution of correct part of model-based map. If None, assumed to be resolution times rmsd_resolution_factor.rmsd_resolution_factor = 0.25 default RMSD is resolution times resolution factorfraction_complete = None Completness of model (if supplied). Used to estimate correct part of model-based map. If None, estimated from max(FSC).auto_sharpen = True Automatically determine sharpening using kurtosis maximization or adjusted surface area. Default is Trueauto_sharpen_methods = no_sharpening b_iso *b_iso_to_d_cut resolution_dependent model_sharpening half_map_sharpening target_b_iso_to_d_cut external_map_sharpening None Methods to use in sharpening. b_iso searches for b_iso to maximize sharpening target (kurtosis or adjusted_sa). b_iso_to_d_cut applies b_iso only up to resolution specified, with fall-over of k_sharpen. Resolution dependent adjusts 3 parameters to sharpen variably over resolution range. Default is b_iso_to_d_cut . target_b_iso_to_d_cut uses target_b_iso_ratio to set b_iso.box_in_auto_sharpen = False Use a representative box of density for initial auto-sharpening instead of the entire map. Default is False.density_select_in_auto_sharpen = True Choose representative box of density for initial auto-sharpening with density_select method (choose region where there is high density). Normally use False for X-ray data and True for cryo-EM.density_select_threshold_in_auto_sharpen = None Threshold for density select choice of box. Default is 0.05. If your map has low overall contrast you might need to make this bigger such as 0.2.allow_box_if_b_iso_set = False Allow box_in_auto_sharpen (if set to True) even if b_iso is set. Default is to set box_n_auto_sharpen=False if b_iso is set.soft_mask = True Use soft mask (smooth change from inside to outside with radius based on resolution of map).use_weak_density = False When choosing box of representative density, use poor density (to get optimized map for weaker density)discard_if_worse = None Discard sharpening if worselocal_sharpening = None Sharpen locally using overlapping regions. NOTE: Best to turn off local_aniso_in_local_sharpening if NCS is present. If local_aniso_in_local_sharpening is True and NCS is present this can distort the map for some NCS copies because an anisotropy correction is applied based on local density in one copy and is transferred without rotation to other copies.local_aniso_in_local_sharpening = None Use local anisotropy in local sharpening. Default is True unless NCS is present.overall_before_local = True Apply overall scaling before local scalingselect_sharpened_map = None Select a single sharpened map to useread_sharpened_maps = No...
- b_iso = None Target B-value for map (sharpening will be applied to yield this value of b_iso). If sharpening method is not supplied, default is to use b_iso_to_d_cut sharpening.
- b_sharpen = None Sharpen with this b-value. Contrast with b_iso that yields a targeted value of b_iso

- b_blur_hires = 200 Blur high_resolution data (higher than d_cut) with this b-value. Contrast with b_sharpen applied to data up to d_cut. Note on defaults: If None and b_sharpen is positive (sharpening) then high-resolution data is left as is (not sharpened). If None and b_sharpen is negative (blurring) high-resolution data is also blurred.
- resolution_dependent_b = None If set, apply resolution_dependent_b (b0 b1 b2). Log10(amplitudes) will start at 1, change to b0 at half of resolution specified, changing linearly, change to b1/2 at resolution specified, and change to b1/2+b2 at d_min_ratio*resolution
- normalize_amplitudes_in_resdep = False Normalize amplitudes in resolution-dependent sharpening
- d_min_ratio = 0.833 Sharpening will be applied using d_min equal to d_min_ratio times resolution. Default is 0.833
- scale_max = 100000 Scale amplitudes from inverse FFT to yield maximum of this value
- input_d_cut = None High-resolution limit for sharpening
- rmsd = None RMSD of model to true model (if supplied). Used to estimate expected fall-of with resolution of correct part of model-based map. If None, assumed to be resolution times rmsd_resolution_factor.
- rmsd_resolution_factor = 0.25 default RMSD is resolution times resolution factor
- fraction_complete = None Completness of model (if supplied). Used to estimate correct part of model-based map. If None, estimated from max(FSC).
- auto_sharpen = True Automatically determine sharpening using kurtosis maximization or adjusted surface area. Default is True
- auto_sharpen_methods = no_sharpening b_iso *b_iso_to_d_cut resolution_dependent model_sharpening half_map_sharpening target_b_iso_to_d_cut external_map_sharpening None Methods to use in sharpening. b_iso searches for b_iso to maximize sharpening target (kurtosis or adjusted_sa). b_iso_to_d_cut applies b_iso only up to resolution specified, with fall-over of k_sharpen. Resolution dependent adjusts 3 parameters to sharpen variably over resolution range. Default is b_iso_to_d_cut . target_b_iso_to_d_cut uses target_b_iso_ratio to set b_iso.
- box_in_auto_sharpen = False Use a representative box of density for initial auto-sharpening instead of the entire map. Default is False.
- density_select_in_auto_sharpen = True Choose representative box of density for initial auto-sharpening with density_select method (choose region where there is high density). Normally use False for X-ray data and True for cryo-EM.
- density_select_threshold_in_auto_sharpen = None Threshold for density select choice of box. Default is 0.05. If your map has low overall contrast you might need to make this bigger such as 0.2.
- allow_box_if_b_iso_set = False Allow box_in_auto_sharpen (if set to True) even if b_iso is set. Default is to set box_n_auto_sharpen=False if b_iso is set.
- soft_mask = True Use soft mask (smooth change from inside to outside with radius based on resolution of map).
- use_weak_density = False When choosing box of representative density, use poor density (to get optimized map for weaker density)
- discard_if_worse = None Discard sharpening if worse
- local_sharpening = None Sharpen locally using overlapping regions. NOTE: Best to turn off local_aniso_in_local_sharpening if NCS is present. If local_aniso_in_local_sharpening is True and NCS is present this can distort the map for some NCS copies because an anisotropy correction is applied

based on local density in one copy and is transferred without rotation to other copies.

- local_aniso_in_local_sharpening = None Use local anisotropy in local sharpening. Default is True unless NCS is present.
- overall_before_local = True Apply overall scaling before local scaling
- select_sharpened_map = None Select a single sharpened map to use
- read_sharpened_maps = None Read in previously-calculated sharpened maps
- write_sharpened_maps = None Write out local sharpened maps
- smoothing_radius = None Sharpen locally using smoothing_radius. Default is 2/3 of mean distance between centers for sharpening
- box_center = None You can specify the center of the box (A units)
- box_size = 30 30 30 You can specify the size of the box (grid units)
- target_n_overlap = 10 You can specify the targeted overlap of boxes in local sharpening
- restrict_map_size = None Restrict box map to be inside full map (required for cryo-EM data). Default is True if use_sg_symmetry=False and False if use_sg_symmetry=True
- remove_aniso = True You can remove anisotropy (overall and locally) during sharpening
- max_box_fraction = 0.5 If box is greater than this fraction of entire map, use entire map. Default is 0.5.
- density_select_max_box_fraction = 0.95 If box is greater than this fraction of entire map, use entire map for density_select. Default is 0.95
- cc_cut = 0.2 Estimate of minimum highly reliable CC in half-map FSC. Used to decide at what CC value to smooth the remaining CC values.
- max_cc_for_rescale = 0.2 Used along with cc_cut and scale_using_last to correct for small errors in FSC estimation at high resolution. If the value of FSC near the high-resolution limit is above max_cc_for_rescale, assume these values are correct and do not correct them.
- scale_using_last = 3 If set, assume that the last scale_using_last bins in the FSC for half-map or model sharpening are about zero (corrects for errors int the half-map process).
- mask_atoms = True Mask atoms when using model sharpening
- mask_atoms_atom_radius = 3 Mask for mask_atoms will have mask_atoms_atom_radius
- value_outside_atoms = None Value of map outside atoms (set to mean to have mean value inside and outside mask be equal)
- k_sharpen = 10 Steepness of transition between sharpening (up to resolution) and not sharpening (d < resolution). Note: for blurring, all data are blurred (regardless of resolution), while for sharpening, only data with d about resolution or lower are sharpened. This prevents making very high-resolution data too strong. Note 2: if k_sharpen is zero or None, then no transition is applied and all data is sharpened or blurred.
- iterate = False You can iterate auto-sharpening. This is useful in cases where you do not specify the solvent content and it is not accurately estimated until sharpening is optimized.
- optimize_b_blur_hires = False Optimize value of b_blur_hires. Only applies for auto_sharpen_methods b_iso_to_d_cut and b_iso. This is normally carried out and helps prevent over-blurring at high resolution if the same map is sharpened more than once.
- optimize_d_cut = None Optimize value of d_cut. Only applies for auto_sharpen_methods b_iso_to_d_cut and b_iso

• adjust_region_weight = True Adjust region_weight to make overall change in surface area equal to overall change in normalized regions over the range of search_b_min to search_b_max using b_iso_to_d_cut.

• region_weight_method = initial_ratio *delta_ratio b_iso Method for choosing region_weights. Initial_ratio uses ratio of surface area to regions at low B value. Delta ratio uses change in this ratio from low to high B. B_iso uses resolution-dependent b_iso (not weights) with the formula b_iso=5.9*d_min**2

• region_weight_factor = 1.0 Multiplies region_weight after calculation with region_weight_method above

• region_weight_buffer = 0.1 Region_weight adjusted to be region_weight_buffer away from minimum or maximum values

• target_b_iso_ratio = 5.9 Target b_iso ratio : b_iso is estimated as target_b_iso_ratio * resolution**2

• target_b_iso_model_scale = 0. For model sharpening, the target_biso is scaled (normally zero).

• signal_min = 3.0 Minimum signal in estimation of optimal b_iso. If not achieved, use any other method chosen.

• search_b_min = None Low bound for b_iso search. Default is -100.

• search_b_max = None High bound for b_iso search. Default is 300.

• search_b_n = None Number of b_iso values to search. Default is 21.

• residual_target = None Target for maximization steps in sharpening. Can be kurtosis or adjusted_sa (adjusted surface area). Default is adjusted_sa.

• sharpening_target = None Overall target for sharpening. Can be kurtosis or adjusted_sa (adjusted surface area). Used to decide which sharpening approach is used. Note that during optimization, residual_target is used (they can be the same.) Default is adjusted_sa.

• require_improvement = None Require improvement in score for sharpening to be applied. Default is True.

• region_weight = None Region weighting in adjusted surface area calculation. Score is surface area minus region_weight times number of regions. Default is set automatically. A smaller value will give more sharpening.

• sa_percent = None Percent of target regions used in calulation of adjusted surface area. Default is 30.

• fraction_occupied = None Fraction of molecular volume targeted to be inside contours. Used to set contour level. Default is 0.20

• n_bins = None Number of resolution bins for sharpening. Default is 20.

• regions_to_keep = None You can specify a limit to the number of regions to keep when generating the asymmetric unit of density.

• max_regions_to_test = None Number of regions to test for surface area in adjusted_sa scoring of sharpening. Default is 30

• eps = None

• k_sol = 0.35 k_sol value for model map calculation. IGNORED (Not applied)

• b_sol = 50 b_sol value for model map calculation. IGNORED (Not applied)

• controlverbose = False Verbose outputresolve_size = None Size of resolve to use. ignore_map_limitations = None Ignore limitations such as map cannot be sharpenedmultiprocessing = *multiprocessing sge lsf pbs condor pbspro slurm Choices are multiprocessing (single machine) or queuing systemsqueue_run_command = None run command for queue jobs. For example qsub.nproc = 1 Number of processors to use

- verbose = False Verbose output
- resolve_size = None Size of resolve to use.
- ignore_map_limitations = None Ignore limitations such as map cannot be sharpened
- multiprocessing = *multiprocessing sge lsf pbs condor pbspro slurm Choices are multiprocessing (single machine) or queuing systems
- queue_run_command = None run command for queue jobs. For example qsub.
- nproc = 1 Number of processors to use
- guiGUI-specific parameter required for output directoryoutput_dir = None
- output_dir = None

Density modification with with phenix.density_modification

density_modification.html

Density modification with with phenix.density_modification

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Author(s)

- phenix.density_modification: Tom Terwilliger

Purpose

phenix.density_modification is a tool to run Resolve to carry out density modification, including the use of NCS symmetry and electron-density distributions. Masks for the solvent boundary and for the region where NCS operators are to be applied can be specified as PDB files with dummy atoms marking the masked regions.

Usage

How phenix.density_modification works

In phenix.density_modification phases are improved by combining any existing phase information with phase probabilities based on the agreement of your electron density map with expectations about that map. This procedure is known as statisical density modification and is carried out by phenix.resolve. The expectations about a map include:

- A flat solvent region
- Matching NCS-related density
- Density distributions (histograms of density) matching expected distributions
- Density in a region where a model has been built matching expected density for the model

The inputs to phenix.density_modification are :

Required:

- F, SIGF: Structure factor amplitudes for your structure.

Optional:

- PHIB, FOM, HLA HLB HLC HLC: Phases and figure of merit and Hendrickson Lattman (HL) coefficients. These normally come from experimental phasing, but they can be from any source. These are optional if you supply a model. The HL coefficients describe the phase probabilities for each phase.
- Model: This is normally a PDB file containing your current model. It can be used as a source of phase information (calculated starting phases) and also as a target for density modification.
- NCS operators: These are supplied as a .ncs_spec file that you can create by running phenix.find_ncs on a model with NCS or using heavy-atom sites and a map or just with a map.
- Solvent boundary file: This is a PDB file that defines the region that is not solvent (the macromolecule) as dummy atoms.
- NCS boundary file: This is a PDB file that defines the region where NCS is to be applied (all the NCS copies) as dummy atoms.

Density modification starting with phases and HL coefficients

This kind of density modification is normally carried out after experimental phasing, but it can also be carried out if you have created HL coefficients from your data and a model.

In these cases you have phases and phase probability information (F, SIGF, PHIB FOM, HLA HLB HLC HLD).

Prior to density modification, the amplitudes (F, SIGF) are normally corrected for anisotropy and sharpened. You can control this by specifying the target overall B value after sharpening. By default the target overall (Wilson) B is 10 times the high-resolution limit of the data (or the resolution that you specify), up to a maximum of 40 (you can change that too).

A starting map is calculated from F, PHIB, FOM. The location of the macromolecule and solvent are estimated in a probabilistic way using the local variation in the map (with a smoothing radius of rad_wang) to discriminate between the two.

Phases are optimized to agree with the HL coefficients and to yield a map that has a flat solvent, NCS (if present), and density distributions matching model maps.

Density modification with NCS

If you supply NCS operators, phenix.density_modification will automatically figure out the region over which this NCS is present. Within this region, it will weight the density for NCS calculations according to how similar the density is at NCS-related locations. If you supply an NCS mask, that is used instead. The density at NCS-related locations is used as part of the target for density modification.

Density modification with a model

If you supply a model, density modification occurs in several steps. First the data are corrected for anisotropy as above. Then the model is used to estimate phases, fom, and HL coefficients. Then density modification is carried out much as described above for experimental phasing. Then a final step is carried out in which model density is used as part of the target for density modification.

To instead create HL coefficients from your data and model and use those with phenix.density_modification, first run phenix.refine to refine your model. Then run phenix.reciprocal_space_arrays with your model and data and it will create HL coefficients for you.

Note on R-values in density modification

The R-value for density modification is quite different from a refinement R. Basically the statistical (maximum-likelihood) density modification procedure allows calculation of each structure factor based on the values of all the other structure factors and the expectations about the map (i.e, flat solvent, distribution of density values in protein, ncs, etc.) These "map-phasing" structure factors are (at least in the first cycle) independent of the starting structure factors and can be compared with the measured data and an R-value obtained. The values of these R-values range from about 0.25 to 0.5 in most cases, and when the map is very good usually the R values are smaller. These R values do not involve any atomic model so they are quite unlike a refinement R.

For discussion of how map-likelihood structure factors are calculated in Resolve, see the Map-likelihood phasing reference below.

Output files from phenix.density_modification

denmod.mtz: Density-modified map coefficients. If you specify

hl_in_output_mtz=True then it will also contain Hendrickson Lattman coefficients

solvent_boundary.map: CCP4-style map file showing the solvent boundary

ncs_boundary.map: CCP4-style map file showing the asymmetric unit of NCS

Examples

Standard run of phenix.density_modification

Running phenix.density_modification from the command line is easy.

```
phenix.phenix.density_modification phaser_1.mtz solvent_content=0.5
```

In this example, phaser_1.mtz (the output of Phaser experimental phasing) normally has F, SIGF, PHIB, FOM, HLA HLB HLC HLD, each of which is identified automatically. Standard density modification using these experimental phases and phase probabilities is carried out.

Run specifying resolution, mask type, and output mtz

```
phenix.phenix.density_modification phaser_1.mtz solvent_content=0.64 \
resolution=3 output_mtz=denmod_std.mtz mask_type=histograms
```

In this example, the high-resolution limit is set to 3 A, the output file is denmod_std.mtz, and the solvent mask is identified using a histogram-based method (mask_type=histograms).

Run specifying a mask for the solvent boundary

```
phenix.phenix.density_modification phaser_1.mtz solvent_content=0.64 \
mask_from_pdb=mask.pdb rad_mask=4
```

In this example, the file mask.pdb has dummy atoms that indicate where the macromolecule is located. All points within 4 A (rad_mask=4) of an atom in mask.pdb are considered to be within the macromolecule region.

Run specifying NCS operators

```
phenix.phenix.density_modification phaser_1.mtz solvent_content=0.64 \
ncs_file=ncs.ncs_spec
```

In this example, the file ncs.ncs_spec (created with phenix.find_ncs or phenix.autosol or phenix.simple_ncs_from_pdb or phenix.autobuild) contains the NCS operators. These will be used to automatically find the region where NCS applies. Then the NCS-related density will be used as part of density modification.

Run specifying NCS operators and the region where they are to apply

```
phenix.phenix.density_modification phaser_1.mtz solvent_content=0.64 \
ncs_file=ncs.ncs_spec ncs_domain_pdb=ncs_mask.pdb rad_mask=5
```

This example is like the previous one, except instead of finding the region where NCS applies automatically, it is defined by the atoms in ncs_mask.pdb. All points within rad_mask (5 A in this example) of an atom in ncs_mask.pdb are considered to be within the region where NCS applies. This is useful if you want to apply NCS only to a part of the region where NCS is present.

Possible Problems

Specific limitations and problems:

Literature

Rapid automatic NCS identification using heavy-atom substructures. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 58, 2213-5 (2002).

- Rapid automatic NCS identification using heavy-atom substructures. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 58, 2213-5 (2002).

Statistical density modification with non-crystallographic symmetry. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 58, 2082-6 (2002).

- Statistical density modification with non-crystallographic symmetry. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 58, 2082-6 (2002).

Maximum-likelihood density modification. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 56, 965-72 (2000).

- Maximum-likelihood density modification. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 56, 965-72 (2000).

Map-likelihood phasing. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 57, 1763-75 (2001).

- Map-likelihood phasing. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 57, 1763-75 (2001).

Additional information

List of all available keywords

input_filesdata_file = None Mtz or other data file with intensities or amplitudesdata_labels = None Optional data labels for Iobs,SIGIobs or Fobs,SIGFobs. Input data can be I, F, or I+/I- or F+/F-.phib_labels = None Optional data label for PHIB. You can force skipping with phib_labels=SKIPfom_labels = None Optional data label for FOM. You can force skipping with fom_labels=SKIPhl_file = None Mtz or other data file with HL coefficients. If None assumed to be the same as data_file. Required unless a pdb_file is supplied.hl_labels = None Optional labels for HL coefficients. You can force skipping with hl_labels=SKIPmap_coeffs_file = None Mtz or other file with map coefficients for starting map. If None assumed to be same as data_filemap_coeffs_labels = None Optional labels for map coefficients. You can force skipping with map_coeffs_labels=SKIPpdb_file = None Optional file with model to be used in density modification. If provided, the density modification will use model density as part of the density modification target. If supplied, HL coefficients are optional.ha_file = None Optional file with heavy-atom sites to be truncated in density modification. Density within rad_mask of a site in this file will be truncated at 2 x the rms of the map. Note: not compatible with mask_from_pdb.mask_from_pdb = None Optional file with dummy atoms marking the region to be considered as the macromolecule in density modification. The region within rad_mask of an atom in this file will be considered within the macromolecule. Note: not compatible with ha_file.add_mask = None Mask will be the union of the standard mask and mask_from_pdbncs_file = None Optional file with NCS information (.ncs_spec file). This file can be obtained with phenix.find_ncs or phenix.simple_ncs_from_pdbncs_domain_pdb = None Optional file with dummy atoms marking the region where NCS is to be applied. Supersedes ncs_domain_pdb in the ncs_file. Points within rad_mask of an atom in this file are considered for NCS. You need to supply a mask covering all NCS copies. NOTE: the output_ncs_mask_file map shows the boundary of the asymmetric unit of NCS (just one copy) while the input ncs_domain_pdb file must contain all NCS copies.resolve_commands_file = None Optional file with any commands for resolve density modification.directoriestemp_dir = temp_denmod Temporary directory. Not normally used. Contents deleted after the run if clean_up=True is specified.crystal_inforesolution = None Resolution for density modificationoutput_filesoutput_mtz = denmod.mtz Output MTZ file with density-modified map coefficientshl_in_output_mtz = True Calculate density modification HL coefficients and write to output mtz filetarget_map_file_name = target_map.ccp4 Output CCP4-style map with target map. only written if write_target_map_and_sigma is setsigma_map_file_name = sigma_map.ccp4 Output CCP4-style map with sigma map. only written if write_target_map_and_sigma is setoutput_mask_file = solvent_boundary.map Output CCP4-style map file with solvent boundary mask.output_ncs_mask_file = ncs_boundary.map Output CCP4-style map file with mask marking region used for NCS. NOTE: the output_ncs_mask_file map shows the boundary of the asymmetric unit of NCS (just one copy) while the input ncs_domain_pdb file must contain all NCS copies.anisotropyremove_aniso = True Remove anisotropy from data and optionally sharpenb_iso = None Target overall B value for anisotropy correction. Ignored if remove_aniso = False. If None, default is minimum of (max_b_iso, B of dataset, target_b_ratio*resolution)max_b_iso = 40. Default maximum overall B value for anisotropy correction. Ignored if remove_aniso = False. Ignored if b_iso is set. If used, default is minimum of (max_b_iso, B of dataset, target_b_ratio*resolution)target_b_ratio = 10. Default ratio of target B value to resolution for anisotropy correction. Ignored if remove_aniso = False. Ignored if b_iso is set. If used, default is minimum of (max_b_iso, B of dataset, ta...

• input_filesdata_file = None Mtz or other data file with intensities or amplitudesdata_labels = None Optional data labels for Iobs,SIGIobs or Fobs,SIGFobs. Input data can be I, F, or I+/I- or F+/F-.phib_labels = None Optional data label for PHIB. You can force skipping with phib_labels=SKIPfom_labels = None Optional data label for FOM. You can force skipping with fom_labels=SKIPhl_file = None Mtz or other data file with HL coefficients. If None assumed to be the same as data_file. Required unless a pdb_file is supplied.hl_labels = None Optional labels for HL coefficients. You can force skipping with hl_labels=SKIPmap_coeffs_file = None Mtz or other file with map coefficients for starting map. If None assumed to be same as data_filemap_coeffs_labels = None Optional labels for map coefficients. You can force skipping with map_coeffs_labels=SKIPpdb_file = None Optional file with model to be used in density modification. If provided, the density modification will use model density as part of the density modification target. If supplied, HL coefficients are optional.ha_file = None Optional file with heavy-atom sites to be truncated in density modification. Density within rad_mask of a site in this file will be truncated at 2 x the rms of the map. Note: not compatible with mask_from_pdb.mask_from_pdb = None Optional file with dummy atoms marking the region to be considered as the macromolecule in density modification. The region within rad_mask of an atom in this file will be considered within the macromolecule. Note: not compatible with ha_file.add_mask = None Mask will be the union of the standard mask and mask_from_pdbncs_file = None Optional file with NCS information (.ncs_spec file). This file can be obtained with phenix.find_ncs or phenix.simple_ncs_from_pdbncs_domain_pdb = None Optional file with dummy atoms marking the region where NCS is to be applied. Supersedes ncs_domain_pdb in the ncs_file. Points within rad_mask of an atom in this file are considered for NCS. You need to supply a mask covering all NCS copies. NOTE: the output_ncs_mask_file map shows the boundary of the asymmetric unit of NCS (just one copy) while the input ncs_domain_pdb file must contain all NCS copies.resolve_commands_file = None Optional file with any commands for resolve density modification.

• data_file = None Mtz or other data file with intensities or amplitudes

• data_labels = None Optional data labels for Iobs,SIGIobs or Fobs,SIGFobs. Input data can be I, F, or I+/I- or F+/F-.

• phib_labels = None Optional data label for PHIB. You can force skipping with phib_labels=SKIP

• fom_labels = None Optional data label for FOM. You can force skipping with fom_labels=SKIP

• hl_file = None Mtz or other data file with HL coefficients. If None assumed to be the same as data_file. Required unless a pdb_file is supplied.

• hl_labels = None Optional labels for HL coefficients. You can force skipping with hl_labels=SKIP

• map_coeffs_file = None Mtz or other file with map coefficients for starting map. If None assumed to be same as data_file

• map_coeffs_labels = None Optional labels for map coefficients. You can force skipping with map_coeffs_labels=SKIP

• pdb_file = None Optional file with model to be used in density modification. If provided, the density modification will use model density as part of the density modification target. If supplied, HL coefficients are optional.

• ha_file = None Optional file with heavy-atom sites to be truncated in density modification. Density within rad_mask of a site in this file will be truncated at 2 x the rms of the map. Note: not compatible with mask_from_pdb.

• mask_from_pdb = None Optional file with dummy atoms marking the region to be considered as the macromolecule in density modification. The region within rad_mask of an atom in this file will be considered within the macromolecule. Note: not compatible with ha_file.

- add_mask = None Mask will be the union of the standard mask and mask_from_pdb
- ncs_file = None Optional file with NCS information (.ncs_spec file). This file can be obtained with phenix.find_ncs or phenix.simple_ncs_from_pdb
- ncs_domain_pdb = None Optional file with dummy atoms marking the region where NCS is to be applied. Supersedes ncs_domain_pdb in the ncs_file. Points within rad_mask of an atom in this file are considered for NCS. You need to supply a mask covering all NCS copies. NOTE: the output_ncs_mask_file map shows the boundary of the asymmetric unit of NCS (just one copy) while the input ncs_domain_pdb file must contain all NCS copies.
- resolve_commands_file = None Optional file with any commands for resolve density modification.
- directoriestemp_dir = temp_denmod Temporary directory. Not normally used. Contents deleted after the run if clean_up=True is specified.
- temp_dir = temp_denmod Temporary directory. Not normally used. Contents deleted after the run if clean_up=True is specified.
- crystal_inforesolution = None Resolution for density modification
- resolution = None Resolution for density modification
- output_filesoutput_mtz = denmod.mtz Output MTZ file with density-modified map coefficientshl_in_output_mtz = True Calculate density modification HL coefficients and write to output mtz filetarget_map_file_name = target_map.ccp4 Output CCP4-style map with target map. only written if write_target_map_and_sigma is setsigma_map_file_name = sigma_map.ccp4 Output CCP4-style map with sigma map. only written if write_target_map_and_sigma is setoutput_mask_file = solvent_boundary.map Output CCP4-style map file with solvent boundary mask.output_ncs_mask_file = ncs_boundary.map Output CCP4-style map file with mask marking region used for NCS. NOTE: the output_ncs_mask_file map shows the boundary of the asymmetric unit of NCS (just one copy) while the input ncs_domain_pdb file must contain all NCS copies.
- output_mtz = denmod.mtz Output MTZ file with density-modified map coefficients
- hl_in_output_mtz = True Calculate density modification HL coefficients and write to output mtz file
- target_map_file_name = target_map.ccp4 Output CCP4-style map with target map. only written if write_target_map_and_sigma is set
- sigma_map_file_name = sigma_map.ccp4 Output CCP4-style map with sigma map. only written if write_target_map_and_sigma is set
- output_mask_file = solvent_boundary.map Output CCP4-style map file with solvent boundary mask.
- output_ncs_mask_file = ncs_boundary.map Output CCP4-style map file with mask marking region used for NCS. NOTE: the output_ncs_mask_file map shows the boundary of the asymmetric unit of NCS (just one copy) while the input ncs_domain_pdb file must contain all NCS copies.
- anisotropyremove_aniso = True Remove anisotropy from data and optionally sharpenb_iso = None Target overall B value for anisotropy correction. Ignored if remove_aniso = False. If None, default is minimum of (max_b_iso, B of dataset, target_b_ratio*resolution)max_b_iso = 40. Default maximum overall B value for anisotropy correction. Ignored if remove_aniso = False. Ignored if b_iso is set. If used, default is minimum of (max_b_iso, B of dataset, target_b_ratio*resolution)target_b_ratio = 10. Default ratio of target B value to resolution for anisotropy correction. Ignored if remove_aniso = False. Ignored if b_iso is set. If used, default is minimum of (max_b_iso, B of dataset, target_b_ratio*resolution)
- remove_aniso = True Remove anisotropy from data and optionally sharpen

- b_iso = None Target overall B value for anisotropy correction. Ignored if remove_aniso = False. If None, default is minimum of (max_b_iso, B of dataset, target_b_ratio*resolution)
- max_b_iso = 40. Default maximum overall B value for anisotropy correction. Ignored if remove_aniso = False. Ignored if b_iso is set. If used, default is minimum of (max_b_iso, B of dataset, target_b_ratio*resolution)
- target_b_ratio = 10. Default ratio of target B value to resolution for anisotropy correction. Ignored if remove_aniso = False. Ignored if b_iso is set. If used, default is minimum of (max_b_iso, B of dataset, target_b_ratio*resolution)
- denmodrad_wang = None Radius for estimation of solvent boundary. Normally chosen automatically.optimize_rad_wang = False Optimize value of rad_wangmask_type = *None histograms probability wang Method for solvent identification. Default is histograms. If you have a SAD dataset with a heavy atom such as Pt or Au then you may wish to choose wang because the histogram method is sensitive to very high peaks. Options are: histograms: compare local rms of map and local skew of map to values from a model map and estimate probabilities. This one is usually the best. probability: compare local rms of map to distribution for all points in this map and estimate probabilities. In a few cases this one is much better than histograms. wang: take points with highest local rms and define as protein.solvent_content = None Solvent content (0 to 1) of the crystal (required)mask_cycles = 5 Mask cycles (overall cycles of density modification)minor_cycles = 10 Minor cycles of density modification for each mask cyclerad_mask = 4 Radius (A) for mask calculation around heavy-atom sites (ha_file), for definition of protein boundary (mask_from_pdb), and for definition of the region used for NCS (ncs_domain_pdb).denmod_with_model = None By default if a pdb_file is supplied it will be used in density modification. You can turn this off with denmod_with_model=Falseremove_waters = True Remove waters from model before use in density modification or mask generationremove_hetatm = True Remove HETATM (ligands, metals etc.) from model before use in density modification or mask generationmap_phasing_ab_average_weight = 0.1 Add in this fraction of average sigma when calculating weightsmap_phasing_ab_weighting = False Use weighting in map_phasing_ab output from sigmaxwrite_target_map_and_sigma = False Write out map files for target rho and sigmamap_phasing_ab = None Use map phasing to get estimates of structure factors as x,y and sigmax,sigmay
- rad_wang = None Radius for estimation of solvent boundary. Normally chosen automatically.
- optimize_rad_wang = False Optimize value of rad_wang
- mask_type = *None histograms probability wang Method for solvent identification. Default is histograms. If you have a SAD dataset with a heavy atom such as Pt or Au then you may wish to choose wang because the histogram method is sensitive to very high peaks. Options are: histograms: compare local rms of map and local skew of map to values from a model map and estimate probabilities. This one is usually the best. probability: compare local rms of map to distribution for all points in this map and estimate probabilities. In a few cases this one is much better than histograms. wang: take points with highest local rms and define as protein.
- solvent_content = None Solvent content (0 to 1) of the crystal (required)
- mask_cycles = 5 Mask cycles (overall cycles of density modification)
- minor_cycles = 10 Minor cycles of density modification for each mask cycle
- rad_mask = 4 Radius (A) for mask calculation around heavy-atom sites (ha_file), for definition of protein boundary (mask_from_pdb), and for definition of the region used for NCS (ncs_domain_pdb).
- denmod_with_model = None By default if a pdb_file is supplied it will be used in density modification. You can turn this off with denmod_with_model=False

- remove_waters = True Remove waters from model before use in density modification or mask generation
- remove_hetatm = True Remove HETATM (ligands, metals etc.) from model before use in density modification or mask generation
- map_phasing_ab_average_weight = 0.1 Add in this fraction of average sigma when calculating weights
- map_phasing_ab_weighting = False Use weighting in map_phasing_ab output from sigmax
- write_target_map_and_sigma = False Write out map files for target rho and sigma
- map_phasing_ab = None Use map phasing to get estimates of structure factors as x,y and sigmax,sigmay
- controln_xyz = None gridding for mapsverbose = False Verbose outputclean_up = True Remove temp_dir when finished
- n_xyz = None gridding for maps
- verbose = False Verbose output
- clean_up = True Remove temp_dir when finished

Docking a model into a cryo-EM map with dock_in_map

dock_in_map.html

Author(s)

- dock_in_map: Tom Terwilliger

Purpose

The routine dock_in_map will automatically dock a model or models into a map.

Usage

How dock_in_map works:

Dock-in-map uses both SSM and convolution-based shape searches to find a part of a map that is similar to a model. The key elements of the search are:

```
An SSM search is carried out first (protein only). This search is
identical to the ssm_match_to_map option in superpose_models. Helices
and strands are found in the map to create a target model, then a brute
force superposition of helices and strands in the search model to those in
the target model is carried out. Potential superpositions are evaluated
by map-model correlation.

If the SSM search fails to find a satisfactory superposition,
an initial search at lower resolution that focuses on overall shape of the
molecule is carried out. This allows a quick search but is less
effective if the map contains more than one molecule. If this fails, the
next methods are tried.

Initial search without rotation. This allows a fast search that can place
a molecule that is just shifted by a translation

Optional initial search based on matching moments of inertia of model and map.
This is also a fast search but requires an accurate map that looks a lot
like the model and does not have extra density.

Full search at the full resolution of the map. This search can be run on
multiple processors to speed it up.
```

Uses

You can use dock_in_map to place any number of copies of any number of unique molecules. You specify the molecules one by one:

```
search_model=model1.pdb
search_model=model2.pdb
search_model=model3.pdb
```

and the number of copies all at once with:

```
search_model_copies="3 1 2"
```

to look for 3 copies of model1.pdb, one of model2.pdb and two of model3.pdb.

You can also use dock_in_map to superimpose density from one map on another map. You just specify the second map with the density:

```
search_map_file=map_file.ccp4
```

and dock_in_map will find the density in this map, convert it to a pseudo-model that corresponds to this density, and then search for the pseudo-model just as any other model. When dock_in_map is done finding the pseudo-model, it applies the transformation that it found to the original density in your search_map_file to create a new map that superimposes on the target map.

Examples

Standard run of dock_in_map:

Running dock_in_map is easy. From the command-line you can type:

```
phenix.dock_in_map 1ss8_A.pdb emd_8750.map resolution=4 nproc=4 \
 pdb_out=placed_model_from_emd_8750.pdb
```

where 1ss8_A.pdb is the model you would like to place, emd_8750.map is a CCP4, mrc or other related map format, and you specify the nominal resolution of the map and the number of processors to use.

For docking density into a map:

phenix.dock_in_map emd_8750.map density.ccp4 resolution=4
superposed_map_file=density_superposed.ccp4

phenix.dock_in_map emd_8750.map density.ccp4 resolution=4

superposed_map_file=density_superposed.ccp4

Possible Problems

If your map is inverted (left-handed), docking and model-building will not work properly. You can often tell if your map is inverted because any helices will be left-handed. If you are unsure, you can run MapBox with invert_hand=True to invert the map and then see if docking works. Note that if your map is inverted, you will want to invert all your maps and start everything from the beginning.

If your map has pseudo-symmetry (like a proteasome) you might need to box one subunit or try ssm_search=False to use a more thorough search in docking.

Specific limitations and problems:

Literature

Additional information

List of all available keywords

job_title = None Job title in PHENIX GUI, not used on command lineinput_filesmap_file = None File with CCP4-style map. May have origin in any location.seq_file = None Optional sequence filesearch_model = None Input PDB file(s) to be placed in the map.search_map_file = None Search map file with CCP4-style map to be superimposed on the reference map. Map will be converted to a pseudo-model that corresponds to density in map. At completion of dock_in_map the search map file will be transformed to match the target map and written out.search_model_copies = None You can specify how many copies for each search model by supplying one value of search_model_copies for each of your search models. Put them in all at once as in search_model_copies="1,3,2".search_map_copies = None You can specify how many copies for each search map by supplying one value of search_map_copies for each of your search models. Put them all in at once as in search_map_copies="3,4,5"fixed_model = None Fixed model marking region not to be considered. Fixed model will be added to placed search model to create new fixed model if append_to_fixed_model is set and multiple searches are done.target_model = None Model (typically helices-strands model) already docked into map and used as a marker.symmetry_file = None Symmetry file (.ncs_spec format or MATRX records) with reconstruction symmetry. Used in identification of unique part of map. Required if allow_symmetry is set and more than one copy is requested. NOTE: symmetry file applies to map file in original positionoutput_filespdb_out = placed_model.pdb Search PDB file, transformed to superpose on the target map file.superposed_map_file = superposed_map.ccp4 Search map file (if just one), transformed to superpose on the target map file.skip_fixed_model_in_pdb_out = True Skip fixed model in output PDB file. Just write the new part.temp_dir = None Temporary directory.crystal_inforesolution = None High-resolution limit for main search. This can be lower resolution than the data. The search is quicker at lower resolution. If your model is poor, try 2-3 A lower resolution than your data (i.e, if your data is 2.5 A, try 5 A).scattering_table = n_gaussian wk1995 it1992 *electron neutron Choice of scattering table for structure factor calculations. Standard for X-ray is n_gaussian, for cryoEM is electron.sequence = None Sequence as text string. Normally supply a sequence file insteadchain_type = *PROTEIN DNA RNA Chain typewrapping = None You can specify whether the map is wrapped (can map values outside bounds to inside with cell translations).searchdock_chains_individually = False Dock each chain (identified by chain_id field) individually. This is useful if the chains might have different orientations in the map compared to the search model, such as in a search model that is a predicted model. Normally used along with create_unique_chain_at_end=True for predicted models to put the model back together.create_unique_chain_at_end = None Take all docked chains and try to create one chain that has no duplicate residue numbers and that has distances between ends of fragments consistent with the number of residues between them. If symmetry has been applied, create N copies representing that symmetry. Default is True if dock_chains_individually = True and False otherwise.minimum_cc_to_keep_domain = 0.2 Do not keep domains in create_unique_chain_at_end if cc is less than minimum_cc_to_keep_domain.weight_sequential_fragments_by_distance = None Try to dock a chain in a way that minimizes the distance between its first residue and the previously-docked residue with the highest residue number less than this, and similarly for the last residue and the next available placed residue. Default is True if create_unique_chain_at_end is set. Normally use only if you have a model with a single chain and you are docking pieces of that chain.choose_better_of_individual_and_group_docking = None Dock entire search model (normally one only) and also ...

- job_title = None Job title in PHENIX GUI, not used on command line
- input_filesmap_file = None File with CCP4-style map. May have origin in any location.seq_file = None Optional sequence filesearch_model = None Input PDB file(s) to be placed in the map.search_map_file = None Search map file with CCP4-style map to be superimposed on the reference map. Map will be converted to a pseudo-model that corresponds to density in map. At completion of dock_in_map the search map file will be transformed to match the target map and written out.search_model_copies = None You can specify how many copies for each search model by supplying one value of search_model_copies for each of your search models. Put them in all at once as in search_model_copies="1,3,2".search_map_copies = None You can specify how many copies for each search map by supplying one value of search_map_copies for each of your search models. Put them all in at once as in search_map_copies="3,4,5"fixed_model = None Fixed model marking region not to be considered. Fixed model will be added to placed search model to create new fixed model if append_to_fixed_model is set and multiple searches are done.target_model = None Model (typically helices-strands model) already docked into map and used as a marker.symmetry_file = None Symmetry

file (.ncs_spec format or MATRX records) with reconstruction symmetry. Used in identification of unique part of map. Required if allow_symmetry is set and more than one copy is requested. NOTE: symmetry file applies to map file in original position

- map_file = None File with CCP4-style map. May have origin in any location.
- seq_file = None Optional sequence file
- search_model = None Input PDB file(s) to be placed in the map.
- search_map_file = None Search map file with CCP4-style map to be superimposed on the reference map. Map will be converted to a pseudo-model that corresponds to density in map. At completion of dock_in_map the search map file will be transformed to match the target map and written out.
- search_model_copies = None You can specify how many copies for each search model by supplying one value of search_model_copies for each of your search models. Put them in all at once as in search_model_copies="1,3,2".
- search_map_copies = None You can specify how many copies for each search map by supplying one value of search_map_copies for each of your search models. Put them all in at once as in search_map_copies="3,4,5"
- fixed_model = None Fixed model marking region not to be considered. Fixed model will be added to placed search model to create new fixed model if append_to_fixed_model is set and multiple searches are done.
- target_model = None Model (typically helices-strands model) already docked into map and used as a marker.
- symmetry_file = None Symmetry file (.ncs_spec format or MATRX records) with reconstruction symmetry. Used in identification of unique part of map. Required if allow_symmetry is set and more than one copy is requested. NOTE: symmetry file applies to map file in original position
- output_filespdb_out = placed_model.pdb Search PDB file, transformed to superpose on the target map file.superposed_map_file = superposed_map.ccp4 Search map file (if just one), transformed to superpose on the target map file.skip_fixed_model_in_pdb_out = True Skip fixed model in output PDB file. Just write the new part.temp_dir = None Temporary directory.
- pdb_out = placed_model.pdb Search PDB file, transformed to superpose on the target map file.
- superposed_map_file = superposed_map.ccp4 Search map file (if just one), transformed to superpose on the target map file.
- skip_fixed_model_in_pdb_out = True Skip fixed model in output PDB file. Just write the new part.
- temp_dir = None Temporary directory.
- crystal_inforesolution = None High-resolution limit for main search. This can be lower resolution than the data. The search is quicker at lower resolution. If your model is poor, try 2-3 A lower resolution than your data (i.e, if your data is 2.5 A, try 5 A).scattering_table = n_gaussian wk1995 it1992 *electron neutron Choice of scattering table for structure factor calculations. Standard for X-ray is n_gaussian, for cryoEM is electron.sequence = None Sequence as text string. Normally supply a sequence file insteadchain_type = *PROTEIN DNA RNA Chain typewrapping = None You can specify whether the map is wrapped (can map values outside bounds to inside with cell translations).
- resolution = None High-resolution limit for main search. This can be lower resolution than the data. The search is quicker at lower resolution. If your model is poor, try 2-3 A lower resolution than your data (i.e, if your data is 2.5 A, try 5 A).

- scattering_table = n_gaussian wk1995 it1992 *electron neutron Choice of scattering table for structure factor calculations. Standard for X-ray is n_gaussian, for cryoEM is electron.
- sequence = None Sequence as text string. Normally supply a sequence file instead
- chain_type = *PROTEIN DNA RNA Chain type
- wrapping = None You can specify whether the map is wrapped (can map values outside bounds to inside with cell translations).
- searchdock_chains_individually = False Dock each chain (identified by chain_id field) individually. This is useful if the chains might have different orientations in the map compared to the search model, such as in a search model that is a predicted model. Normally used along with create_unique_chain_at_end=True for predicted models to put the model back together.create_unique_chain_at_end = None Take all docked chains and try to create one chain that has no duplicate residue numbers and that has distances between ends of fragments consistent with the number of residues between them. If symmetry has been applied, create N copies representing that symmetry. Default is True if dock_chains_individually = True and False otherwise.minimum_cc_to_keep_domain = 0.2 Do not keep domains in create_unique_chain_at_end if cc is less than minimum_cc_to_keep_domain.weight_sequential_fragments_by_distance = None Try to dock a chain in a way that minimizes the distance between its first residue and the previously-docked residue with the highest residue number less than this, and similarly for the last residue and the next available placed residue. Default is True if create_unique_chain_at_end is set. Normally use only if you have a model with a single chain and you are docking pieces of that chain.choose_better_of_individual_and_group_docking = None Dock entire search model (normally one only) and also by chain...pick better-fitting of the two for each chainlow_res_search = True Try to fit by searching at low resolution firstdock_with_mr = True Try using MR to dock firstssm_search = None Try to fit by searching for secondary structure. Default is False for dock_in_map and True for dock_and_rebuild if dock_with_mr is not used.refine_cycles = 3 rigid-body refinement cyclesssm_search_min_cc = 0.30 Stop ssm search if this cc achievedbackup_resolution_cutoff_searches = 2 If initial fitting does not work, try up to this many times increasing resolution by backup_resolution_ratio each timebackup_resolution_ratio = 1.67 If initial fitting does not work, try up to backup_resolution_cutoff_searches times increasing the resolution by backup_resolution_ratio each timeresolution_radius_scale = 0.5 Resolution for low-res search will be resolution_radius_scale times the radius of gyration of the search model.align_moments = False Try to fit by aligning moments of inertia if size of density region and molecule are similarmax_radius_ratio = 2. Radius of gyration of molecule and density must be within max_radius_ratio of each otherradius_scale = 1.5 Mask for density and atoms will be radius_scale times the radius of gyration of modeluse_symmetry = True If search_model_copies or search_map_copies are the same for each model and greater than one and allow_symmetry is set, use symmetry in the map to place all copies after the first one.skip_if_low_cc = True Skip solution before rigid-body refinement if map-model CC is less than half of min_cc.rigid_body_refinement = True Run rigid-body refinement on final modelrigid_body_refinement_single_unit = True Run rigid-body refinement with just one unit (do not break up into chains)rigid_body_refinement_split_method = *chain_id segid When splitting up molecule for rigid-body refinement (if rigid_body_refinement_single_unit=False), use either chain_id or segid to split up moleculerigid_body_refinement_resolution = None Run rigid-body refinement at this resolution if specifiedappend_to_fixed_model = True Append placed search model to fixed model (if any) after searchmin_cc = 0.4 If quick run, stop if minimum CC is achieved in local search. Also always skip if starting CC_mask is less than 1/4 min_cc.run_in_boxes = True Run on sub-boxes and combine at the endtarget_box_size = 60 Try to get boxes about this big on a side (grid units)target_boxes = None Try to get this many boxes. Default is nproc unless this makes box size much smaller than target_box_sizebox_to_run = None Run only this boxbox_overlap_scale = 1 box overlap (ov...

- dock_chains_individually = False Dock each chain (identified by chain_id field) individually. This is useful if the chains might have different orientations in the map compared to the search model, such as in a search model that is a predicted model. Normally used along with create_unique_chain_at_end=True for predicted models to put the model back together.
- create_unique_chain_at_end = None Take all docked chains and try to create one chain that has no duplicate residue numbers and that has distances between ends of fragments consistent with the number of residues between them. If symmetry has been applied, create N copies representing that symmetry. Default is True if dock_chains_individually = True and False otherwise.
- minimum_cc_to_keep_domain = 0.2 Do not keep domains in create_unique_chain_at_end if cc is less than minimum_cc_to_keep_domain.
- weight_sequential_fragments_by_distance = None Try to dock a chain in a way that minimizes the distance between its first residue and the previously-docked residue with the highest residue number less than this, and similarly for the last residue and the next available placed residue. Default is True if create_unique_chain_at_end is set. Normally use only if you have a model with a single chain and you are docking pieces of that chain.
- choose_better_of_individual_and_group_docking = None Dock entire search model (normally one only) and also by chain...pick better-fitting of the two for each chain
- low_res_search = True Try to fit by searching at low resolution first
- dock_with_mr = True Try using MR to dock first
- ssm_search = None Try to fit by searching for secondary structure. Default is False for dock_in_map and True for dock_and_rebuild if dock_with_mr is not used.
- refine_cycles = 3 rigid-body refinement cycles
- ssm_search_min_cc = 0.30 Stop ssm search if this cc achieved
- backup_resolution_cutoff_searches = 2 If initial fitting does not work, try up to this many times increasing resolution by backup_resolution_ratio each time
- backup_resolution_ratio = 1.67 If initial fitting does not work, try up to backup_resolution_cutoff_searches times increasing the resolution by backup_resolution_ratio each time
- resolution_radius_scale = 0.5 Resolution for low-res search will be resolution_radius_scale times the radius of gyration of the search model.
- align_moments = False Try to fit by aligning moments of inertia if size of density region and molecule are similar
- max_radius_ratio = 2. Radius of gyration of molecule and density must be within max_radius_ratio of each other
- radius_scale = 1.5 Mask for density and atoms will be radius_scale times the radius of gyration of model
- use_symmetry = True If search_model_copies or search_map_copies are the same for each model and greater than one and allow_symmetry is set, use symmetry in the map to place all copies after the first one.
- skip_if_low_cc = True Skip solution before rigid-body refinement if map-model CC is less than half of min_cc.
- rigid_body_refinement = True Run rigid-body refinement on final model
- rigid_body_refinement_single_unit = True Run rigid-body refinement with just one unit (do not break up into chains)

- rigid_body_refinement_split_method = *chain_id segid When splitting up molecule for rigid-body refinement (if rigid_body_refinement_single_unit=False), use either chain_id or segid to split up molecule
- rigid_body_refinement_resolution = None Run rigid-body refinement at this resolution if specified
- append_to_fixed_model = True Append placed search model to fixed model (if any) after search
- min_cc = 0.4 If quick run, stop if minimum CC is achieved in local search. Also always skip if starting CC_mask is less than 1/4 min_cc.
- run_in_boxes = True Run on sub-boxes and combine at the end
- target_box_size = 60 Try to get boxes about this big on a side (grid units)
- target_boxes = None Try to get this many boxes. Default is nproc unless this makes box size much smaller than target_box_size
- box_to_run = None Run only this box
- box_overlap_scale = 1 box overlap (overlap of boxes) will be box_overlap_scale times the density radius
- edge_ratio = 10 box edge box_overlap times edge_ratio
- density_radius = None Radius for density to be cut out and compared. Default is 6 times the resolution.
- model_radius = 3 Radius for removing density near fixed_model
- zero_value = 0 Value to set map in regions overlapping fixed model
- density_peaks = 20 Number of NCS-related peaks of density to check
- delta_phi = 20 Angular spacing of search
- max_rot = None Maximum rotations to try
- rotz_only = None Rotate only around Z
- single_positions_to_try = 10 Number of offset positions to try in optimizing orientation. Positions along the chain are selected as centers and a local fit near each position is carried out. The resulting offsets relative to the original placement are used to optimize the overall orientation and position.
- max_position_shift_frac = 0.05 Maximum fractional positional shift in single_positions run
- min_relative_cc = 0.67 Minimum local CC relative to original CC to keep a local search. This is a way to reject local searches that are completely wrong.
- sieve_fit = None Use sieve_fit fraction of single positions in fitting. If None, use all
- ncs_copies_max = None Maximum number of matching models to write. If more than one they will be written as MODEL records in the output PDB file. You can get them individually afterwards with phenix.pdbtools placed_model.pdb keep="model 1" etc.
- start_rot = None Three numbers rotx, rotz, rotx defining the starting rotation of the search model. Normally used along with delta_phi=1000 or max_rot=1 to generate exactly one defined rotation.
- search_center = None Optional coordinates in search model for centering search. Note this is different from target_search_center which is the location xyz in the map to look.
- search_center_selection = None Optional selection defining coordinates in search model for centering search.
- target_search_center = None Optional coordinates in reference map where search_center should be approximately located after superimposing maps. Used to eliminate possible superpositions that place the search center elsewhere. Overlap scores decreased based on distance to target_search_center/density_radius.

- model_search_position = None Optional coordinates (usually part of search model) for matching to target_search_position after transformation. These can be specified in addition to target_search_center. Can have multiple model_search_position and target_search_position pairs by specifying each multiple times in order. NOTE: Not compatible with map search
- target_search_position = None Target positions for model_search_position after transformation.
- search_position_radius = None Radius for comparison of target_search_position and transformed search_position values. Default is density_radius. If specified, must be a single value or the same number of values as entries in model_search_position and target_search_position
- rot_id_n = None Number of rotation groups. Along with rot_id_group, allows defining groups of rotations to be carried out in one run. .short_caption = Number of rotation groups
- rot_id_group = None rotation group to include. See rot_id_n.
- map_box = True Run map_box to extract useful part of map before search
- fix_search_position = False You can choose to not move your model center of mass to the origin by fixing the search position
- search_box_size = None You can choose the size of the search box for local searches. Normally set by default corresponding to 3 times density_radius
- lower_bounds = None You can select a part of your map for analysis with lower_bounds and upper_bounds.
- upper_bounds = None You can select a part of your map for analysis with lower_bounds and upper_bounds.
- keep_search_order = None Keep search order as input
- remove_water = False Remove waters and other hetero atoms from input files
- controlnproc = 1 Number of processors to userandom_seed = 171731 Random seedverbose = False Verbose outputquick = True Try translation search first and stop if CC is at least min_cc
- nproc = 1 Number of processors to use
- random_seed = 171731 Random seed
- verbose = False Verbose output
- quick = True Try translation search first and stop if CC is at least min_cc
- guiGUI-specific parameter required for output directoryoutput_dir = None
- output_dir = None

Creating a local resolution map with local_resolution

local_resolution.html

Author(s)

- local_resolution: Tom Terwilliger

Purpose

The routine local_resolution is a tool for creating a local resolution map showing the local resolution of a cryo-EM map and having the same gridding as the cryo-EM map

Usage and how local_resolution works:

Local resolution is calculated in small boxes cut out from the cryo-EM map, where the density near the edges of each box is masked so that it gradually diminishes to zero at the edges. Then the values of local resolution are smoothed and placed at grid points of a full-size map matching the original map.

The local resolution map can be used to color a map in display programs such as ChimeraX. In ChimeraX you can simply load in your original map and the local resolution map and then specify:

color sample #1 map #2 palette rainbow

to color map #1 based on the resolution values in map #2.

Procedure used by local_resolution

The basic inputs to local_resolution are:

```
Two unmasked half-maps
```

Optional inputs are:

```
nominal resolution
```

Examples

Standard run of local_resolution:

You can use local_resolution to get the local resolution in a cryo-EM map:

```
phenix.local_resolution half_map_A.mrc half_map_B.mrc
```

Possible Problems

If the half-maps are not actually independent the procedure will not work well

Specific limitations and problems:

If you have a very large map, this procedure can be really slow. If your map is large but is mostly empty, it may be helpful to run map_box to select just the region of your two half-maps where your molecule is located before running local_resolution.

Literature

Additional information

List of all available keywords

job_title = None Job title in PHENIX GUI, not used on command lineinput_filesmap_modelfull_map = None Input full map filehalf_map = None Input half map filesmodel = None Input model fileoutputlocal_resolution_map = default Local resolution map file nameadjust_local_resolution_coloring = False Adjust local resolution coloring based on limits of resolution. Default is to use a standard coloring scheme.full_map = default Full map file name. Same as input full map if supplied, otherwise result of default run of local_aniso_sharpen on half mapsoverwrite = True Overwrite files with same namesfile_name = None Not usedfilename = None Not usedserial = None Not usedlocal_resolutionmethod = boxes *fft Method for local resolution. Boxes means calculate resolution in boxes and display smoothed values. FFT means use FFT-based method to estimate resolution in one stepn_bins = None Number of bins for analysesn_bins_default_boxes = 2000 Default bins for boxes methodn_bins_default_fft = 20 Default bins for fft methodmin_bin_width = 20 Minimum number of Fourier coefficients per bin. Overrides value of n_bins.max_resolution_ratio = 10 Maximum resolution as ratio to minimum. Maximum resolution will be truncated at this valuen_boxes = None Number of boxesbox_size_grid_units = None Size of core region of boxes (not including region where mask is applied, in grid unitssmoothing_radius_ratio = 1 Ratio of smoothing radius to resolution (FFT only)crystal_inforesolution = None Nominal resolution of mapcontrolmultiprocessing = *multiprocessing sge lsf pbs condor pbspro slurm Choices are multiprocessing (single machine) or queuing systems Not implementedqueue_run_command = None run command for queue jobs. For example qsub. Not implementednproc = 1 Number of processors to use. NOTE: by default multiple processors will only be used in the map-to-model step (this is because multiprocessing requires writing out nproc sets of huge files and it can be very slow with distributed queues.). You can override this with force_nproc = True.guiGUI-specific parameter required for output directoryoutput_dir = None

- job_title = None Job title in PHENIX GUI, not used on command line
- input_filesmap_modelfull_map = None Input full map filehalf_map = None Input half map filesmodel = None Input model file
- map_modelfull_map = None Input full map filehalf_map = None Input half map filesmodel = None Input model file
- full_map = None Input full map file
- half_map = None Input half map files
- model = None Input model file
- outputlocal_resolution_map = default Local resolution map file nameadjust_local_resolution_coloring = False Adjust local resolution coloring based on limits of resolution. Default is to use a standard coloring scheme.full_map = default Full map file name. Same as input full map if supplied, otherwise result of default run of local_aniso_sharpen on half mapsoverwrite = True Overwrite files with same namesfile_name = None Not usedfilename = None Not usedserial = None Not used
- local_resolution_map = default Local resolution map file name
- adjust_local_resolution_coloring = False Adjust local resolution coloring based on limits of resolution. Default is to use a standard coloring scheme.
- full_map = default Full map file name. Same as input full map if supplied, otherwise result of default run of local_aniso_sharpen on half maps

- overwrite = True Overwrite files with same names
- file_name = None Not used
- filename = None Not used
- serial = None Not used
- local_resolutionmethod = boxes *fft Method for local resolution. Boxes means calculate resolution in boxes and display smoothed values. FFT means use FFT-based method to estimate resolution in one stepn_bins = None Number of bins for analysesn_bins_default_boxes = 2000 Default bins for boxes methodn_bins_default_fft = 20 Default bins for fft methodmin_bin_width = 20 Minimum number of Fourier coefficients per bin. Overrides value of n_bins.max_resolution_ratio = 10 Maximum resolution as ratio to minimum. Maximum resolution will be truncated at this valuen_boxes = None Number of boxesbox_size_grid_units = None Size of core region of boxes (not including region where mask is applied, in grid unitssmoothing_radius_ratio = 1 Ratio of smoothing radius to resolution (FFT only)
- method = boxes *fft Method for local resolution. Boxes means calculate resolution in boxes and display smoothed values. FFT means use FFT-based method to estimate resolution in one step
- n_bins = None Number of bins for analyses
- n_bins_default_boxes = 2000 Default bins for boxes method
- n_bins_default_fft = 20 Default bins for fft method
- min_bin_width = 20 Minimum number of Fourier coefficients per bin. Overrides value of n_bins.
- max_resolution_ratio = 10 Maximum resolution as ratio to minimum. Maximum resolution will be truncated at this value
- n_boxes = None Number of boxes
- box_size_grid_units = None Size of core region of boxes (not including region where mask is applied, in grid units
- smoothing_radius_ratio = 1 Ratio of smoothing radius to resolution (FFT only)
- crystal_inforesolution = None Nominal resolution of map
- resolution = None Nominal resolution of map
- controlmultiprocessing = *multiprocessing sge lsf pbs condor pbspro slurm Choices are multiprocessing (single machine) or queuing systems Not implementedqueue_run_command = None run command for queue jobs. For example qsub. Not implementednproc = 1 Number of processors to use. NOTE: by default multiple processors will only be used in the map-to-model step (this is because multiprocessing requires writing out nproc sets of huge files and it can be very slow with distributed queues.). You can override this with force_nproc = True.
- multiprocessing = *multiprocessing sge lsf pbs condor pbspro slurm Choices are multiprocessing (single machine) or queuing systems Not implemented
- queue_run_command = None run command for queue jobs. For example qsub. Not implemented
- nproc = 1 Number of processors to use. NOTE: by default multiple processors will only be used in the map-to-model step (this is because multiprocessing requires writing out nproc sets of huge files and it can be very slow with distributed queues.). You can override this with force_nproc = True.
- guiGUI-specific parameter required for output directoryoutput_dir = None
- output_dir = None

Optimizing a map with using map_sharpening

map_sharpening.html

Author(s)

- map_sharpening: Tom Terwilliger

Purpose

The routine map_sharpening is a tool for optimizing a map by applying a resolution-dependent scaling factor. The scaling can be isotropic or anisotropic and can be local or over the entire map. Scaling can be carried out be smoothed shell-scaling or by B-factor sharpening, applying an overall B-factor, optionally downweighting all data past a resolution cutoff.

Input maps

Your input maps can be full-size or boxed (for example using Map Box). For local anisotropic sharpening, boxed maps are recommended as the procedure can take a long time for full size maps. In other cases boxed maps are quicker to work with but are not required.

Should I use shell-sharpening or B-factor sharpening?

If you have two half-maps (cryo-EM data), normally you should use shell-sharpening with an anisotropic correction (the default procedure).

If you have a single map (cryo-EM) or map coefficients (X-ray) then you should use B-factor sharpening, as the shell-sharpening procedure cannot run on a single map. In some cases B-factor sharpening starting with a single (full) map can give a more interpretable map than shell sharpening with half-maps, so it is fine to try this procedure as well.

If you have a map and model, either method is fine. Normally shell-sharpening (the default) is recommended.

If you want to supply a sharpening B-factor and apply it to your map, or if you want to obtain a target B-value, use B-factor sharpening.

If you want to match the resolution fall-off of your map to a target map, use B-factor sharpening.

Shell sharpening

Shell sharpening can be carried out locally or on a map as a whole.

The basis for shell sharpening is an analysis of the resolution-dependent fall-off of amplitudes of Fourier coefficients for the map, along with an analysis of the resolution-dependent fall-off of correlation between two half-maps, or between a map and a model-based map. The analysis is carried out in shells of resolution.

When carried out locally, a full map is divided into small boxes. The density near the edges of each box is masked so that it gradually diminishes to zero at the edges. Each small box is treated as a full map to identify its optimal sharpening. Then the optimal sharpening parameters from the small boxes are applied to the full map in a way that has no edge effects and smoothly varies from one place to another in the map.

To identify the optimal anisotropic sharpening of a map based on the information in two half-maps, two analyses are done. The first is an analysis of the resolution-dependent fall-off of rms amplitudes of Fourier coefficients representing the map. This is examined as a function of direction in reciprocal space, and is similar to the calculation normally done to apply an anisotropy correction to a map. This analysis shows the anisotropy of the map itself.

The second is an analysis of the correlation between Fourier coefficients for the two half-maps. This is also done as a function of resolution and direction in reciprocal space. This analysis shows the anisotropy of the errors in the map.

For purposes of this analysis, the optimal map is the one that has the maximal expected correlation to an idealized version of the true map. This idealized map is a map that would be obtained from a model where all the atoms are point atoms (B values of about zero).

For a map with zero error (all correlations at all resolutions and directions equal to 1), the optimal map will be one that has no anisotropy and the same resolution dependence as the idealized map. For such a map, first the anisotropy in the map is removed, then an overall resolution-dependence matching that of the idealized map is imposed by simple multiplication with a resolution_dependent scale factor.

For a map with errors, the map coefficients obtained from the previous step are modified by a local scale factor that reflects the expected signal-to-noise in that map coefficient. The scale factor for a particular map coefficient is given by 1/(1 + E**2), where E is the normalized expected error in that map coefficient. This scale factor will ordinarily be anisotropic and resolution-dependent.

Output from anisotropic shell sharpening

The map_sharpening tool will provide a summary of the anisotropy of your map if you use shell scaling and leave anisotropic_sharpen=True. Here are some of the values that are calculated for a test case, with annotations indicated by >>:

```
NOTE: All values apply after removal of overall U of:
 (0.81, 0.84, 0.78, -0.03, -0.01, 0.04)

&gt;&gt; This overall U value has large and similar values for the first 3
&gt;&gt; numbers, and small values for the last 3. This indicates that this
&gt;&gt; map is relatively isotropic.
&gt;&gt; In this output, the overall U listed above is removed from all other
&gt;&gt; U values before printing them out.

&gt;&gt; All values are reported in normalized units (U values). See
&gt;&gt; https://www.iucr.org/__data/iucr/cifdic_html/1/cif_core.dic/Iatom_site_U_iso_or_equiv.html

 Estimated anisotropic fall-off of the data relative to ideal
(Positive means amplitudes fall off more in this direction)
 ( X, Y, Z, XY, XZ, YZ)
(0.002, 0.089, 0.037, 0.031, 0.042, -0.002)

&gt;&gt; This is the fall-off of the data, after removal of the overall U values,
&gt;&gt; relative to a standard of values calculated from a beta-galactosidase
&gt;&gt; model without a bulk solvent correction (available from the phenix
&gt;&gt; python module cctbx.development.approx_amplitude_vs_resolution)

 Anisotropy of the data
(Positive means amplitudes fall off more in this direction)
 ( X, Y, Z, XY, XZ, YZ)
(-0.119, 0.167, -0.133, -0.013, 0.031, 0.064)

&gt;&gt; This is the anisotropy of the data, after removal of overall U values

 Anisotropy of the uncertainties
(Positive means uncertainties decrease more in this direction)
 ( X, Y, Z, XY, XZ, YZ)
(0.160, -0.032, -0.035, 0.031, 0.054, -0.010)
```

```
&gt;&gt; This is the anisotropy of the uncertainties in scale factors,
&gt;&gt; after removal of overall U values

 Anisotropy of the scale factors
(Positive means scale factors increase more in this direction)
 ( X, Y, Z, XY, XZ, YZ)
(-0.022, -0.121, -0.088, -0.050, -0.040, -0.019)

&gt;&gt; This is the anisotropy of the scale factors,
&gt;&gt; after removal of overall U values
```

B-factor sharpening

In contrast to the shell-scaling approach above, B-factor sharpening will identify an overall B-factor (exponential function) to apply to the Fourier coefficients representing a map. This overall scale factor can optionally be modified to strongly down-weight data beyond a resolution cutoff.

Normally in B-factor sharpening, the optimization is carried out to identify sharpening or blurring that maximizes the clarity of the map, as represented by the adjusted surface area. This adjusted surface are is based on the surface area of a contour map obtained by setting a contour level that encloses the expected volume of the molecule. That area is then reduced proportionally to the number of distinct regions enclosed in that contour map.

Optionally B-factor sharpening can instead be carried out by half-map sharpening, maximizing the expected quality of the average of the two half maps in each range of resolution.

B-factor sharpening was known as auto_sharpen in previous versions of Phenix. The tutorial below refers to auto-sharpening but it is the same as B-factor sharpening with the map sharpening tool.

Kurtosis is a standard statistical measure that reflects the peakiness of the map.

The adjusted surface area is a combination of the surface area of contours in the map at a particular threshold and of the number of distinct regions enclosed by the top 30% (default) of those contours. The threshold is chosen by default to be one where the volume enclosed by the contours is 20% of the non-solvent volume in the map. The weighting between the surface area (to be maximized) and number of regions enclosed (to be minimized) is chosen empirically (default region_weight=20).

Several resolution-dependent functions are tested, and the one that gives the best adjusted surface area (or kurtosis) is chosen. In each case the map is transformed to obtain Fourier coefficients. The amplitudes of these coefficients are then adjusted, keeping the phases constant. The available functions for modifying the amplitudes are:

```
No sharpening (map is left as is)

Sharpening b-factor applied over entire resolution range (b_sharpen
applied to achieve an effective isotropic overall b-value of b_iso).

Sharpening b-factor applied up to resolution specified with the
resolution=xxx keyword, then blurred beyond this resolution (with
transition specified by the keyword k_sharpen, b_iso_to_d_cut). If
the sharpening b_sharpen is negative (blurring the map),
the blurring is applied over the entire resolution range.

Resolution-dependent sharpening factor with three parameters.
First the resolution-dependence of the map is removed by normalizing the
amplitudes. Then a scale factor S is to the data, where
log10(S) is determined by coefficients b[0],b[1],b[2] and a resolution
d_cut (typically d_cut is the nominal resolution of the map).
The value of log10(S) varies smoothly from 0 at resolution=infinity, to b[0]
at d_cut/2, to b[1] at d_cut, and to b[1]+b[2] at the highest resolution
in the map. The value of b[1] is limited to being no larger than b[0] and the
```

```
value of b[1]+b[2] is limited to be no larger than b[1].
```

You can also choose to specify the sharpening/blurring parameters for your map and they will simply be applied to the map. For example you can apply a sharpening B-value (b_sharpen) to sharpen the map, or you can specify a target overall B-value (b_iso) to obtain after sharpening.

Box of density in sharpening in B-factor sharpening

Normally phenix.auto_sharpen will determine the optimal sharpening by examining the density in a box cut out of your map, then apply this to the entire map.

Local sharpening in B-factor sharpening

You can choose to apply autosharpening locally if you want. In this case the auto-sharpening parameters are determined in many boxes cut out of the map, and corresponding sharpened maps are calculated. The map that is produced is a weighted map where the density at a particular point comes most from the sharpened map based on a box near that point.

Note that this takes a long time for large maps. It is usually best to box your map with MapBox before running local sharpening.

Half-map-based sharpening in B-factor sharpening

You can identify the sharpening parameters using two half-maps if you want. The resolution-dependent correlation of density in the two half-maps is used to identify the optimal resolution-dependent weighting of the map. This approach requires a target resolution which is used to set the overall fall-off with resolution for an ideal map. That fall-off for an ideal map is then multiplied by an estimated resolution-dependent correlation of density in the map with the true map (the estimation comes from the half-map correlations).

Model-based sharpening in B-factor sharpening

You can identify instead the sharpening parameters using your map and a model. This approach requires a guess of the RMSD between the model and the true model. The resolution-dependent correlation of model and map density is used as in the half-map approach above to identify the weighting of Fourier coefficients.

Using crystallographic maps in B-factor sharpening

You can use phenix.auto_sharpen with a crystallographic map (represented as map coefficients).

Shifting the map to the origin in B-factor sharpening

Most crystallographic maps have the origin at the corner of the map (grid point [0,0,0]), while most cryo-EM maps have the orgin in the middle of the map. An output map with the origin shifted to the corner of the map is optionally written out.

GUI

A Graphical User Interface is available. The Coot and ChimeraX buttons load the input map and sharpened map, along with the model if supplied.

If you run shell sharpening in the GUI, by default a local resolution map is calculated at the end. The ChimeraX button will then load the sharpened map and color it according to the local resolution.

Output files from map_sharpening

sharpened_map.ccp4: Sharpened map.

sharpened_map_coeffs.mtz: Sharpened map, shifted to place the origin on grid point (0,0,0) and sharpened, represented as map coefficients.

Examples

You can use map_sharpening with either two half-maps or a map and a model.

Standard run of map_sharpening with two half-maps and using shell scaling:

To run shell sharpening using map_sharpening with two half maps, you can say:

```
phenix.map_sharpening half_map_A.mrc half_map_B.mrc seq_file=seq.fa
```

If you wish, you can specify a nominal resolution. The sequence file is to let the program figure out the volume of the molecule that is present.

Standard run of map_sharpening with map and model and using shell scaling:

To run shell map_sharpening with a map and model, you can say:

```
phenix.map_sharpening map.mrc model.pdb resolution=3 seq_file=seq.fa
```

The resolution is again optional.

You can specify whether anisotropy or local sharpening are to be applied:

```
local_sharpen=True
anisotropic_sharpen=True
```

Run of map_sharpening with map alone and using B-factor sharpening:

To run B-factor sharpening with a single map, you can say:

```
phenix.map_sharpening sharpening_method=b-factor map.mrc \
 resolution=3 seq_file=seq.fa
```

The resolution is required here as the resolution of the map is not well-defined.

What to do next after map sharpening

You can use your sharpened cryo-EM map anywhere that you need your clearest map (refinement, model-building, ligand fitting, validation).

Often the next step after map_sharpening will be model-building with predict_and_build or map_to_model. Alternatively you might carry out docking with EmPlacement or DockInMap. You might also try density modification with resolve_cryo_em and compare that map with the sharpened one.

Possible Problems

If a full-size map is supplied and local anisotropic sharpening (using shell-sharpening) is applied, the procedure can take a very long time. It is recommended that the map be boxed around the model or around density beforehand using Map Box.

If half-maps are not actually independent half-map sharpening will not work well.

If the model is very poor model-sharpening will not work well

For model-based shell sharpening, if local sharpening is used, the sharpening is only applied in the region of the model

Specific limitations and problems:

Maps produced with the extract-unique option of map_box should not be sharpened with B-sharpening. These maps are closely masked around the density of a single molecule and are set to zero in much of the map, so the information about noise in the map that normally is available is missing and the B-sharpening method of auto-sharpen does not work properly.

Literature

Automated map sharpening by maximization of detail and connectivity. T.C. Terwilliger, P.V. Afonine, Sobolev, OV, and P.D. Adams. Acta Cryst. D74, 545-559 (2018).

- Automated map sharpening by maximization of detail and connectivity. T.C. Terwilliger, P.V. Afonine, Sobolev, OV, and P.D. Adams. Acta Cryst. D74, 545-559 (2018).

Video Tutorial for B-factor sharpening (called auto_sharpen in this video)

The tutorial video is available on the Phenix YouTube channel and covers the following topics:

- basic overview
- how to run phenix.auto_sharpen (now phenix.map_sharpening) via the GUI

Additional information

List of all available keywords

job_title = None Job title in PHENIX GUI, not used on command lineinput_filesseq_file = None Sequence file. Include all copies of each sequence so that the full contents of the map can be estimated. (Fasta format or sequences separated by blank lines)ncs_file = None File with NCS information (typically point-group NCS with the center specified). Typically in PDB format. Can also be a .ncs_spec file from phenix. Created automatically if symmetry is specified.external_map_file = None External map to be used to scale map_file (power vs resolution will be matched). Applies to b_factor_sharpening only.mtz_in = None MTZ file with coefficients for a map mtz_in_virtual = None Used internallymap_coeff_labels = None If map coefficients cannot be identified automatically from your MTZ file, you can specify the label or labels for them. (Please separate labels with blank space, MTZ columns grouped together separated by commas with no blanks.) You can specify: map_coeff_labels (e.g., FWT,PHWT) amplitudes and phases (e.g., FP,SIGFP PHIB) or amplitudes, phases, weights (e.g., FP,SIGFP PHIB FOM)map_modelfull_map = None Input full map filehalf_map = None Input half map filesmodel = None Input model fileoutputsharpened_map_coeffs_file = default Sharpened map coefficients file namesharpened_map_file_1 = default Sharpened half map 1 file namesharpened_map_file_2 = default Sharpened half map 2 file namelocal_resolution_map_file = default Local resolution map file nameoverwrite = True Overwrite files with same namesfile_name = None Not usedfilename = None Not usedserial = None Not usedoutput_scale_factor = None Scale factor to be applied to output map just before writing. Normally the output map will have a mean of zero and SD of 1. This may lead to the maximum in the map being much greater than 1. You can adjust the output SD with this scale factor.local_resolution_smoothing_radius_ratio = 1 Ratio of smoothing radius to resolution (local resolution only)sharpeningsharpening_method = *shells b-factor Sharpening method. Shell sharpening applies a scale factor that is calculated in shells of resolution and smoothed. B-factor sharpening applies a Wilson B factor (exponential function) as a function of resolution.local_sharpen = False Sharpen locally (alternative is global sharpening). Note: can take a long timeanisotropic_sharpen = True Use anisotropic sharpening. Can be combined with local sharpening. Applies to shell scaling onlymodel_sharpen = False Model sharpening. Requires model supplied.n_bins = None Number of bins for sharpening (default 200 overall and 20 local)n_boxes = None Number of boxes. Alternative to box size.box_size_grid_units = None Size of core region of boxes (not including region where mask is applied, in grid units. Alternative to boxes for local sharpeningsharpen_all_maps = True Sharpen and write out half-maps in addition to the full map (applies to half-map sharpening with shell scaling only)local_resolution_map = True Calculate local resolution map at end (only applies if two half-maps are supplied and analyzed with shell scaling).b_factor_sharpeningmap_modificationb_iso = None Target B-value for map (sharpening will be applied to yield this value of b_iso). If sharpening method is not supplied, default is to use b_iso_to_d_cut sharpening.b_sharpen = None Sharpen with this b-value. Contrast with b_iso that yields a targeted value of b_isob_blur_hires = 200 Blur high_resolution data (higher than d_cut) with this b-value. Contrast with b_sharpen applied to data up to d_cut. Note on defaults: If None and b_sharpen is positive (sharpening) then high-resolution data is left as is (not sharpened). If None and b_sharpen is negative (blurring) high-resolution data is also blurred.resolution_dependent_b = None If set, apply resolution_dependent_b (b0 b1 b2). Log10(amplitudes) will start at 1, change to b0 at half of resolution specified, changing linearly, change to b1/2 at resolution specified, and change to b1/2+b2 at d_min_ratio*r...

- job_title = None Job title in PHENIX GUI, not used on command line
- input_filesseq_file = None Sequence file. Include all copies of each sequence so that the full contents of the map can be estimated. (Fasta format or sequences separated by blank lines)ncs_file = None File with NCS information (typically point-group NCS with the center specified). Typically in PDB format. Can also be a .ncs_spec file from phenix. Created automatically if symmetry is specified.external_map_file = None External map to be used to scale map_file (power vs resolution will be matched). Applies to b_factor_sharpening only.mtz_in = None MTZ file with coefficients for a map mtz_in_virtual = None Used internallymap_coeff_labels = None If map coefficients cannot be identified automatically from your MTZ file, you can specify the label or labels for them. (Please separate labels with blank space, MTZ columns grouped together separated by commas with no blanks.) You can specify: map_coeff_labels (e.g., FWT,PHWT) amplitudes and phases (e.g., FP,SIGFP PHIB) or amplitudes, phases, weights (e.g., FP,SIGFP PHIB FOM)map_modelfull_map = None Input full map filehalf_map = None Input half map filesmodel = None Input model file
- seq_file = None Sequence file. Include all copies of each sequence so that the full contents of the map can be estimated. (Fasta format or sequences separated by blank lines)
- ncs_file = None File with NCS information (typically point-group NCS with the center specified). Typically in PDB format. Can also be a .ncs_spec file from phenix. Created automatically if symmetry is specified.
- external_map_file = None External map to be used to scale map_file (power vs resolution will be matched). Applies to b_factor_sharpening only.
- mtz_in = None MTZ file with coefficients for a map
- mtz_in_virtual = None Used internally
- map_coeff_labels = None If map coefficients cannot be identified automatically from your MTZ file, you can specify the label or labels for them. (Please separate labels with blank space, MTZ columns grouped together separated by commas with no blanks.) You can specify: map_coeff_labels (e.g., FWT,PHWT) amplitudes and phases (e.g., FP,SIGFP PHIB) or amplitudes, phases, weights (e.g., FP,SIGFP PHIB

FOM)

• map_modelfull_map = None Input full map filehalf_map = None Input half map filesmodel = None Input model file

• full_map = None Input full map file

• half_map = None Input half map files

• model = None Input model file

• outputsharpened_map_coeffs_file = default Sharpened map coefficients file namesharpened_map_file_1 = default Sharpened half map 1 file namesharpened_map_file_2 = default Sharpened half map 2 file namelocal_resolution_map_file = default Local resolution map file nameoverwrite = True Overwrite files with same namesfile_name = None Not usedfilename = None Not usedserial = None Not usedoutput_scale_factor = None Scale factor to be applied to output map just before writing. Normally the output map will have a mean of zero and SD of 1. This may lead to the maximum in the map being much greater than 1. You can adjust the output SD with this scale factor.local_resolution_smoothing_radius_ratio = 1 Ratio of smoothing radius to resolution (local resolution only)

• sharpened_map_coeffs_file = default Sharpened map coefficients file name

• sharpened_map_file_1 = default Sharpened half map 1 file name

• sharpened_map_file_2 = default Sharpened half map 2 file name

• local_resolution_map_file = default Local resolution map file name

• overwrite = True Overwrite files with same names

• file_name = None Not used

• filename = None Not used

• serial = None Not used

• output_scale_factor = None Scale factor to be applied to output map just before writing. Normally the output map will have a mean of zero and SD of 1. This may lead to the maximum in the map being much greater than 1. You can adjust the output SD with this scale factor.

• local_resolution_smoothing_radius_ratio = 1 Ratio of smoothing radius to resolution (local resolution only)

• sharpeningsharpening_method = *shells b-factor Sharpening method. Shell sharpening applies a scale factor that is calculated in shells of resolution and smoothed. B-factor sharpening applies a Wilson B factor (exponential function) as a function of resolution.local_sharpen = False Sharpen locally (alternative is global sharpening). Note: can take a long timeanisotropic_sharpen = True Use anisotropic sharpening. Can be combined with local sharpening. Applies to shell scaling onlymodel_sharpen = False Model sharpening. Requires model supplied.n_bins = None Number of bins for sharpening (default 200 overall and 20 local)n_boxes = None Number of boxes. Alternative to box size.box_size_grid_units = None Size of core region of boxes (not including region where mask is applied, in grid units. Alternative to boxes for local sharpeningsharpen_all_maps = True Sharpen and write out half-maps in addition to the full map (applies to half-map sharpening with shell scaling only)local_resolution_map = True Calculate local resolution map at end (only applies if two half-maps are supplied and analyzed with shell scaling).

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• local_sharpen = False Sharpen locally (alternative is global sharpening). Note: can take a long time

- anisotropic_sharpen = True Use anisotropic sharpening. Can be combined with local sharpening. Applies to shell scaling only
- model_sharpen = False Model sharpening. Requires model supplied.
- n_bins = None Number of bins for sharpening (default 200 overall and 20 local)
- n_boxes = None Number of boxes. Alternative to box size.
- box_size_grid_units = None Size of core region of boxes (not including region where mask is applied, in grid units. Alternative to boxes for local sharpening
- sharpen_all_maps = True Sharpen and write out half-maps in addition to the full map (applies to half-map sharpening with shell scaling only)
- local_resolution_map = True Calculate local resolution map at end (only applies if two half-maps are supplied and analyzed with shell scaling).
- b_factor_sharpeningmap_modificationb_iso = None Target B-value for map (sharpening will be applied to yield this value of b_iso). If sharpening method is not supplied, default is to use b_iso_to_d_cut sharpening.b_sharpen = None Sharpen with this b-value. Contrast with b_iso that yields a targeted value of b_isob_blur_hires = 200 Blur high_resolution data (higher than d_cut) with this b-value. Contrast with b_sharpen applied to data up to d_cut. Note on defaults: If None and b_sharpen is positive (sharpening) then high-resolution data is left as is (not sharpened). If None and b_sharpen is negative (blurring) high-resolution data is also blurred.resolution_dependent_b = None If set, apply resolution_dependent_b (b0 b1 b2). Log10(amplitudes) will start at 1, change to b0 at half of resolution specified, changing linearly, change to b1/2 at resolution specified, and change to b1/2+b2 at d_min_ratio*resolutionnormalize_amplitudes_in_resdep = False Normalize amplitudes in resolution-dependent sharpeningd_min_ratio = 0.833 Sharpening will be applied using d_min equal to d_min_ratio times resolution. Default is 0.833scale_max = 100000 Scale amplitudes from inverse FFT to yield maximum of this valueinput_d_cut = None High-resolution limit for sharpeningrmsd = None RMSD of model to true model (if supplied). Used to estimate expected fall-of with resolution of correct part of model-based map. If None, assumed to be resolution times rmsd_resolution_factor.rmsd_resolution_factor = 0.25 default RMSD is resolution times resolution factorfraction_complete = None Completness of model (if supplied). Used to estimate correct part of model-based map. If None, estimated from max(FSC).auto_sharpen = True Automatically determine sharpening using kurtosis maximization or adjusted surface area. Default is Trueauto_sharpen_methods = no_sharpening b_iso *b_iso_to_d_cut resolution_dependent model_sharpening half_map_sharpening target_b_iso_to_d_cut external_map_sharpening None Methods to use in sharpening. b_iso searches for b_iso to maximize sharpening target (kurtosis or adjusted_sa). b_iso_to_d_cut applies b_iso only up to resolution specified, with fall-over of k_sharpen. Resolution dependent adjusts 3 parameters to sharpen variably over resolution range. Default is b_iso_to_d_cut . target_b_iso_to_d_cut uses target_b_iso_ratio to set b_iso.box_in_auto_sharpen = False Use a representative box of density for initial auto-sharpening instead of the entire map. Default is False.density_select_in_auto_sharpen = True Choose representative box of density for initial auto-sharpening with density_select method (choose region where there is high density). Normally use False for X-ray data and True for cryo-EM.density_select_threshold_in_auto_sharpen = None Threshold for density select choice of box. Default is 0.05. If your map has low overall contrast you might need to make this bigger such as 0.2.allow_box_if_b_iso_set = False Allow box_in_auto_sharpen (if set to True) even if b_iso is set. Default is to set box_n_auto_sharpen=False if b_iso is set.soft_mask = True Use soft mask (smooth change from inside to outside with radius based on resolution of map).use_weak_density = False When choosing box of representative density, use poor density (to get optimized map for weaker density)discard_if_worse = None Discard sharpening if worselocal_sharpening = None Sharpen locally

using overlapping regions. NOTE: Best to turn off local_aniso_in_local_sharpening if NCS is present. If local_aniso_in_local_sharpening is True and NCS is present this can distort the map for some NCS copies because an anisotropy correction is applied based on local density in one copy and is transferred without rotation to other copies.local_aniso_in_local_sharpening = None Use local anisotropy in local sharpening. Default is True unless NCS is present.overall_before_local = True Apply overall scaling before local scalingselect_sharpened_map = None Select a single sharpened map to useread_...

- map_modificationb_iso = None Target B-value for map (sharpening will be applied to yield this value of b_iso). If sharpening method is not supplied, default is to use b_iso_to_d_cut sharpening.b_sharpen = None Sharpen with this b-value. Contrast with b_iso that yields a targeted value of b_isob_blur_hires = 200 Blur high_resolution data (higher than d_cut) with this b-value. Contrast with b_sharpen applied to data up to d_cut. Note on defaults: If None and b_sharpen is positive (sharpening) then high-resolution data is left as is (not sharpened). If None and b_sharpen is negative (blurring) high-resolution data is also blurred.resolution_dependent_b = None If set, apply resolution_dependent_b (b0 b1 b2). Log10(amplitudes) will start at 1, change to b0 at half of resolution specified, changing linearly, change to b1/2 at resolution specified, and change to b1/2+b2 at d_min_ratio*resolutionnormalize_amplitudes_in_resdep = False Normalize amplitudes in resolution-dependent sharpeningd_min_ratio = 0.833 Sharpening will be applied using d_min equal to d_min_ratio times resolution. Default is 0.833scale_max = 100000 Scale amplitudes from inverse FFT to yield maximum of this valueinput_d_cut = None High-resolution limit for sharpeningrmsd = None RMSD of model to true model (if supplied). Used to estimate expected fall-of with resolution of correct part of model-based map. If None, assumed to be resolution times rmsd_resolution_factor.rmsd_resolution_factor = 0.25 default RMSD is resolution times resolution factorfraction_complete = None Completness of model (if supplied). Used to estimate correct part of model-based map. If None, estimated from max(FSC).auto_sharpen = True Automatically determine sharpening using kurtosis maximization or adjusted surface area. Default is Trueauto_sharpen_methods = no_sharpening b_iso *b_iso_to_d_cut resolution_dependent model_sharpening half_map_sharpening target_b_iso_to_d_cut external_map_sharpening None Methods to use in sharpening. b_iso searches for b_iso to maximize sharpening target (kurtosis or adjusted_sa). b_iso_to_d_cut applies b_iso only up to resolution specified, with fall-over of k_sharpen. Resolution dependent adjusts 3 parameters to sharpen variably over resolution range. Default is b_iso_to_d_cut . target_b_iso_to_d_cut uses target_b_iso_ratio to set b_iso.box_in_auto_sharpen = False Use a representative box of density for initial auto-sharpening instead of the entire map. Default is False.density_select_in_auto_sharpen = True Choose representative box of density for initial auto-sharpening with density_select method (choose region where there is high density). Normally use False for X-ray data and True for cryo-EM.density_select_threshold_in_auto_sharpen = None Threshold for density select choice of box. Default is 0.05. If your map has low overall contrast you might need to make this bigger such as 0.2.allow_box_if_b_iso_set = False Allow box_in_auto_sharpen (if set to True) even if b_iso is set. Default is to set box_n_auto_sharpen=False if b_iso is set.soft_mask = True Use soft mask (smooth change from inside to outside with radius based on resolution of map).use_weak_density = False When choosing box of representative density, use poor density (to get optimized map for weaker density)discard_if_worse = None Discard sharpening if worselocal_sharpening = None Sharpen locally using overlapping regions. NOTE: Best to turn off local_aniso_in_local_sharpening if NCS is present. If local_aniso_in_local_sharpening is True and NCS is present this can distort the map for some NCS copies because an anisotropy correction is applied based on local density in one copy and is transferred without rotation to other copies.local_aniso_in_local_sharpening = None Use local anisotropy in local sharpening. Default is True unless NCS is present.overall_before_local = True Apply overall scaling before local scalingselect_sharpened_map = None Select a single sharpened map to useread_sharpened_maps = No...

- b_iso = None Target B-value for map (sharpening will be applied to yield this value of b_iso). If sharpening method is not supplied, default is to use b_iso_to_d_cut sharpening.
- b_sharpen = None Sharpen with this b-value. Contrast with b_iso that yields a targeted value of b_iso
- b_blur_hires = 200 Blur high_resolution data (higher than d_cut) with this b-value. Contrast with b_sharpen applied to data up to d_cut. Note on defaults: If None and b_sharpen is positive (sharpening) then high-resolution data is left as is (not sharpened). If None and b_sharpen is negative (blurring) high-resolution data is also blurred.
- resolution_dependent_b = None If set, apply resolution_dependent_b (b0 b1 b2). Log10(amplitudes) will start at 1, change to b0 at half of resolution specified, changing linearly, change to b1/2 at resolution specified, and change to b1/2+b2 at d_min_ratio*resolution
- normalize_amplitudes_in_resdep = False Normalize amplitudes in resolution-dependent sharpening
- d_min_ratio = 0.833 Sharpening will be applied using d_min equal to d_min_ratio times resolution. Default is 0.833
- scale_max = 100000 Scale amplitudes from inverse FFT to yield maximum of this value
- input_d_cut = None High-resolution limit for sharpening
- rmsd = None RMSD of model to true model (if supplied). Used to estimate expected fall-of with resolution of correct part of model-based map. If None, assumed to be resolution times rmsd_resolution_factor.
- rmsd_resolution_factor = 0.25 default RMSD is resolution times resolution factor
- fraction_complete = None Completness of model (if supplied). Used to estimate correct part of model-based map. If None, estimated from max(FSC).
- auto_sharpen = True Automatically determine sharpening using kurtosis maximization or adjusted surface area. Default is True
- auto_sharpen_methods = no_sharpening b_iso *b_iso_to_d_cut resolution_dependent model_sharpening half_map_sharpening target_b_iso_to_d_cut external_map_sharpening None Methods to use in sharpening. b_iso searches for b_iso to maximize sharpening target (kurtosis or adjusted_sa). b_iso_to_d_cut applies b_iso only up to resolution specified, with fall-over of k_sharpen. Resolution dependent adjusts 3 parameters to sharpen variably over resolution range. Default is b_iso_to_d_cut . target_b_iso_to_d_cut uses target_b_iso_ratio to set b_iso.
- box_in_auto_sharpen = False Use a representative box of density for initial auto-sharpening instead of the entire map. Default is False.
- density_select_in_auto_sharpen = True Choose representative box of density for initial auto-sharpening with density_select method (choose region where there is high density). Normally use False for X-ray data and True for cryo-EM.
- density_select_threshold_in_auto_sharpen = None Threshold for density select choice of box. Default is 0.05. If your map has low overall contrast you might need to make this bigger such as 0.2.
- allow_box_if_b_iso_set = False Allow box_in_auto_sharpen (if set to True) even if b_iso is set. Default is to set box_n_auto_sharpen=False if b_iso is set.
- soft_mask = True Use soft mask (smooth change from inside to outside with radius based on resolution of map).
- use_weak_density = False When choosing box of representative density, use poor density (to get optimized map for weaker density)
- discard_if_worse = None Discard sharpening if worse

- local_sharpening = None Sharpen locally using overlapping regions. NOTE: Best to turn off local_aniso_in_local_sharpening if NCS is present. If local_aniso_in_local_sharpening is True and NCS is present this can distort the map for some NCS copies because an anisotropy correction is applied based on local density in one copy and is transferred without rotation to other copies.
- local_aniso_in_local_sharpening = None Use local anisotropy in local sharpening. Default is True unless NCS is present.
- overall_before_local = True Apply overall scaling before local scaling
- select_sharpened_map = None Select a single sharpened map to use
- read_sharpened_maps = None Read in previously-calculated sharpened maps
- write_sharpened_maps = None Write out local sharpened maps
- smoothing_radius = None Sharpen locally using smoothing_radius. Default is 2/3 of mean distance between centers for sharpening
- box_center = None You can specify the center of the box (A units)
- box_size = 30 30 30 You can specify the size of the box (grid units)
- target_n_overlap = 10 You can specify the targeted overlap of boxes in local sharpening
- restrict_map_size = None Restrict box map to be inside full map (required for cryo-EM data). Default is True if use_sg_symmetry=False and False if use_sg_symmetry=True
- remove_aniso = True You can remove anisotropy (overall and locally) during sharpening
- max_box_fraction = 0.5 If box is greater than this fraction of entire map, use entire map. Default is 0.5.
- density_select_max_box_fraction = 0.95 If box is greater than this fraction of entire map, use entire map for density_select. Default is 0.95
- cc_cut = 0.2 Estimate of minimum highly reliable CC in half-map FSC. Used to decide at what CC value to smooth the remaining CC values.
- max_cc_for_rescale = 0.2 Used along with cc_cut and scale_using_last to correct for small errors in FSC estimation at high resolution. If the value of FSC near the high-resolution limit is above max_cc_for_rescale, assume these values are correct and do not correct them.
- scale_using_last = 3 If set, assume that the last scale_using_last bins in the FSC for half-map or model sharpening are about zero (corrects for errors int the half-map process).
- mask_atoms = True Mask atoms when using model sharpening
- mask_atoms_atom_radius = 3 Mask for mask_atoms will have mask_atoms_atom_radius
- value_outside_atoms = None Value of map outside atoms (set to mean to have mean value inside and outside mask be equal)
- k_sharpen = 10 Steepness of transition between sharpening (up to resolution) and not sharpening (d < resolution). Note: for blurring, all data are blurred (regardless of resolution), while for sharpening, only data with d about resolution or lower are sharpened. This prevents making very high-resolution data too strong. Note 2: if k_sharpen is zero or None, then no transition is applied and all data is sharpened or blurred.
- iterate = False You can iterate auto-sharpening. This is useful in cases where you do not specify the solvent content and it is not accurately estimated until sharpening is optimized.
- optimize_b_blur_hires = False Optimize value of b_blur_hires. Only applies for auto_sharpen_methods b_iso_to_d_cut and b_iso. This is normally carried out and helps prevent over-blurring at high resolution if the same map is sharpened more than once.

- optimize_d_cut = None Optimize value of d_cut. Only applies for auto_sharpen_methods b_iso_to_d_cut and b_iso
- adjust_region_weight = True Adjust region_weight to make overall change in surface area equal to overall change in normalized regions over the range of search_b_min to search_b_max using b_iso_to_d_cut.
- region_weight_method = initial_ratio *delta_ratio b_iso Method for choosing region_weights. Initial_ratio uses ratio of surface area to regions at low B value. Delta ratio uses change in this ratio from low to high B. B_iso uses resolution-dependent b_iso (not weights) with the formula b_iso=5.9*d_min**2
- region_weight_factor = 1.0 Multiplies region_weight after calculation with region_weight_method above
- region_weight_buffer = 0.1 Region_weight adjusted to be region_weight_buffer away from minimum or maximum values
- target_b_iso_ratio = 5.9 Target b_iso ratio : b_iso is estimated as target_b_iso_ratio * resolution**2
- target_b_iso_model_scale = 0. For model sharpening, the target_biso is scaled (normally zero).
- signal_min = 3.0 Minimum signal in estimation of optimal b_iso. If not achieved, use any other method chosen.
- search_b_min = None Low bound for b_iso search. Default is -100.
- search_b_max = None High bound for b_iso search. Default is 300.
- search_b_n = None Number of b_iso values to search. Default is 21.
- residual_target = None Target for maximization steps in sharpening. Can be kurtosis or adjusted_sa (adjusted surface area). Default is adjusted_sa.
- sharpening_target = None Overall target for sharpening. Can be kurtosis or adjusted_sa (adjusted surface area). Used to decide which sharpening approach is used. Note that during optimization, residual_target is used (they can be the same.) Default is adjusted_sa.
- require_improvement = None Require improvement in score for sharpening to be applied. Default is True.
- region_weight = None Region weighting in adjusted surface area calculation. Score is surface area minus region_weight times number of regions. Default is set automatically. A smaller value will give more sharpening.
- sa_percent = None Percent of target regions used in calulation of adjusted surface area. Default is 30.
- fraction_occupied = None Fraction of molecular volume targeted to be inside contours. Used to set contour level. Default is 0.20
- n_bins = None Number of resolution bins for sharpening. Default is 20.
- regions_to_keep = None You can specify a limit to the number of regions to keep when generating the asymmetric unit of density.
- max_regions_to_test = None Number of regions to test for surface area in adjusted_sa scoring of sharpening. Default is 30
- eps = None
- k_sol = 0.35 k_sol value for model map calculation. IGNORED (Not applied)
- b_sol = 50 b_sol value for model map calculation. IGNORED (Not applied)
- b_factor_sharpening_output_filesshifted_map_file = shifted_map.ccp4 Input map file shifted to new origin.sharpened_map_file = None Sharpened input map file. In the same location as input

map.shifted_sharpened_map_file = None Input map file shifted to place origin at 0,0,0 and sharpened.sharpened_map_coeffs_file = None Sharpened input map (shifted to new origin if original origin was not 0,0,0), written out as map coefficientsoutput_weight_map_pickle_file = weight_map_pickle_file.pkl Output weight map pickle fileoutput_directory = None Directory where output files are to be written applied.

• shifted_map_file = shifted_map.ccp4 Input map file shifted to new origin.

• sharpened_map_file = None Sharpened input map file. In the same location as input map.

• shifted_sharpened_map_file = None Input map file shifted to place origin at 0,0,0 and sharpened.

• sharpened_map_coeffs_file = None Sharpened input map (shifted to new origin if original origin was not 0,0,0), written out as map coefficients

• output_weight_map_pickle_file = weight_map_pickle_file.pkl Output weight map pickle file

• output_directory = None Directory where output files are to be written applied.

• b_factor_sharpening_crystal_infois_crystal = None Defines whether this is a crystal (or cryo-EM). Default is True if use_sg_symmetry=True and False otherwise.resolution = None Optional nominal resolution of the map.solvent_content = None Optional solvent fraction of the cell.solvent_content_iterations = 3 Iterations of solvent fraction estimation. Used for ID of solvent content in boxed maps.molecular_mass = None Molecular mass of molecule in Da. Used as alternative method of specifying solvent content.ncs_copies = None You can specify ncs copies and seq file to define solvent contentwang_radius = None Wang radius for solvent identification. Default is 1.5* resolutionbuffer_radius = None Buffer radius for mask smoothing. Default is resolutionpseudo_likelihood = None Use pseudo-likelihood method for half-map sharpening. (In development)

• is_crystal = None Defines whether this is a crystal (or cryo-EM). Default is True if use_sg_symmetry=True and False otherwise.

• resolution = None Optional nominal resolution of the map.

• solvent_content = None Optional solvent fraction of the cell.

• solvent_content_iterations = 3 Iterations of solvent fraction estimation. Used for ID of solvent content in boxed maps.

• molecular_mass = None Molecular mass of molecule in Da. Used as alternative method of specifying solvent content.

• ncs_copies = None You can specify ncs copies and seq file to define solvent content

• wang_radius = None Wang radius for solvent identification. Default is 1.5* resolution

• buffer_radius = None Buffer radius for mask smoothing. Default is resolution

• pseudo_likelihood = None Use pseudo-likelihood method for half-map sharpening. (In development)

• crystal_infowrapping = None You can specify whether the map is wrapped (can map values outside bounds to inside with cell translations). Always true for crystallographic maps.

• wrapping = None You can specify whether the map is wrapped (can map values outside bounds to inside with cell translations). Always true for crystallographic maps.

• controlmultiprocessing = *multiprocessing sge lsf pbs condor pbspro slurm Choices are multiprocessing (single machine) or queuing systems Not implementedqueue_run_command = None run command for queue jobs. For example qsub. Not implementednproc = 1 Number of processors to use. NOTE: by default multiple processors will only be used in the map-to-model step (this is because multiprocessing requires writing out nproc sets of huge files and it can be very slow with distributed queues.). You can override this with force_nproc = True.ignore_symmetry_conflicts = False You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with box_map for example.verbose = False Verbose output

- multiprocessing = *multiprocessing sge lsf pbs condor pbspro slurm Choices are multiprocessing (single machine) or queuing systems Not implemented
- queue_run_command = None run command for queue jobs. For example qsub. Not implemented
- nproc = 1 Number of processors to use. NOTE: by default multiple processors will only be used in the map-to-model step (this is because multiprocessing requires writing out nproc sets of huge files and it can be very slow with distributed queues.). You can override this with force_nproc = True.
- ignore_symmetry_conflicts = False You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with box_map for example.
- verbose = False Verbose output
- guiGUI-specific parameter required for output directoryoutput_dir = None
- output_dir = None

Identifying symmetry in a map with map_symmetry

map_symmetry.html

Author(s)

- map_symmetry: Tom Terwilliger

Purpose

The routine map_symmetry will identify the symmetry in a map and create a file with the symmetry operations. Normally this is used to identify the symmetry used in processing a cryo-EM map.

Usage

How map_symmetry works:

Map-symmetry uses the density at a set of randomly-chosen points in a map to identify the symmetry in a map from a set of pre-defined symmetry types that include point-group symmetry as well as helical symmetry. For each possible symmetry, the correlation of values at symmetry-related points is calculated. The score for the symmetry type is the square root of the number of elements in the symmetry multiplied by the map correlation for that symmetry.

By default, map-symmetry assumes that the principal axes of symmetry in a map are along x, y, or z, and that helical symmetry is helical along z. The software has a database of 45 common symmetry elements and point groups, along with alternative settings of some of these. The user can specify a specific symmetry type to examine (e.g., I for icosahedral or D7 for a 7-fold axis along z perpendicular to a 2-fold axis along x or along y), or all elements in the database can be examined.

You can also use map-symmetry to find general (point-group or non-point-group) symmetry using the keyword "find_ncs_directly=True". In this case a density search will be carried out using spheres of density in the regions where the map varies the most as templates. Any similar density (with any translations or rotations) will indicate map symmetry.

For helical symmetry, a Fourier transform of the density is examined along the c* axis to find periodicity along z likely to correspond to the helical translation. These likely translations (and integer fractions of them) are tested along with possible rotations to check for helical symmetry.

Examples

Standard run of map_symmetry:

Running map_symmetry is easy. From the command-line you can type:

```
phenix.map_symmetry emd_8750.map symmetry=D7
```

where emd_8750.map is a map (CCP4, mrc or other related format), and in this case you are specifying that only D7 symmetry should be considered. An output file containing the symmetry (also known as NCS in Phenix) operators will be created (map_symmetry.ncs_spec).

You can leave the symmetry as default (symmetry=ALL), or you can specify symmetry with the name of a symmetry (symmetry=D7) or with a file with symmetry operators in the same format as map_symmetry.ncs_spec (symmetry_file=sym.ncs_spec).

Note that to find helical symmetry, you have to explicitly set the keyword include_helical_symmetry=True.

What to do after map_symmetry

Map symmetry will provide you with a symmetry file that shows the symmetry of your cryo-EM map. You can use this symmetry file in many later steps in cryo-EM structure determination, such as in density modification and model-building. You can use it as well to identify the asymmetric unit of your map by supplying it along with your map to map_box. You can also use this symmetry file to construct a full model from just the asymmetric unit of your model using apply_ncs.

Possible Problems

Specific limitations and problems:

Helical symmetry is assumed to be along the z-axis.

Notes on map symmetry in Phenix:

Helical symmetry is more difficult to analyze than other symmetries. (a) Having the helical parameters (rotation about z, rise) is very helpful because finding these is not so easy and takes some time with map_symmetry. (b) Helical symmetry has several different forms as well. There is 1-start (normal) and 2-start (two sets of helices, not handled automatically in Phenix). Helical symmetry is usually identified using a Fourier transform along Z. (c) As helical symmetry is not common and takes longer to find automatically, finding helical symmetry is off by default and requires setting the keyword include_helical_symmetry to True to look for it. (d) Only one-start helical symmetry is found automatically. For other symmetries, a symmetry file (.ncs_spec) would need to be supplied.

For other symmetries, having the information about symmetry type (D, O, etc) is slightly useful, but generally it is quick to identify these symmetries by direct testing. Note that knowing that symmetry is D2 (for example) does not actually define the matrices. There are two settings for D2 (and many others), one with the two-fold along x and one with the two-fold along the x=y, z=0 line. It is even more complicated for icosahedral symmetries.

Phenix uses the method mmtbx/ncs/ncs/generate_ncs_ops to generate NCS operators for all the common symmetries that are found in cyo-EM maps. If there are symmetries that we do not recognize, this is the place to put them.

In Phenix map analysis and model building and manipulation code, the mmtbx.ncs.ncs object is used to handle the symmetry relationships for cryo-EM symmetry. There is another NCS object that is used inside the model object.

Literature

Additional information

List of all available keywords

job_title = None Job title in PHENIX GUI, not used on command lineinput_filesmap_file = None File with CCP4-style map.symmetry_file = None Optional symmetry file (.ncs_spec format or BIOMT records) with reconstruction symmetryoutput_filessymmetry_out = symmetry_from_map.ncs_spec Output file (.ncs_spec format) describing symmetry in the maptemp_dir = temp_dir Temp directorycrystal_inforesolution = None High-resolution limit for main search. This can be lower resolution than the data. The search is quicker at lower resolution. .short_caption = High-resolution limituse_sg_symmetry = False If you set use_sg_symmetry=True then the symmetry of the space group will be used. For example in P1 a point at one end of the unit cell is next to a point on the other end. Normally for cryo-EM data this should be set to False and for crystal data it should be set to True.reconstruction_symmetrysymmetry = ANY Symmetry used in reconstruction. For example D7, C3, C2 I (icosahedral),T (tetrahedral), or ANY (try everything and use the highest symmetry found). Not needed if symmetry is supplied.must_be_consistent_with_space_group_number = None Searches for symmetry must be compatible with this space group number.include_helical_symmetry = False You can include or exclude searches for helical symmetrysymmetry_center = None Center (in A) for Symmetry operators (if ncs is found automatically). If set to None, guess is the center of the cell (see use_unit_cell_center) and then if that fails, found automatically as the center of the density in the map.check_grid_offset = True Check optimizing map center by moving up to 1 grid unit in each directionuse_unit_cell_center = True Use center of full unit cell as guess of center of cellfind_ncs_directly = False You can use find_ncs_from_density to find arbitrary NCS. Try this if your NCS is not proper NCS and not helical.optimize_center = False Optimize position of symmetry center. Also checks for center at (0,0,0) vs center of maphelical_rot_deg = None helical rotation about z in degreeshelical_trans_z_angstrom = None helical translation along z in Angstrom unitsmax_helical_optimizations = 2 Number of optimizations of helical parameters when finding NCSmax_helical_ops_to_check = 5 Number of helical operations in each direction to check when finding NCSmax_helical_rotations_to_check = None Number of helical rotations to check when finding NCStwo_fold_along_x = None Specifies D or I or T two-fold is along x (True) or y (False). If None, both are tried.smallest_object = None Dimension of smallest object to consider when finding symmetry. Default is 5 * resolutionscore_basis = ncs_score cc *None Symmetry score basis. Normally ncs_score (sqrt(n)* cc) is used except for identification of helical symmetryscale_weight_fractional_translation = 1.05 Give slight increase in weighting in helical symmetry search to translations that are a fraction (1/2, 1/3) of the d-spacing of the peak of intensity in the fourier transform of the density.random_points = 300 Number of random points in map to examine in finding symmetryidentify_ncs_id = True If Symmetry is not point-group symmetry, try each possible operator when evaluating Symmetry and choose the one that results in the most uniform density at Symmetry-related points.min_ncs_cc = 0.80 Minimum Symmetry CC to keep operators when identifying automaticallyn_rescore = 5 Number of Symmetry operators to rescoreop_max = 14 If ncs_type is ANY, try up to op_max-fold symmetriestol_r = 0.02 tolerance in rotations for point group or helical symmetryabs_tol_t = 2 tolerance in translations (A) for point group or helical symmetrymax_helical_operators = None Maximum helical operators (if extending existing helical operators)rel_tol_t = .05 tolerance in translations (fractional) for point group or helical symmetryrequire_helical_or_point_group_symmetry = False normally helical or point-group symmetry (or none) is expected. However in some cases (helical + rotational symmetry for example) th...

- job_title = None Job title in PHENIX GUI, not used on command line
- input_filesmap_file = None File with CCP4-style map.symmetry_file = None Optional symmetry file (.ncs_spec format or BIOMT records) with reconstruction symmetry
- map_file = None File with CCP4-style map.
- symmetry_file = None Optional symmetry file (.ncs_spec format or BIOMT records) with reconstruction symmetry
- output_filessymmetry_out = symmetry_from_map.ncs_spec Output file (.ncs_spec format) describing symmetry in the maptemp_dir = temp_dir Temp directory

- symmetry_out = symmetry_from_map.ncs_spec Output file (.ncs_spec format) describing symmetry in the map
- temp_dir = temp_dir Temp directory
- crystal_inforesolution = None High-resolution limit for main search. This can be lower resolution than the data. The search is quicker at lower resolution. .short_caption = High-resolution limituse_sg_symmetry = False If you set use_sg_symmetry=True then the symmetry of the space group will be used. For example in P1 a point at one end of the unit cell is next to a point on the other end. Normally for cryo-EM data this should be set to False and for crystal data it should be set to True.
- resolution = None High-resolution limit for main search. This can be lower resolution than the data. The search is quicker at lower resolution. .short_caption = High-resolution limit
- use_sg_symmetry = False If you set use_sg_symmetry=True then the symmetry of the space group will be used. For example in P1 a point at one end of the unit cell is next to a point on the other end. Normally for cryo-EM data this should be set to False and for crystal data it should be set to True.
- reconstruction_symmetrysymmetry = ANY Symmetry used in reconstruction. For example D7, C3, C2 I (icosahedral),T (tetrahedral), or ANY (try everything and use the highest symmetry found). Not needed if symmetry is supplied.must_be_consistent_with_space_group_number = None Searches for symmetry must be compatible with this space group number.include_helical_symmetry = False You can include or exclude searches for helical symmetrysymmetry_center = None Center (in A) for Symmetry operators (if ncs is found automatically). If set to None, guess is the center of the cell (see use_unit_cell_center) and then if that fails, found automatically as the center of the density in the map.check_grid_offset = True Check optimizing map center by moving up to 1 grid unit in each directionuse_unit_cell_center = True Use center of full unit cell as guess of center of cellfind_ncs_directly = False You can use find_ncs_from_density to find arbitrary NCS. Try this if your NCS is not proper NCS and not helical.optimize_center = False Optimize position of symmetry center. Also checks for center at (0,0,0) vs center of maphelical_rot_deg = None helical rotation about z in degreeshelical_trans_z_angstrom = None helical translation along z in Angstrom unitsmax_helical_optimizations = 2 Number of optimizations of helical parameters when finding NCSmax_helical_ops_to_check = 5 Number of helical operations in each direction to check when finding NCSmax_helical_rotations_to_check = None Number of helical rotations to check when finding NCStwo_fold_along_x = None Specifies D or I or T two-fold is along x (True) or y (False). If None, both are tried.smallest_object = None Dimension of smallest object to consider when finding symmetry. Default is 5 * resolutionscore_basis = ncs_score cc *None Symmetry score basis. Normally ncs_score (sqrt(n)* cc) is used except for identification of helical symmetryscale_weight_fractional_translation = 1.05 Give slight increase in weighting in helical symmetry search to translations that are a fraction (1/2, 1/3) of the d-spacing of the peak of intensity in the fourier transform of the density.random_points = 300 Number of random points in map to examine in finding symmetryidentify_ncs_id = True If Symmetry is not point-group symmetry, try each possible operator when evaluating Symmetry and choose the one that results in the most uniform density at Symmetry-related points.min_ncs_cc = 0.80 Minimum Symmetry CC to keep operators when identifying automaticallyn_rescore = 5 Number of Symmetry operators to rescoreop_max = 14 If ncs_type is ANY, try up to op_max-fold symmetriestol_r = 0.02 tolerance in rotations for point group or helical symmetryabs_tol_t = 2 tolerance in translations (A) for point group or helical symmetrymax_helical_operators = None Maximum helical operators (if extending existing helical operators)rel_tol_t = .05 tolerance in translations (fractional) for point group or helical symmetryrequire_helical_or_point_group_symmetry = False normally helical or point-group symmetry (or none) is expected. However in some cases (helical + rotational symmetry for example) this is not needed and is not the case.

• symmetry = ANY Symmetry used in reconstruction. For example D7, C3, C2 I (icosahedral),T (tetrahedral), or ANY (try everything and use the highest symmetry found). Not needed if symmetry is supplied.

• must_be_consistent_with_space_group_number = None Searches for symmetry must be compatible with this space group number.

• include_helical_symmetry = False You can include or exclude searches for helical symmetry

• symmetry_center = None Center (in A) for Symmetry operators (if ncs is found automatically). If set to None, guess is the center of the cell (see use_unit_cell_center) and then if that fails, found automatically as the center of the density in the map.

• check_grid_offset = True Check optimizing map center by moving up to 1 grid unit in each direction

• use_unit_cell_center = True Use center of full unit cell as guess of center of cell

• find_ncs_directly = False You can use find_ncs_from_density to find arbitrary NCS. Try this if your NCS is not proper NCS and not helical.

• optimize_center = False Optimize position of symmetry center. Also checks for center at (0,0,0) vs center of map

• helical_rot_deg = None helical rotation about z in degrees

• helical_trans_z_angstrom = None helical translation along z in Angstrom units

• max_helical_optimizations = 2 Number of optimizations of helical parameters when finding NCS

• max_helical_ops_to_check = 5 Number of helical operations in each direction to check when finding NCS

• max_helical_rotations_to_check = None Number of helical rotations to check when finding NCS

• two_fold_along_x = None Specifies D or I or T two-fold is along x (True) or y (False). If None, both are tried.

• smallest_object = None Dimension of smallest object to consider when finding symmetry. Default is 5 * resolution

• score_basis = ncs_score cc *None Symmetry score basis. Normally ncs_score (sqrt(n)* cc) is used except for identification of helical symmetry

• scale_weight_fractional_translation = 1.05 Give slight increase in weighting in helical symmetry search to translations that are a fraction (1/2, 1/3) of the d-spacing of the peak of intensity in the fourier transform of the density.

• random_points = 300 Number of random points in map to examine in finding symmetry

• identify_ncs_id = True If Symmetry is not point-group symmetry, try each possible operator when evaluating Symmetry and choose the one that results in the most uniform density at Symmetry-related points.

• min_ncs_cc = 0.80 Minimum Symmetry CC to keep operators when identifying automatically

• n_rescore = 5 Number of Symmetry operators to rescore

• op_max = 14 If ncs_type is ANY, try up to op_max-fold symmetries

• tol_r = 0.02 tolerance in rotations for point group or helical symmetry

• abs_tol_t = 2 tolerance in translations (A) for point group or helical symmetry

• max_helical_operators = None Maximum helical operators (if extending existing helical operators)

• rel_tol_t = .05 tolerance in translations (fractional) for point group or helical symmetry

- require_helical_or_point_group_symmetry = False normally helical or point-group symmetry (or none) is expected. However in some cases (helical + rotational symmetry for example) this is not needed and is not the case.
- controlnproc = 1 Number of processors to userandom_seed = 171731 Random seedquick = True Run quickly if possibleverbose = False Verbose output
- nproc = 1 Number of processors to use
- random_seed = 171731 Random seed
- quick = True Run quickly if possible
- verbose = False Verbose output
- guiGUI-specific parameter required for output directoryoutput_dir = None
- output_dir = None

Model-building into cryo-EM and low-resolution maps with map_to_model

map_to_model.html

Author(s)

- map_to_model: Tom Terwilliger

Purpose

The routine map_to_model will interpret a map (cryo-EM, low-resolution X-ray) and try to build an atomic model, fully automatically.

Video Tutorial

The tutorial video is available on the Phenix YouTube channel and covers the following topics:

- basic overview
- how to run phenix.map_to_model via the GUI

GUI

A Graphical User Interface is available.

Usage

How map_to_model works:

If you have a CCP4-style (mrc, etc) map or just mtz map coefficients and a sequence file, you can use map_to_model to build a model into your map. The tool map_to_model will identify what kind of chains to build based on your sequence file. It will find where your molecule is in the map and cut out and work with just that part of the density.

To run map_to_model successfully, you'll need a map with resolution of about 4.5 A or better. The higher the resolution, the more complete the model that you will typically get.

It is easiest to supply map_to_model with a map that has already been sharpened optimally (for example with phenix.map_sharpening). It is also easiest if you cut out just the unique part of your map before supplying it to map_to_model (you can use phenix.map_box to do this with the extract_unique=True option). After you are all done you can reconstruct the entire molecule using phenix.apply_ncs.

(Alternatively you can have map_to_model do everything for you (map_sharpening, symmetry average, cut out the unique part, reconstruct the entire molecule). This is a lot slower so normally it is best to prepare just the right part of the map in advance.)

You have a choice in map_to_model of running quickly (quick=True) or more thoroughly. Normally quick=True should be fine, particularly for building protein chains. In this mode, map_to_model runs the phenix.trace_and_build tool to trace the chain, identify positions of side chains, build a model for each segment of density, and refine the entire model. This can be done quickly with multiple processors, requiring perhaps 15 minutes to build 300 residues with 4 processors.

If you choose thorough building (quick=False), the map_to_model will try several different methods of building your model. One method is to build in small regions of the map. The map_to_model tool will cut the density in your map into small pieces of connected density and try to build model into each one. It will merge all the pieces into a compact model, refine it, and superimpose final model on the original map. Another method is the trace_and_build method used in the quick version. A third is standard resolve model-building. A fourth is finding helices and strands.

If your structure has RNA or DNA, model-building for these chains is done separately from protein. The type of chain or chains to be built are chosen based on your sequence file. If multiple chain types are considered, the entire map is interpreted with each chain type, then the best-fitting non-overlapping chains are chosen.

How to use map_to_model and trace_and_build to build and complete a model

The recommended way to build a model using phenix is first to run map_to_model and get a starting model, and then to try and build additional model and fix errors using trace_and_build and fix_insertions_deletions. Note that map_to_model will show you all the chains that are built along with their density, so you can see even before it is done whether there are changes in chain tracing that you would like to make.

Once map_to_model is done, you can try to connect parts of the model, extend chains, or build new chains using trace_and_build. map_to_model writes out each segment as a separate chain, so you can specify chains to be connected with trace_and_build. For example, if you think chains A and B should be connected, you can use pdbtools to create a working model working_AB.pdb with just chains A and B. Then you can run trace_and_build with working_AB.pdb and your map, specifying connect_chains and turning off extend_chains and find_chains. Then trace_and_build will attempt to connect your chains A and B and create a new model with a single chain. You can then replace chains A and B in your model with the new chain.

You can also try to fix insertions and deletions in your model by running fix_insertions_deletions with your model, your sequence file, and your map.

You can try to improve the sequence assignment with sequence_from_map. This is run automatically at the end of map_to_model and trace_and_build but if you create new chains you might want to try and assign them to sequence separately.

If you want to create a model with full symmetry, you can do this with map_to_model as well. You supply the full map, your model of the unique part of the structure as the starting_model, and the symmetry matrices that you can obtain from the map_symmetry tool. You can turn off all the model-building methods so that the only thing that happens is assembly and refinement. Then map_to_model will apply the symmetry, remove any pieces that overlap due to symmetry, and refine the entire model.

Applying magnification to the map

Cryo-EM maps often have a scale that is not precisely defined by the experiment. map_to_model allows application of a scale factor (magnification) to the grid of the map. Normally this scale factor will be close to 1. If the magnification is specified and is not equal to 1, it will be applied to the input map and a magnification map will be written out to the output directory and will be used as if it were the original input file from then on. Additionally any input symmetry information will be adjusted by the same magnification factor (translations and centers are scaled by the magnification factor, rotations are unchanged). Any input models are not modified.

Shifting the map to the origin

Most crystallographic maps have the origin at the corner of the map (grid point [0,0,0]), while most cryo-EM maps have the orgin in the middle of the map. To make a consistent map, any maps with an origin not at the corner are shifted to put the origin at grid point [0,0,0]. This map is the shifted map that is used for further steps in model-building. At the conclusion of model-building, the model is shifted back to superimpose on the original map.

Finding the region containing the molecule

If requested, (density_select=True), the region of the map containing density is cut out of the entire map. This is particularly useful if the original map is very large and the molecule only takes up a small part of the map. This portion of the map is then shifted to place the origin at grid point [0,0,0]. (At the conclusion of model-building, the final model is shifted back to superimpose on the original map.) The region containing density is chosen as a box containing all the points above a threshold, typically 5% of the maximum in the map.

Map sharpening/blurring

If requested, (auto_sharpen=True) the resolution dependence of the map will be adjusted to maximize the clarity of the map. It is generally preferable to do this in advance using the phenix.map_sharpening tool however.

If you run phenix.map_sharpening, You can choose to use map kurtosis or the adjusted surface area of the map (default) for this purpose.

Kurtosis is a standard statistical measure that reflects the peakiness of the map.

The adjusted surface area is a combination of the surface area of contours in the map at a particular threshold and of the number of distinct regions enclosed by the top 30% (default) of those contours. The threshold is chosen by default to be one where the volume enclosed by the contours is 20% of the non-solvent volume in the map. The weighting between the surface area (to be maximized) and number of regions enclosed (to be minimized) is chosen empirically (default region_weight=20).

Several resolution-dependent functions are tested, and the one that gives the best adjusted surface area (or kurtosis) is chosen. In each case the map is transformed to obtain Fourier coefficients. The amplitudes of these coefficients are then adjusted, keeping the phases constant. The available functions for modifying the amplitudes are:

```
No sharpening (map is left as is)

Sharpening b-factor applied over entire resolution range (b_sharpen
applied to achieve an effective isotropic overall b-value of b_iso).

Sharpening b-factor applied up to resolution specified with the
resolution=xxx keyword, then blurring applied beyond this resolution (with
transition specified by the keyword k_sharpen, b_iso_to_d_cut). If
sharpening value is less than zero (map is to be blurred),
the blurring is applied over the entire resolution range.

Resolution-dependent sharpening factor with three parameters.
First the resolution-dependence of the map is removed by normalizing the
amplitudes. Then a scale factor S is to the data, where
log10(S) is determined by coefficients b[0],b[1],b[2] and a resolution
d_cut (typically d_cut is the nominal resolution of the map).
The value of log10(S) varies smoothly from 0 at resolution=infinity, to b[0]
at d_cut/2, to b[1] at d_cut, and to b[1]+b[2] at the highest resolution
in the map. The value of b[1] is limited to being no larger than b[0] and the
value of b[1]+b[2] is limited to be no larger than b[1].
```

```
Sharpening using a half-dataset correlation.
The resolution-dependent correlation of density in two half-maps
is used to identify the optimal resolution-dependent weighting of
the map. This approach requires a target resolution which is used
to set the overall fall-off with resolution for an ideal map. That
fall-off for an ideal map is then multiplied by an estimated
resolution-dependent correlation of density in the map with the true
map (the estimation comes from the half-map correlations).

Model-based sharpening.
You can identify the sharpening parameters using your map and a
model. This approach requires a guess of the RMSD between the model and
the true model. The resolution-dependent correlation of model and map
density is used as in the half-map approach above to identify the
weighting of Fourier coefficients.
```

Local sharpening

Any of the sharpening methods can be applied locally instead of globally over the entire map. For local sharpening, the map is cut into overlapping boxes and sharpening parameter are determined for that box and applied. For parts of boxes that overlap, the distances from a grid point to the centers of the corresponding boxes are used to weight the values for those boxes.

Finding the asymmetric unit of the map

Normally you should supply map_to_model with the unique part (asymmetric unit of the map. You can get this using the phenix.map_box tool with the extract_unique=True command. However if you want map_to_model to do this automatically, you can supply symmetry matrices describing the symmetry used to average the map (if any), then map_to_model will try to define a region of the map that represents the asymmetric unit of the map. Application of the symmetry operators to the asymmetric unit will generate the entire map, and application to a model built into the asymmetric unit will generate the entire model.

You can also supply the type of symmetry (e.g., C3, D7, etc) and map_to_model will try to find that symmetry in the map. You can even search for all plausible symmetry (ANY). For helical symmetry either a symmetry file or the rotation and translation information is required, however.

Normally identification of the asymmetric unit and segmentation of the map (below) are done as a single step, yielding an asymmetric unit and a set of contiguous regions of density within that asymmetric unit. The asymmetric unit will be written out as a map to the segmentation_dir directory, superimposed on the shifted map (so that they can be viewed together in Coot).

Segmentation of the map

By default (segment=True) the map or asymmetric unit of the map will be segmented (cut into small pieces) into regions of connected density. This is done by choosing a threshold of density and identifying contiguous regions where all grid points are above this threshold. The threshold is chosen to yield regions that have a size corresponding to about 50 residues. The regions of density are written out to the segmentation_dir directory and are superimposed on the shifted map (if you load the shifted map in Coot and a region map in Coot, they should superimpose.)

Model-building

Models are built in several ways by map_to_model and then the best-fitting, non-overlapping models are chosen. If the quick method for map_to_model is selected and the chain type is protein, then the default is to run just the trace_and_build model-building method. If quick=False, all methods are tried. The main methods used for model-building are:

```
Chain tracing followed by model-building (trace_and_build). Continuous
density is identified in the map, choosing the longest paths if there
are choices. Then C-alpha positions are identified from the presence of
side chain density, an all-atom model is constructed and refined.

Standard RESOLVE model-building for PROTEIN/RNA/DNA for the entire
asymmetric unit (or the entire molecule if no symmetry was used ).

Helices (RNA) or helices/strands (PROTEIN) for entire asymmetric unit

tracing chain (RNA/PROTEIN/DNA) for each segmented region, with various
values of map sharpening applied

RESOLVE model-building for each segmented region, with various values of
map sharpening applied
```

Intermediate models are refined with phenix.real_space_refine and are written out relative to the shifted map with origin at [0,0,0]. You can view these intermediate models, the shifted map, and the shifted map containing just the asymmetric unit , and any region maps in Coot and they should all superimpose.

Once all intermediate models are built, all models of each chain type are combined, taking the best-fitting model for each part of the map. Then all chain types are combined, once again taking the best-fitting model for each part of the map. The models are refined again.

Then (if present) symmetry is applied to the model and the full model is refined. Finally the best model, with symmetry applied if present, is shifted to match the original map and is written out.

Iterative map improvement with model-building and sharpening

A procedure is available for carrying out multiple cycles multiple cycles of model-building and map sharpening, using the model from each cycle in the sharpening process. In practice this may help in borderline cases but it is not recommended for normal use. The map is sharpened to make it as similar as possible to density calculated from the current model and the new map is used for the next cycle of model building.

Using map_to_model to just build parts of the model that are missing

You can use map_to_model to build missing parts of your model in two steps:

```
Use map_box to mask your map around your partial model. Use the keywords
mask_atoms=True and invert_mask=True to remove the density near your model.

Take your masked map and supply a sequence file corresponding to the part
that remains, and run map_to_model to try and build the remaining parts
of your model
```

Note that this is appropriate for cases where an entire chain or domain is missing, usually not for fitting loops (those are best handled with fit_loops).

Examples

Standard run of map_to_model:

Running map_to_model is easy. From the command-line you can type:

```
phenix.map_to_model my_map.map seq.fa resolution=3
```

where my_map.map is a CCP4, mrc or other related map format, seq.fa is a sequence file. Normally you should sharpen your map with phenix.map_sharpening and cut out the unique part of the map with phenix.map_box (using the extract_unique=True keyword) before supplying it to map_to_model.

Output files from map_to_model

map_to_model.pdb: A PDB file with the resulting model, superimposed on

the original map (or on the magnified map if magnification is applied).

Standard run of map_to_model, specifying symmetry type:

Running map_to_model with symmetry is also easy. From the command-line you can type:

```
phenix.map_to_model my_map.map seq.fa symmetry=D7 resolution=3
```

Here map_to_model will look for D7 (7-fold symmetry along the c-axis and 2-fold symmetry along a or b) in the map and apply it if it is found. It will also write out the matrices corresponding to this symmetry.

Using carry_on to continue with a partially-finished run

If you have a completed or partially-completed run of map_to_model and you want to run again but you do not want to re-run all the steps, you can just carry on from where you left off by using the keyword carry_on=True.

Using carry_on to break up your run into small pieces

You can use carry_on=True to progressively build up your model from pieces or to run many jobs in parallel.

First you run map_to_model with stop_after_segment_and_split=True to split up your map and write a file (the info_pickle_file) that has all the necessary information to put everything together.

Then you can create script files to run each small step in the analysis. You can do that with the keyword split_up_job=True. This will put your scripts in a commands/ subdirectory and you can run them with your queueing system. When they are done, you can use carry_on=True to put everything together.

You can also specify a source or other command to be placed in all your scripts with the keyword source_command="xxx".

The scripts created by the split_up_job command use a set of keywords like this (here only RNA will be built, building will be done in just region number 26, secondary structure searches are disabled and overall model-building is disabled:

```
do_only_one_thing=True
chain_type=RNA
build_in_regions=True
input_map_id_start=26
input_map_id_end=26

include_helices_strands_only=False
include_phase_and_build=False
```

Another run might look like this, where overall model-building is done starting with model number 4:

```
do_only_one_thing=True
chain_type=PROTEIN
build_in_regions=False
include_helices_strands_only=False
include_phase_and_build=True
model_start=4
```

You can run all the combinations that you would like in parallel.

Special treatment of structures with very high symmetry

If your structure has very high symmetry (by default, 20 or more symmetry operators), then segment_and_split_map will try to cut out a piece of your map and work with that instead of working with the whole map. You can control this with the keyword select_au_box=True (or None, which will give the default behavior). If you use select_au_box then a box that contains about n_au_box asymmetric units of the map (default of 5) will be cut out of your map. That map will be worked with along with the symmetry operators that apply to it for the remainder of the analysis. At the end of the analysis, just the region cut out will be built and placed in the original reference frame. This process can greatly speed up building and save on memory.

Building just the asymmetric unit of your model to save memory

Another way to save on memory is to run just the segment_and_split step of map_to_model on a computer with a lot of memory, then use the small map created by segment_and_split for subsequent stages, and then finally take the resulting model for this small map, put it into map_to_model on the computer with a lot of memory and use it to create a final model. Here are the steps for doing this:

```
In a clean directory,
run phenix.map_to_model with the keyword
stop_after_segment_and_split_map=True on your map, including symmetry
information. Run this on a computer with a huge amount of memory.
Your segmentation information will be in segmented_maps/

Take the file segmented_maps/box_map_au.ccp4, which contains just the
asymmetric unit of density from your original map, and in a new
directory (perhaps box_map_directory/)
run phenix.map_to_model with box_map_au.ccp4 and the unique
part of your sequence file. This can be run (with luck) on a computer
with a more moderate amount of memory. The final model will be
box_map_directory/map_to_model/map_to_model.pdb;
it should match the box_map_au.ccp4 map.

Go back to the original directory where you split up your original map.
Run phenix.map_to_model with carry_on=True and specifying three keywords:

starting_model=box_map_directory/map_to_model/map_to_model.pdb
starting_model_is_from_box_map_au=True
build_new_model=False

Now map_to_model will start with your model from the small map, shift it
to match the original map, apply any symmetry you used originally,
refine the model, and write out a model map_to_model/map_to_model.pdb
```

How long does map_to_model take to run

Map to model can take quite a while to run. For a small structure (200 amino acids), expect a couple hours. For a large one (1000 amino acids or protein and RNA or DNA), expect a day or more.

What to do after running map_to_model

In a typical case, the Map to model tool should yield a partial model that fits well with your cryo-EM map. The next steps are usually to refine and validate the model with RealSpaceRefine, and to rebuild or build parts of the model that are incomplete or that fit poorly using Coot or Isolde.

Possible Problems

If you have a very large structure it is possible that your computer may not have enough memory to run map_to_model and that one or more sub-processes might crash.

The easiest solution is to run phenix.map_box with the extract_unique=True option beforehand so that you are running map_to_model on just the unique part of the structure.

You can also just cut out boxes of density from your structure with the phenix.map_box tool and work on each one individually. An easy way to do this is to make a dummy model (any model) that kind of fills the space that you are interested in (doesn't have to really fill it, it just has to have atoms that show how big a box would have to be to hold them all). Then run the "map_box" function in the Phenix GUI with your map and this dummy model. You get a boxed map that you can work with.

You can also set the keyword coarse_grid=True to use a coarse grid in RESOLVE and save memory. You can also cut back the resolution which will save memory. Otherwise, you can try on a computer with even more memory.

If your queueing system crashes during a run or one or more sub-processes crashes, then you might end up with models built for some stages of building and others not. You can continue on with the keyword carry_on=True in this case. (see above in the section on combining partial runs).

Specific limitations and problems:

If your map is inverted (left-handed), docking and model-building will not work properly. You can often tell if your map is inverted because any helices will be left-handed. If you are unsure, you can run MapBox with invert_hand=True to invert the map and then see if docking works. Note that if your map is inverted, you will want to invert all your maps and start everything from the beginning.

Literature

A fully automatic method yielding initial models from high-resolution electron cryo-microscopy maps. T.C. Terwilliger, P.D. Adams, P.V. Afonine, and O.V. Sobolev. Nature Methods 15, 905-9080 (2019).

- A fully automatic method yielding initial models from high-resolution electron cryo-microscopy maps. T.C. Terwilliger, P.D. Adams, P.V. Afonine, and O.V. Sobolev. Nature Methods 15, 905-9080 (2019).

Additional information

List of all available keywords

job_title = None Job title in PHENIX GUI, not used on command lineinput_filesmap_coeffs_file = None File with map coefficientsmap_coeffs_labels = None Optional label specifying which columns of of map coefficients to usemap_file = None File with CCP4-style map. This is the input map and its position defines the starting reference frame. The map may be shifted in later steps but the final model will be in the same reference frame as this input map.seq_file = None Sequence filencs_file = None Input symmetry file for use with segment_and_split_map. Refers to the symmetry in the reference frame of the input map.model_file = None Optional input PDB files to be used as starting point for further model building. This model or models should be in the same reference frame as the input map unless starting_model_is_from_box_map_au is set.starting_model_is_from_box_map_au = False The starting model was built into box_map_au.ccp4, the asymmetric unit map created by segment_and_split_map. This model is offset from the original cell. This keyword will shift the starting model to match the original map before using it.sharpen_with_starting_model = True If starting_model is supplied, use it in identifying optimal resolution-dependent map sharpening. Also defines whether model-based sharpening will be used on cycles after the first.include_starting_model = True If starting_model is supplied, use it in model-building.target_ncs_au_file = None Optional PDB file to partially define the ncs asymmetric unit of the map. The coordinates in this file will be used to mark part of the ncs au and all points nearby that are not part of another ncs au will be added. The model should be positioned relative to the input map. If starting_model is supplied and include_starting_model=True then it will be used for this purpose as well.partial_model = None Optional input PDB file with protein/RNA/DNA chains for one step in model-building. If supplied, no model-building for this step will be carried out. Must match offset map (normally not the same as the original map.) This model will replace the model normally built at the step specified by partial_model_type and should match its position. Multiple partial models may be specified. You need to specify what type each one is with the keyword partial_model_type (the order of partial_model entries must match the order of partial_model_type entries). NOTE: This is intended as a way to use the models from a previous run, not to input your own model. For your own models, use starting_model instead.partial_model_type = init_PROTEIN init_RNA init_DNA helices_strands_only_PROTEIN build_rna_helices_RNA standard_PROTEIN standard_RNA standard_DNA *final_PROTEIN final_RNA final_DNA Model type for a partial model. The model will be used instead of generating this kind of model. One required for each partial_model. The partial_model_type keyword specifies both the model-building step (init, helices_strands_only, build_rna_helices, standard, final) and the chain_type (RNA, DNA, PROTEIN).trace_chain_pdb_in = None Input PDB file to use instead of tracing chains. Suitable only for the case where one region is being analyzed. The PDB file should match the original map (not the offset region).info_pickle_file = None Pickle file from segment_and_split_map. You can read in this file and and choose which map file to use. default is in segmentation_directory with file name of segment_and_split_map_info.pkl .input_map_id_start = None first ID (number 1 to N) of the map to analyze from info_pickle fileinput_map_id_end = None last ID (number 1 to N) of the map to analyze from info_pickle fileseed_ca_in = None Input PDB file with possible CA positions to assist in tracing chainsseed_ca_only = False Only use sites from seed_ca_in (do not find additional sites)model_pickle_file = None Pickle file with working models. You can read in this file and and start with the models in it. Only suitable if just one region of the map is to be analyzed (same...

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match offset map (normally not the same as the original map.) This model will replace the model normally built at the step specified by partial_model_type and should match its position. Multiple partial models may be specified. You need to specify what type each one is with the keyword partial_model_type (the order of partial_model entries must match the order of partial_model_type entries). NOTE: This is intended as a way to use the models from a previous run, not to input your own model. For your own models, use starting_model instead.partial_model_type = init_PROTEIN init_RNA init_DNA helices_strands_only_PROTEIN build_rna_helices_RNA standard_PROTEIN standard_RNA standard_DNA *final_PROTEIN final_RNA final_DNA Model type for a partial model. The model will be used instead of generating this kind of model. One required for each partial_model. The partial_model_type keyword specifies both the model-building step (init, helices_strands_only, build_rna_helices, standard, final) and the chain_type (RNA, DNA, PROTEIN).trace_chain_pdb_in = None Input PDB file to use instead of tracing chains. Suitable only for the case where one region is being analyzed. The PDB file should match the original map (not the offset region).info_pickle_file = None Pickle file from segment_and_split_map. You can read in this file and and choose which map file to use. default is in segmentation_directory with file name of segment_and_split_map_info.pkl .input_map_id_start = None first ID (number 1 to N) of the map to analyze from info_pickle fileinput_map_id_end = None last ID (number 1 to N) of the map to analyze from info_pickle fileseed_ca_in = None Input PDB file with possible CA positions to assist in tracing chainsseed_ca_only = False Only use sites from seed_ca_in (do not find additional sites)model_pickle_file = None Pickle file with working models. You can read in this file and and start with the models in it. Only suitable if just one region of the map is to be analyzed (same as for trace_chain_pdb_in)pdb_to_shift = None You can supply a mo...

- map_coeffs_file = None File with map coefficients
- map_coeffs_labels = None Optional label specifying which columns of of map coefficients to use
- map_file = None File with CCP4-style map. This is the input map and its position defines the starting reference frame. The map may be shifted in later steps but the final model will be in the same reference frame as this input map.
- seq_file = None Sequence file
- ncs_file = None Input symmetry file for use with segment_and_split_map. Refers to the symmetry in the reference frame of the input map.
- model_file = None Optional input PDB files to be used as starting point for further model building. This model or models should be in the same reference frame as the input map unless starting_model_is_from_box_map_au is set.
- starting_model_is_from_box_map_au = False The starting model was built into box_map_au.ccp4, the asymmetric unit map created by segment_and_split_map. This model is offset from the original cell. This keyword will shift the starting model to match the original map before using it.
- sharpen_with_starting_model = True If starting_model is supplied, use it in identifying optimal resolution-dependent map sharpening. Also defines whether model-based sharpening will be used on cycles after the first.
- include_starting_model = True If starting_model is supplied, use it in model-building.
- target_ncs_au_file = None Optional PDB file to partially define the ncs asymmetric unit of the map. The coordinates in this file will be used to mark part of the ncs au and all points nearby that are not part of another ncs au will be added. The model should be positioned relative to the input map. If starting_model is supplied and include_starting_model=True then it will be used for this purpose as well.

• partial_model = None Optional input PDB file with protein/RNA/DNA chains for one step in model-building. If supplied, no model-building for this step will be carried out. Must match offset map (normally not the same as the original map.) This model will replace the model normally built at the step specified by partial_model_type and should match its position. Multiple partial models may be specified. You need to specify what type each one is with the keyword partial_model_type (the order of partial_model entries must match the order of partial_model_type entries). NOTE: This is intended as a way to use the models from a previous run, not to input your own model. For your own models, use starting_model instead.

• partial_model_type = init_PROTEIN init_RNA init_DNA helices_strands_only_PROTEIN build_rna_helices_RNA standard_PROTEIN standard_RNA standard_DNA *final_PROTEIN final_RNA final_DNA Model type for a partial model. The model will be used instead of generating this kind of model. One required for each partial_model. The partial_model_type keyword specifies both the model-building step (init, helices_strands_only, build_rna_helices, standard, final) and the chain_type (RNA, DNA, PROTEIN).

• trace_chain_pdb_in = None Input PDB file to use instead of tracing chains. Suitable only for the case where one region is being analyzed. The PDB file should match the original map (not the offset region).

• info_pickle_file = None Pickle file from segment_and_split_map. You can read in this file and and choose which map file to use. default is in segmentation_directory with file name of segment_and_split_map_info.pkl .

• input_map_id_start = None first ID (number 1 to N) of the map to analyze from info_pickle file

• input_map_id_end = None last ID (number 1 to N) of the map to analyze from info_pickle file

• seed_ca_in = None Input PDB file with possible CA positions to assist in tracing chains

• seed_ca_only = False Only use sites from seed_ca_in (do not find additional sites)

• model_pickle_file = None Pickle file with working models. You can read in this file and and start with the models in it. Only suitable if just one region of the map is to be analyzed (same as for trace_chain_pdb_in)

• pdb_to_shift = None You can supply a model and then set the keyword shift_type to specify which map this model is to to match. You can reverse the shift with reverse_shift=True. Requires the info_pickle_file file. If specified this is the only thing that happens (stops after writing out the file to shifted_pdb.)

• reverse_shift = False Specifies that the shift is from shifted_map or box_map_au reference frame back TO the original map reference frame (default is FROM the original map reference frame to the shifted_map or box_map_au map.)

• output_filesparams_out = map_to_model_params.eff Output parameters filepdb_out = map_to_model.pdb Output PDB file. It will be placed in both the output_directory and the final_output_directory.pdb_out_unique = map_to_model_unique.pdb Output PDB file (ncs AU). Just asymmetric unit of modelsharpened_map_file = sharpened_map.ccp4 Output sharpened map file. This will be the map that is used in model-building. It will be superimposed on the original map.pdb_out_reverse = map_to_model_reverse.pdb Optional output reversed PDB file (traced opposite direction as pdb_out). Created if include_reverse is set.protein_mask_file = None Output ccp4-style map with mask for region of macromoleculeregion_model_suffix = None Suffix (before .pdb) for region modelsshifted_pdb = shifted_pdb.pdb Name of output file when using pdb_to_shift.output_directory = None Place where most files will go. Default is map_to_model_xxxfinal_output_directory = None The map_to_model.pdb file and sharpened map files will go here in addition to the output_directory.segmentation_directory = None Place where segmented maps will go if segment=True. Default is segmented_maps_xxxdisplay_in_coot = False Show intermediate models and maps in Coot

- params_out = map_to_model_params.eff Output parameters file
- pdb_out = map_to_model.pdb Output PDB file. It will be placed in both the output_directory and the final_output_directory.
- pdb_out_unique = map_to_model_unique.pdb Output PDB file (ncs AU). Just asymmetric unit of model
- sharpened_map_file = sharpened_map.ccp4 Output sharpened map file. This will be the map that is used in model-building. It will be superimposed on the original map.
- pdb_out_reverse = map_to_model_reverse.pdb Optional output reversed PDB file (traced opposite direction as pdb_out). Created if include_reverse is set.
- protein_mask_file = None Output ccp4-style map with mask for region of macromolecule
- region_model_suffix = None Suffix (before .pdb) for region models
- shifted_pdb = shifted_pdb.pdb Name of output file when using pdb_to_shift.
- output_directory = None Place where most files will go. Default is map_to_model_xxx
- final_output_directory = None The map_to_model.pdb file and sharpened map files will go here in addition to the output_directory.
- segmentation_directory = None Place where segmented maps will go if segment=True. Default is segmented_maps_xxx
- display_in_coot = False Show intermediate models and maps in Coot
- crystal_infochain_types = PROTEIN DNA RNA Chain type (PROTEIN/DNA/RNA). You can choose one or more. If not specified; it will be guessed from your sequence file. Multiple chain types are fine.minimum_fraction = 0.001 If chain types automatically determined, ignore any with fraction less than this.scattering_table = n_gaussian wk1995 it1992 *electron neutron Choice of scattering table for structure factor calculations. Standard for X-ray is n_gaussian, for cryoEM is electron.is_crystal = None Defines whether this is a crystal (or cryo-EM). Default is True if map_coefficients are supplied and False if a map is supplied. If True and no symmetry operators are supplied, segmentation yields the asymmetric unit of the crystal. Additionally the final model will represent the asymmetric unit of the crystal (not the entire map). Normally set is_crystal and use_sg_symmetry together.use_sg_symmetry = None If you set use_sg_symmetry=True then the symmetry of the space group will be used. For example in P1 a point at one end of the unit cell is next to a point on the other end. Normally for cryo-EM data this should be set to False. Default is True if map_coefficients are supplied and False if a map is supplied.minimum_resolution = 2.5 If resolution is finer than minimum_resolution, use minimum_resolution. This speeds up model-building.resolution = None High-resolution limit. Data will be truncated at this resolution. If a map is supplied, it will be Fourier filtered at this resolution (multiplied by d_min_ratio). Required if input is a map and only_original_map is not set.solvent_content = None Solvent fraction of the cell. If this is density cut out from a bigger cell, you can specify the fraction of the volume of this cell that is taken up by the macromolecule. Normally set automatically. Values go from 0 to 1.solvent_content_iterations = 3 Iterations of solvent fraction estimationtruncate_at_d_min = None Truncate data at resolution specified. Default is True if map_coeffs are supplied, False if map is suppliedmap_inside_box = True Place centers of all chains inside (0,1). This is required during normal operation.space_group = None Space group (normally read from the data file)unit_cell = None Unit Cell (normally read from the data file)chain_type = PROTEIN DNA RNA Do not set. Use chain_types instead.sequence = None Sequence as text string. Normally supply a sequence file insteadorigin_frac = None Saves origin shiftsolvent_fraction = None Same as solvent_content. Only used on input and copied to solvent content.
- chain_types = PROTEIN DNA RNA Chain type (PROTEIN/DNA/RNA). You can choose one or more. If not specified; it will be guessed from your sequence file. Multiple chain types are fine.

• minimum_fraction = 0.001 If chain types automatically determined, ignore any with fraction less than this.

• scattering_table = n_gaussian wk1995 it1992 *electron neutron Choice of scattering table for structure factor calculations. Standard for X-ray is n_gaussian, for cryoEM is electron.

• is_crystal = None Defines whether this is a crystal (or cryo-EM). Default is True if map_coefficients are supplied and False if a map is supplied. If True and no symmetry operators are supplied, segmentation yields the asymmetric unit of the crystal. Additionally the final model will represent the asymmetric unit of the crystal (not the entire map). Normally set is_crystal and use_sg_symmetry together.

• use_sg_symmetry = None If you set use_sg_symmetry=True then the symmetry of the space group will be used. For example in P1 a point at one end of the unit cell is next to a point on the other end. Normally for cryo-EM data this should be set to False. Default is True if map_coefficients are supplied and False if a map is supplied.

• minimum_resolution = 2.5 If resolution is finer than minimum_resolution, use minimum_resolution. This speeds up model-building.

• resolution = None High-resolution limit. Data will be truncated at this resolution. If a map is supplied, it will be Fourier filtered at this resolution (multiplied by d_min_ratio). Required if input is a map and only_original_map is not set.

• solvent_content = None Solvent fraction of the cell. If this is density cut out from a bigger cell, you can specify the fraction of the volume of this cell that is taken up by the macromolecule. Normally set automatically. Values go from 0 to 1.

• solvent_content_iterations = 3 Iterations of solvent fraction estimation

• truncate_at_d_min = None Truncate data at resolution specified. Default is True if map_coeffs are supplied, False if map is supplied

• map_inside_box = True Place centers of all chains inside (0,1). This is required during normal operation.

• space_group = None Space group (normally read from the data file)

• unit_cell = None Unit Cell (normally read from the data file)

• chain_type = PROTEIN DNA RNA Do not set. Use chain_types instead.

• sequence = None Sequence as text string. Normally supply a sequence file instead

• origin_frac = None Saves origin shift

• solvent_fraction = None Same as solvent_content. Only used on input and copied to solvent content.

• reconstruction_symmetrysymmetry = None Symmetry used in reconstruction. For example D7, C3, C2 I (icosahedral),T (tetrahedral), or ANY (try everything and use the highest symmetry found). Ignored if ncs_file is supplied or if asymmetric_map is specified or if a map obtained with extract_unique is supplied.find_symmetry = None Find symmetry in the map (alternative to supplying a symmetry file or specifying symmetry or supplying a map of unique part or specifying asymmetric_map=True).unique_part_only = True If symmetry is used, only the unique part of the map will be analyzed. (Do not refine full model and do not write out full model. Otherwise the same as usual).asymmetric_map = None Specifies that this is an asymmetric map and no symmetry is to be supplied or found. Alternative to supplying a symmetry file or symmetry.ncs_center = None Center (in A) for symmetry operators (if symmetry is found automatically). If set to None, first guess is the center of the cell and then if that fails, found automatically as the center of the density in the map.optimize_center = None Optimize position of symmetry center. Default is False if ncs_center is supplied or center of map is used and True if it is found automatically).helical_rot_deg = None helical rotation about z in degreeshelical_trans_z_angstrom = None helical translation along z in Angstrom unitstwo_fold_along_x = None Specifies if D or I two-fold is along x (True) or y (False). If None, both are tried.random_points = None Number of random points in map to examine in finding symmetryn_rescore = None Number of symmetry operators to rescoreop_max = None If symmetry is ANY, try up to op_max-fold symmetriestol_r = 0.02 tolerance in rotations for point group or helical symmetryabs_tol_t = 2 tolerance in translations (A) for point group or helical symmetryrel_tol_t = 0.05 tolerance in translations (fractional) for point group or helical symmetrymax_helical_operators = None Maximum helical operators (if extending existing helical operators)require_helical_or_point_group_symmetry = None Normally helical or point-group symmetry (or none) is expected. However in some cases (helical + rotational symmetry for example) this is not needed and is not the case. If set to True then the program will stop just before symmetry is applied if symmetry is present but helical or point-group symmetry is not present.

• symmetry = None Symmetry used in reconstruction. For example D7, C3, C2 I (icosahedral),T (tetrahedral), or ANY (try everything and use the highest symmetry found). Ignored if ncs_file is supplied or if asymmetric_map is specified or if a map obtained with extract_unique is supplied.

• find_symmetry = None Find symmetry in the map (alternative to supplying a symmetry file or specifying symmetry or supplying a map of unique part or specifying asymmetric_map=True).

• unique_part_only = True If symmetry is used, only the unique part of the map will be analyzed. (Do not refine full model and do not write out full model. Otherwise the same as usual).

• asymmetric_map = None Specifies that this is an asymmetric map and no symmetry is to be supplied or found. Alternative to supplying a symmetry file or symmetry.

• ncs_center = None Center (in A) for symmetry operators (if symmetry is found automatically). If set to None, first guess is the center of the cell and then if that fails, found automatically as the center of the density in the map.

• optimize_center = None Optimize position of symmetry center. Default is False if ncs_center is supplied or center of map is used and True if it is found automatically).

• helical_rot_deg = None helical rotation about z in degrees

• helical_trans_z_angstrom = None helical translation along z in Angstrom units

• two_fold_along_x = None Specifies if D or I two-fold is along x (True) or y (False). If None, both are tried.

• random_points = None Number of random points in map to examine in finding symmetry

• n_rescore = None Number of symmetry operators to rescore

• op_max = None If symmetry is ANY, try up to op_max-fold symmetries

• tol_r = 0.02 tolerance in rotations for point group or helical symmetry

• abs_tol_t = 2 tolerance in translations (A) for point group or helical symmetry

• rel_tol_t = 0.05 tolerance in translations (fractional) for point group or helical symmetry

• max_helical_operators = None Maximum helical operators (if extending existing helical operators)

• require_helical_or_point_group_symmetry = None Normally helical or point-group symmetry (or none) is expected. However in some cases (helical + rotational symmetry for example) this is not needed and is not the case. If set to True then the program will stop just before symmetry is applied if symmetry is present but helical or point-group symmetry is not present.

• map_modificationmagnification = None Magnification to apply to input map. Input map grid will be scaled by magnification factor before anything else is done.b_iso = None Target B-value for map (sharpening will be applied to yield this value of b_iso)b_sharpen = None Sharpen with this b-value. Contrast with b_iso that yield a targeted value of b_isob_blur_hires = 200 Blur high_resolution data (higher than d_cut) with this b-value. Contrast with b_sharpen applied to data up to d_cut. Note on defaults: If None and b_sharpen is positive (sharpening) then high-resolution data is left as is (not sharpened). If None and b_sharpen is negative (blurring) high-resolution data is also blurred.resolution_dependent_b_sharpen = None If set, apply resolution_dependent_b_sharpen (b0 b1 b2). Log10(amplitudes) will start at 1, change to b0 at half of resolution specified, changing linearly, change to b1 at resolution specified, and change to b2 at high-resolution limit of mapnormalize_amplitudes_in_resdep = False Normalize amplitudes in resolution-dependent sharpeningd_min_ratio = 0.833 Sharpening will be applied using d_min equal to d_min_ratio times resolution. If None, box of reflections with the same grid as the map used.input_d_cut = None High-resolution limit for sharpeningrmsd = None RMSD of model to true model (if supplied). Used to estimate expected fall-of with resolution of correct part of model-based map. If None, assumed to be resolution times rmsd_resolution_factor.rmsd_resolution_factor = 0.25 default RMSD is resolution times resolution factorauto_sharpen = False Apply auto-sharpening in segment_and_split_map (requires segment=True).auto_sharpen_regions = False Automatically determine sharpening for each region (in addition to the overall map).auto_sharpen_methods = no_sharpening b_iso *b_iso_to_d_cut resolution_dependent model_sharpening half_map_sharpening target_b_iso_to_d_cut None Methods to use in sharpening. b_iso searches for b_iso to maximize sharpening target (kurtosis or adjusted_sa). b_iso_to_d_cut applies b_iso only up to resolution specified, with fall-over of k_sharpen. Resolution dependent adjusts 3 parameters to sharpen variably over resolution range. Default is b_iso_to_d_cut . target_b_iso_to_d_cut uses target_b_iso_ratio to set b_iso.local_sharpening = None Sharpen locally using overlapping regions. NOTE: Best to turn off local_aniso_in_local_sharpening if symmetry is present. If local_aniso_in_local_sharpening is True and symmetry is present this can distort the map for some symmetry copies because an anisotropy correction is applied based on local density in one copy and is transferred without rotation to other copies.local_aniso_in_local_sharpening = None Use local anisotropy in local sharpening. Default is True unless symmetry is present.overall_before_local = True Apply overall scaling before local scalingbox_in_auto_sharpen = False Use a representative box of density for initial auto-sharpening instead of the entire map.density_select_in_auto_sharpen = False Choose representative box of density for initial auto-sharpening with density_select method (choose region where there is high density). Normally set to False (instead in map_to_model use density_select=True which cuts out the map before auto-sharpening)allow_box_if_b_iso_set = False Allow box_in_auto_sharpen (if set to True) even if b_iso is set. Default is to set box_n_auto_sharpen=False if b_iso is set.soft_mask = None Use soft mask (smooth change from inside to outside with radius based on resolution of map). Default False unless thorough is Truemax_box_fraction = None If box is greater than this fraction of entire map, use entire map.density_select_max_box_fraction = None If box is greater than this fraction of entire map, use entire map for density_select. Default is 0.95use_weak_density = False When choosing box of representative density, use poor density (to get optimized map for weaker density)k_sharpen = None Steepness of transition between...

- magnification = None Magnification to apply to input map. Input map grid will be scaled by magnification factor before anything else is done.
- b_iso = None Target B-value for map (sharpening will be applied to yield this value of b_iso)
- b_sharpen = None Sharpen with this b-value. Contrast with b_iso that yield a targeted value of b_iso
- b_blur_hires = 200 Blur high_resolution data (higher than d_cut) with this b-value. Contrast with b_sharpen applied to data up to d_cut. Note on defaults: If None and b_sharpen is positive (sharpening) then high-resolution data is left as is (not sharpened). If None and b_sharpen is negative (blurring)

high-resolution data is also blurred.

• resolution_dependent_b_sharpen = None If set, apply resolution_dependent_b_sharpen (b0 b1 b2). Log10(amplitudes) will start at 1, change to b0 at half of resolution specified, changing linearly, change to b1 at resolution specified, and change to b2 at high-resolution limit of map

• normalize_amplitudes_in_resdep = False Normalize amplitudes in resolution-dependent sharpening

• d_min_ratio = 0.833 Sharpening will be applied using d_min equal to d_min_ratio times resolution. If None, box of reflections with the same grid as the map used.

• input_d_cut = None High-resolution limit for sharpening

• rmsd = None RMSD of model to true model (if supplied). Used to estimate expected fall-of with resolution of correct part of model-based map. If None, assumed to be resolution times rmsd_resolution_factor.

• rmsd_resolution_factor = 0.25 default RMSD is resolution times resolution factor

• auto_sharpen = False Apply auto-sharpening in segment_and_split_map (requires segment=True).

• auto_sharpen_regions = False Automatically determine sharpening for each region (in addition to the overall map).

• auto_sharpen_methods = no_sharpening b_iso *b_iso_to_d_cut resolution_dependent model_sharpening half_map_sharpening target_b_iso_to_d_cut None Methods to use in sharpening. b_iso searches for b_iso to maximize sharpening target (kurtosis or adjusted_sa). b_iso_to_d_cut applies b_iso only up to resolution specified, with fall-over of k_sharpen. Resolution dependent adjusts 3 parameters to sharpen variably over resolution range. Default is b_iso_to_d_cut . target_b_iso_to_d_cut uses target_b_iso_ratio to set b_iso.

• local_sharpening = None Sharpen locally using overlapping regions. NOTE: Best to turn off local_aniso_in_local_sharpening if symmetry is present. If local_aniso_in_local_sharpening is True and symmetry is present this can distort the map for some symmetry copies because an anisotropy correction is applied based on local density in one copy and is transferred without rotation to other copies.

• local_aniso_in_local_sharpening = None Use local anisotropy in local sharpening. Default is True unless symmetry is present.

• overall_before_local = True Apply overall scaling before local scaling

• box_in_auto_sharpen = False Use a representative box of density for initial auto-sharpening instead of the entire map.

• density_select_in_auto_sharpen = False Choose representative box of density for initial auto-sharpening with density_select method (choose region where there is high density). Normally set to False (instead in map_to_model use density_select=True which cuts out the map before auto-sharpening)

• allow_box_if_b_iso_set = False Allow box_in_auto_sharpen (if set to True) even if b_iso is set. Default is to set box_n_auto_sharpen=False if b_iso is set.

• soft_mask = None Use soft mask (smooth change from inside to outside with radius based on resolution of map). Default False unless thorough is True

• max_box_fraction = None If box is greater than this fraction of entire map, use entire map.

• density_select_max_box_fraction = None If box is greater than this fraction of entire map, use entire map for density_select. Default is 0.95

• use_weak_density = False When choosing box of representative density, use poor density (to get optimized map for weaker density)

- k_sharpen = None Steepness of transition between sharpening (up to resolution) and not sharpening (d < resolution). Note: for blurring, all data are blurred (regardless of resolution), while for sharpening, only data with d about resolution or lower are sharpened. This prevents making very high-resolution data too strong. Note 2: if k_sharpen is zero or None, then no transition is applied and all data is sharpened or blurred. Note 3: only used if b_iso is set.
- iterate = False You can iterate auto-sharpening. This is useful in cases where you do not specify the solvent content and it is not accurately estimated until sharpening is optimized.
- optimize_b_blur_hires = False Optimize value of b_blur_hires. Only applies for auto_sharpen_methods b_iso_to_d_cut and b_iso. This is normally carried out and helps prevent over-blurring at high resolution if the same map is sharpened more than once.
- optimize_d_cut = None Optimize value of d_cut. Only applies for auto_sharpen_methods b_iso_to_d_cut and b_iso
- region_weight_method = initial_ratio *delta_ratio b_iso Method for choosing region_weights. Initial_ratio uses ratio of surface area to regions at low B value. Delta ratio uses change in this ratio from low to high B. B_iso uses resolution-dependent b_iso (not weights) with the formula b_iso=5.9*d_min**2
- search_b_min = None Low bound for b_iso search.
- search_b_max = None High bound for b_iso search.
- search_b_n = None Number of b_iso values to search.
- residual_target = None Target for maximization steps in sharpening. Can be kurtosis or adjusted_sa (adjusted surface area)
- sharpening_target = None Overall target for sharpening. Can be kurtosis or adjusted_sa (adjusted surface area). Used to decide which sharpening approach is used. Note that during optimization, residual_target is used (they can be the same.)
- target_b_iso_ratio = 5.9 Target b_iso ratio : b_iso is estimated as target_b_iso_ratio * resolution**2
- region_weight = 40 Region weighting in adjusted surface area calculation. Score is surface area minus region_weight times number of regions. Default is 40.
- discard_if_worse = None Discard sharpening if worse
- sa_percent = 30. Percent of target regions used in calulation of adjusted surface area. Default is 30.
- fraction_occupied = 0.20 Fraction of molecular volume targeted to be inside contours. Used to set contour level. Default is 0.20
- n_bins = 20 Number of resolution bins for sharpening. Default is 20.
- max_regions_to_test = 30 Number of regions to test for surface area in adjusted_sa scoring of sharpening
- eps = None
- strategykeep_longest_model = None Keep longest model (most residues)build_new_model = True If set to False, no new model building is done...just read in any existing input models and put them together and refine.cycles = 1 You can run cycles of model-building interpersed with model-based map sharpening.cycle_auto_sharpen = True Apply auto-sharpening on all cycles after the first, even if auto_sharpen is False. This applies model-based sharpening.density_select = None Run map_box with density_select=True to cut out the region in the input map that contains density. Useful if the input map is much larger than the structure. Default is True if an unmodified map is supplied and False if is_crystal is set or map_coeffs are supplied or the map has been extracted with extract_unique.density_select_threshold = None Choose region where density is this fraction of

maximum or greatersoft_mask_in_density_select = False Use soft mask (smooth change from inside to outside with radius based on resolution of map) when cutting out map with map_boxget_half_height_width = None Use 4 times half-width at half-height as estimate of max sizebox_ncs_au = None Box the map containing just the au of the map. Default is True if an unmodified map is supplied and False if is_crystal is set or map_coeffs are supplied or the map has been extracted with extract_unique.segment = True Run segment_and_split_map to break up map into pieces for model-building (recommended for larger structures). Required if map origin is not at (0,0,0). Not used in trace_and_build. You can also run segment_and_split_map separately and then run model-building (there are more parameters available for segment_and_split_map in that case).build_in_regions = None Run building on individual segmented regionsinclude_trace_and_build = True If segment is True, run trace_and_build on the entire asymmetric unit of the map.include_phase_and_build = None If segment is True, run phase_and_build on the entire asymmetric unit of the map.include_helices_strands_only = None If True, run build_one_model on the entire asymmetric unit of the map with helices_stands (protein) or with build_rna_helices (RNA).max_segments = None Maximum number of segments to keep (after joining and insertion). Default is take all with min_segment_length or more residuesmin_segments = 1 Minimum number of segments to keep (after joining and insertion) before skipping if segment length is too shortmin_segment_length = None Minimum residues in a segment to keep in trace_and_build (after joining and insertion). NOTE: default of 7 is set to build as much model as possible. Default in trace_and_build is 15, suitable for optimal results with sequence_from_map. Default if quick is 15, otherwise 7.split_and_join = None Split fragments at low density and then rejoin. Criterion is sd_ratio_pare_model. Default False if quick, True otherwise.trace_outside_model = None Trace outside model. After initial tracing, mask out region found and look for additional segments. Default False if quick, True otherwise.retry_long_branches = None If a long branch is found, try to retrace chaincorrect_segments = True Correct segments. Try to fix errors using fixed model as a template.vary_sharpening = None Variable sharpening values. Zero is always added. For example 75 -75 -125 will sharpen by B=75 then blur by b=75 and b=125. May increase residues built but decrease accuracy. Default in trace_and_build and sequence_from_map is None. Default is 75 -75 -125.vary_high_density_from_model = None Try high_density_from_model True and False. Default is True unless quick=Truehigh_density_from_model = None Create high density in map along path of main chain atoms in fixed model in trace_and_build. Also sets keep_main_chain_path. NOTE: Default is try True/False unless quick=True. True may build as much model as possible. Default in trace_and_build is False, suitable for optimal results with sequence_from_map.fix_insertions_deletions = None Retrace chains to fix...

- keep_longest_model = None Keep longest model (most residues)
- build_new_model = True If set to False, no new model building is done...just read in any existing input models and put them together and refine.
- cycles = 1 You can run cycles of model-building interspersed with model-based map sharpening.
- cycle_auto_sharpen = True Apply auto-sharpening on all cycles after the first, even if auto_sharpen is False. This applies model-based sharpening.
- density_select = None Run map_box with density_select=True to cut out the region in the input map that contains density. Useful if the input map is much larger than the structure. Default is True if an unmodified map is supplied and False if is_crystal is set or map_coeffs are supplied or the map has been extracted with extract_unique.
- density_select_threshold = None Choose region where density is this fraction of maximum or greater
- soft_mask_in_density_select = False Use soft mask (smooth change from inside to outside with radius based on resolution of map) when cutting out map with map_box

- get_half_height_width = None Use 4 times half-width at half-height as estimate of max size
- box_ncs_au = None Box the map containing just the au of the map. Default is True if an unmodified map is supplied and False if is_crystal is set or map_coeffs are supplied or the map has been extracted with extract_unique.
- segment = True Run segment_and_split_map to break up map into pieces for model-building (recommended for larger structures). Required if map origin is not at (0,0,0). Not used in trace_and_build. You can also run segment_and_split_map separately and then run model-building (there are more parameters available for segment_and_split_map in that case).
- build_in_regions = None Run building on individual segmented regions
- include_trace_and_build = True If segment is True, run trace_and_build on the entire asymmetric unit of the map.
- include_phase_and_build = None If segment is True, run phase_and_build on the entire asymmetric unit of the map.
- include_helices_strands_only = None If True, run build_one_model on the entire asymmetric unit of the map with helices_stands (protein) or with build_rna_helices (RNA).
- max_segments = None Maximum number of segments to keep (after joining and insertion). Default is take all with min_segment_length or more residues
- min_segments = 1 Minimum number of segments to keep (after joining and insertion) before skipping if segment length is too short
- min_segment_length = None Minimum residues in a segment to keep in trace_and_build (after joining and insertion). NOTE: default of 7 is set to build as much model as possible. Default in trace_and_build is 15, suitable for optimal results with sequence_from_map. Default if quick is 15, otherwise 7.
- split_and_join = None Split fragments at low density and then rejoin. Criterion is sd_ratio_pare_model. Default False if quick, True otherwise.
- trace_outside_model = None Trace outside model. After initial tracing, mask out region found and look for additional segments. Default False if quick, True otherwise.
- retry_long_branches = None If a long branch is found, try to retrace chain
- correct_segments = True Correct segments. Try to fix errors using fixed model as a template.
- vary_sharpening = None Variable sharpening values. Zero is always added. For example 75 -75 -125 will sharpen by B=75 then blur by b=75 and b=125. May increase residues built but decrease accuracy. Default in trace_and_build and sequence_from_map is None. Default is 75 -75 -125.
- vary_high_density_from_model = None Try high_density_from_model True and False. Default is True unless quick=True
- high_density_from_model = None Create high density in map along path of main chain atoms in fixed model in trace_and_build. Also sets keep_main_chain_path. NOTE: Default is try True/False unless quick=True. True may build as much model as possible. Default in trace_and_build is False, suitable for optimal results with sequence_from_map.
- fix_insertions_deletions = None Retrace chains to fix insertions and deletions in trace_and_build. Default is False unless quick=False.
- max_nproc_for_build_one_model = 30 Use only up to this number of processors for a single run of phase_and_build. Typically there is little time advantage to using more than about 30 processors and if you use too many the time could increase and quality decrease due to splitting up the model-building too much.

- nmodels_for_phase_and_build = None Number of models to build in phase_and_build. Default 2 unless thorough is True (then 20)
- score_with_cc_in_phase_and_build = True Use map-model CC to score in phase_and build
- include_reverse = False Create reversed model where all chains are traced in opposite direction to pdb_out
- reverse_only = False Only create reversed model where all chains are traced in opposite direction to pdb_out (requires input model).
- thoroughness = quick *medium thorough extra_thorough If quick is True, then run more quickly. Suitable for easy cases with protein only. If medium, standard run. If thorough, run quick and medium and take best result. If extra_thorough , then run with more cycles and more attempts to build. Suitable for challenging cases.
- quick_debug = False Run very quickly just to test routines
- pdb_in_only = None Only use pdb_in (do not build anything new)
- assign_sequence = None Run assign_sequence to match model to sequence. Default is True if sequence_from_map_before_fit_loops is False (use one or the other).
- rerun_with_avail_seq = True Rerun sequence alignment with parts of the sequence already used removed. With a large structure this can take a long time.
- split_using_alignment = False At end, split fragments based on alignment. Default is False
- rearrange_before_assign_sequence = None Run replace_side_chains with reassign_sequence before assign_sequence. Default False unless thorough.
- sequence_from_map_before_fit_loops = True Run sequence_from_map before fit_loops
- create_gap_length = None Create gaps of this length if segments are nearly overlapping. A gap of 2 is suitable. This is to help fit_loops. Not default as it does not seem to help.
- fit_loops_before_assign_sequence = True Run fit_loops before assign_sequence
- extend_with_resolve = None Run resolve model extension. Default False unless thorough
- min_percent_assigned_for_assign_sequence = 25 Skip assign_sequence if percentage sequence assigned is lower than min_percent_placed_for_assign_sequence
- segmentationvalue_outside_mask = None Value to assign to density outside masks in segment_and_split_mapmin_relative_helical_cc_to_keep = 0.90 For helical symmetry, keep copies within this range of max at endadd_neighbors = True Add neighboring regions around the NCS au. Turns off exclude_points_in_ncs_copies also.select_au_box = None Select box containing at least one representative region of the map. Also select just symmetry operators relevant to that box. Default is true if number of operators is at least n_ops_to_use_au_boxn_ops_to_use_au_box = None If number of operators is this big or more and select_au_box is None, set it to True.n_au_box = None Number of symmetry copies to try and get inside au_boxlower_bounds = None You can select a part of your map for analysis with lower_bounds and upper_bounds.upper_bounds = None You can select a part of your map for analysis with lower_bounds and upper_bounds.
- value_outside_mask = None Value to assign to density outside masks in segment_and_split_map
- min_relative_helical_cc_to_keep = 0.90 For helical symmetry, keep copies within this range of max at end
- add_neighbors = True Add neighboring regions around the NCS au. Turns off exclude_points_in_ncs_copies also.

- select_au_box = None Select box containing at least one representative region of the map. Also select just symmetry operators relevant to that box. Default is true if number of operators is at least n_ops_to_use_au_box
- n_ops_to_use_au_box = None If number of operators is this big or more and select_au_box is None, set it to True.
- n_au_box = None Number of symmetry copies to try and get inside au_box
- lower_bounds = None You can select a part of your map for analysis with lower_bounds and upper_bounds.
- upper_bounds = None You can select a part of your map for analysis with lower_bounds and upper_bounds.
- trace_and_buildfind_chains = True Try to create new chains in build_all_loopsextend_chains = True Try to extend chains in build_all_loopsconnect_chains = True Try to connect chains in build_all_loopsfind_helices_strands = True Find helices/strands before tracing chain if no starting model is suppliedtnb_refine_cycles = None Refine cycles in trace_and_build (1 for quick, otherwise 3)n_short_min = None N short min. Length of group of CA to optimize together (min)n_short_max = None N short max. Length of group of CA to optimize together (max)n_short_delta = None N short delta. Length of group of CA to optimize together (delta)allow_reverse = True If two chains are opposite direction but connected, reverse the one with lower score (usually shorter) and connect them In trace_and_build. Also consider both directions of fragments from model_file.length_variants_to_try = None You can try several lengths of fragments (number of CA). Not compatible with allow_reverse=False. Typical values are 1 or 5final_connect_chains = None Connect chains at very end after merging fragments. Uses full map. Default False if quick, True otherwisemean_density_ratio = None Guess of mean density at coordinates of atoms is mean density in map plus SD of density in map, corrected for solvent content, times mean_density_ratio
- find_chains = True Try to create new chains in build_all_loops
- extend_chains = True Try to extend chains in build_all_loops
- connect_chains = True Try to connect chains in build_all_loops
- find_helices_strands = True Find helices/strands before tracing chain if no starting model is supplied
- tnb_refine_cycles = None Refine cycles in trace_and_build (1 for quick, otherwise 3)
- n_short_min = None N short min. Length of group of CA to optimize together (min)
- n_short_max = None N short max. Length of group of CA to optimize together (max)
- n_short_delta = None N short delta. Length of group of CA to optimize together (delta)
- allow_reverse = True If two chains are opposite direction but connected, reverse the one with lower score (usually shorter) and connect them In trace_and_build. Also consider both directions of fragments from model_file.
- length_variants_to_try = None You can try several lengths of fragments (number of CA). Not compatible with allow_reverse=False. Typical values are 1 or 5
- final_connect_chains = None Connect chains at very end after merging fragments. Uses full map. Default False if quick, True otherwise
- mean_density_ratio = None Guess of mean density at coordinates of atoms is mean density in map plus SD of density in map, corrected for solvent content, times mean_density_ratio
- region_sharpeninginclude_original_map = False You can include the original map without Fourier filtering in local maps with this keyword. To use only the original map use

only_original_map=Trueonly_original_map = False You can include just the original map without Fourier filtering in local maps. Applies if the starting point is a map (not map coefficients).region_b_sharpen = None B-factor for sharpening of local maps. Normally set to None and several values are tested. (Positive is sharpen, negative is blur) Note: if resolution is specified map will be Fourier filtered to that resolution even if it is not sharpened.region_b_sharpen_low = -105 Lowest value of region_b_sharpen to use. Applies if region_b_sharpen_delta is set. Negative is blur, positive is sharpen.region_b_sharpen_high = 0 Ending value of region_b_sharpen. Applies if region_b_sharpen_delta is set Negative is blur, positive is sharpen.region_b_sharpen_delta = 15 Incremental value of region_b_sharpen. If set, region_b_sharpen will be applied with values from region_b_sharpen_low to region_b_sharpen_high in increments of region_b_sharpen_delta. Ignored if region_b_sharpen is set.

• include_original_map = False You can include the original map without Fourier filtering in local maps with this keyword. To use only the original map use only_original_map=True

• only_original_map = False You can include just the original map without Fourier filtering in local maps. Applies if the starting point is a map (not map coefficients).

• region_b_sharpen = None B-factor for sharpening of local maps. Normally set to None and several values are tested. (Positive is sharpen, negative is blur) Note: if resolution is specified map will be Fourier filtered to that resolution even if it is not sharpened.

• region_b_sharpen_low = -105 Lowest value of region_b_sharpen to use. Applies if region_b_sharpen_delta is set. Negative is blur, positive is sharpen.

• region_b_sharpen_high = 0 Ending value of region_b_sharpen. Applies if region_b_sharpen_delta is set Negative is blur, positive is sharpen.

• region_b_sharpen_delta = 15 Incremental value of region_b_sharpen. If set, region_b_sharpen will be applied with values from region_b_sharpen_low to region_b_sharpen_high in increments of region_b_sharpen_delta. Ignored if region_b_sharpen is set.

• resolve_model_buildingrun_resolve_model_building = None Run resolve model-building algorithm on maps. If None, set to True for RNA/DNA and False for protein.thorough_resolve_model_building = True Use thorough resolve model-buildingrun_helices_strands_only = None Run resolve helices-strands-only algorithms on maps. If None set to True for RNA and False for protein. This applies to segmented maps only if they exist. Compare with include_helices_strands_only which applies to the entire asymmetric unit

• run_resolve_model_building = None Run resolve model-building algorithm on maps. If None, set to True for RNA/DNA and False for protein.

• thorough_resolve_model_building = True Use thorough resolve model-building

• run_helices_strands_only = None Run resolve helices-strands-only algorithms on maps. If None set to True for RNA and False for protein. This applies to segmented maps only if they exist. Compare with include_helices_strands_only which applies to the entire asymmetric unit

• trace_chainrun_trace_chain = True Run trace-chain model-building algorithm on mapsrho_cut_min = 2.0 Minimum density (rho/sigma) at coordinates of potential CA atoms in trace_chain, after normalization for solvent fraction. For constant actual local rms in a map, the sigma (overall rms) of the map is proportional to the sqrt(1-solvent_fraction). Therefore rho_cut_min is adjusted by sqrt(0.5)/sqrt(1-solvent_fraction) to place it on a constant scale relative to a map with standard local rms.target_angle = 180 Target angle for CA-CA-CA (set to 180 to maximize chain length)rho_cut_min_low = 1. Starting value of rho_cut_min. Applies if rho_cut_min_delta is set (rho_cut_min is ignored in this case)rho_cut_min_high = 5 Ending value of rho_cut_min. Applies if rho_cut_min_delta is setrho_cut_min_delta = None Incremental value of rho_cut_min. If set, rho_cut_min will be ignoredrat_pair_min = 0.5 Minimum ratio of density at midpoint between points to trace chain between themdist_ca_tol_max = 0.80 Maximum tolerance for CA-CA distances. Normally 0.8 A for thorough run and 1.3 A for quickdist_ca_tol_start = 0.10 Minimum tolerance for CA-CA distances. rad_sep_trace = 0.75 Dummy atom separation in trace_chain Usual 0.6 A for thorough run and 0.75 for quick Increased automatically if resolution is greater than 3 A Value of rad_mask_trace in resolve will be rad_sep_trace*2n_atoms_total_scale = 3.33 Ratio of estimated atoms in au to standard estimatetarget_p_ratio = 3 Target ratio of atoms to peaks in trace_chaintarget_n_ratio = 3 Target ratio of nonamers to peaks in trace_chainmax_triple_ratio = 10 Maximum ratio of triples to pairs in trace_chainmax_pent_ratio = 10 Maximum ratio of pentamers to pairs in trace_chainatom_target_ratio = 0.45 Target ratio of CA to look for to expected atoms in structure Standard is 0.45, quick is 0.35build_both_directions = False Build chains both directions and choose after mergingmin_end_correl = 0.5 Minimum correlation of direction estimated from two ends to use end matching as criterion for keeping a chainadd_side_chains = True Add in side chains at trace_chain step

• run_trace_chain = True Run trace-chain model-building algorithm on maps

• rho_cut_min = 2.0 Minimum density (rho/sigma) at coordinates of potential CA atoms in trace_chain, after normalization for solvent fraction. For constant actual local rms in a map, the sigma (overall rms) of the map is proportional to the sqrt(1-solvent_fraction). Therefore rho_cut_min is adjusted by sqrt(0.5)/sqrt(1-solvent_fraction) to place it on a constant scale relative to a map with standard local rms.

• target_angle = 180 Target angle for CA-CA-CA (set to 180 to maximize chain length)

• rho_cut_min_low = 1. Starting value of rho_cut_min. Applies if rho_cut_min_delta is set (rho_cut_min is ignored in this case)

• rho_cut_min_high = 5 Ending value of rho_cut_min. Applies if rho_cut_min_delta is set

• rho_cut_min_delta = None Incremental value of rho_cut_min. If set, rho_cut_min will be ignored

• rat_pair_min = 0.5 Minimum ratio of density at midpoint between points to trace chain between them

• dist_ca_tol_max = 0.80 Maximum tolerance for CA-CA distances. Normally 0.8 A for thorough run and 1.3 A for quick

• dist_ca_tol_start = 0.10 Minimum tolerance for CA-CA distances.

• rad_sep_trace = 0.75 Dummy atom separation in trace_chain Usual 0.6 A for thorough run and 0.75 for quick Increased automatically if resolution is greater than 3 A Value of rad_mask_trace in resolve will be rad_sep_trace*2

• n_atoms_total_scale = 3.33 Ratio of estimated atoms in au to standard estimate

• target_p_ratio = 3 Target ratio of atoms to peaks in trace_chain

• target_n_ratio = 3 Target ratio of nonamers to peaks in trace_chain

• max_triple_ratio = 10 Maximum ratio of triples to pairs in trace_chain

• max_pent_ratio = 10 Maximum ratio of pentamers to pairs in trace_chain

• atom_target_ratio = 0.45 Target ratio of CA to look for to expected atoms in structure Standard is 0.45, quick is 0.35

• build_both_directions = False Build chains both directions and choose after merging

• min_end_correl = 0.5 Minimum correlation of direction estimated from two ends to use end matching as criterion for keeping a chain

• add_side_chains = True Add in side chains at trace_chain step

• crossingcombine_models = True Merge models from trace_chain to form single modelstandard_merge = None Merge by reading all chains and running resolve merging.extend_in_merge = False Extend fragments of first and second model during merging of modelsuse_cc_in_combine_extend = None You can choose to use the correlation of density rather than density at atomic positions to score models in the merge_second_model or merge_both_models step. This may be useful at lower resolution (> 3 A)merge_remainder = True Merge remainder in merge_by_segment_correlationmerge_by_segment_correlation = True Merge using segment correlation if merge_with_combine_models is selected. Can be used along with merge_remainder=True/False and standard_merge=True/Falsemerge_second_model = True In merging models, cut up second model to fill gaps in first Normally used internally.remove_poor_fragments = True After merging models with RNA and protein, remove poor protein chains with remove_poor_fragmentsremove_poor_fragments_minimum_rna = 0.20 Minimum RNA:protein residues in sequence file to run remove_poor_fragmentswang_radius = 5 Smoothing radius for solvent identificationcc_min = 0.40 Minimum map-model correlation to keep a segment after refinementassignment_weight = 0.20 If set, increase score of segments assigned to sequence.cc_min_rna = 0.25 Minimum map-model correlation (RNA/DNA building)overlap_tolerance = None Minimum distance between N/P/C1prime in different chains. Used to reject symmetry-related chains. Default is 3 A for RNA/DNA and 2 A for proteinncs_clash_threshold = 2. Threshold for considering two atoms too close after applying symmetryremove_outside_cell = None Remove chains with atoms outside cell

- combine_models = True Merge models from trace_chain to form single model
- standard_merge = None Merge by reading all chains and running resolve merging.
- extend_in_merge = False Extend fragments of first and second model during merging of models
- use_cc_in_combine_extend = None You can choose to use the correlation of density rather than density at atomic positions to score models in the merge_second_model or merge_both_models step. This may be useful at lower resolution (> 3 A)
- merge_remainder = True Merge remainder in merge_by_segment_correlation
- merge_by_segment_correlation = True Merge using segment correlation if merge_with_combine_models is selected. Can be used along with merge_remainder=True/False and standard_merge=True/False
- merge_second_model = True In merging models, cut up second model to fill gaps in first Normally used internally.
- remove_poor_fragments = True After merging models with RNA and protein, remove poor protein chains with remove_poor_fragments
- remove_poor_fragments_minimum_rna = 0.20 Minimum RNA:protein residues in sequence file to run remove_poor_fragments
- wang_radius = 5 Smoothing radius for solvent identification
- cc_min = 0.40 Minimum map-model correlation to keep a segment after refinement
- assignment_weight = 0.20 If set, increase score of segments assigned to sequence.
- cc_min_rna = 0.25 Minimum map-model correlation (RNA/DNA building)
- overlap_tolerance = None Minimum distance between N/P/C1prime in different chains. Used to reject symmetry-related chains. Default is 3 A for RNA/DNA and 2 A for protein
- ncs_clash_threshold = 2. Threshold for considering two atoms too close after applying symmetry
- remove_outside_cell = None Remove chains with atoms outside cell
- refinementrefine = True Refine fragments after each building cyclerefine_adp = False Refine individual atomic displacement factors (adp)number_of_sa_models = None Number of refinement sa_models.

Default 2 (20 if thorough)number_of_trials = None Number of refinement trials. Default 2 (20 if thorough)number_of_macro_cycles = None Number of refinement macro_cycles. Default 2 (5 if thorough)number_of_build_cycles = None Number of build cycles. Default is 2 (2 if thorough)

• refine = True Refine fragments after each building cycle

• refine_adp = False Refine individual atomic displacement factors (adp)

• number_of_sa_models = None Number of refinement sa_models. Default 2 (20 if thorough)

• number_of_trials = None Number of refinement trials. Default 2 (20 if thorough)

• number_of_macro_cycles = None Number of refinement macro_cycles. Default 2 (5 if thorough)

• number_of_build_cycles = None Number of build cycles. Default is 2 (2 if thorough)

• rebuildingrebuild = True Rebuild model. Only for protein.rebuild_cycles = None Number of rebuilding cycles. Set to zero to skip rebuilding. Default is 1 (2 if thorough)trace_loops = True Rebuild model by iterative loop tracing. Only for protein.trace_loops_target_time = 60 Target time for completing a single trace_loops job. You can decrease it to speed it up (and miss some rebuilding) or increase it and possibly get some more successful rebuilding.loop_lib = False Rebuild model using loop library. Only for protein.rebuild_length = 8 Length of segments to be rebuiltrebuild_length_worst = 6 Length of segments to be rebuilt in rebuilding worst sectionsoffset_length = 3 Offset of segments to be rebuiltrebuild_length_loop_lib = 3 Length of segments to be rebuilt with loop library (max=3)offset_length_loop_lib = 1 Offset of segments to be rebuilt with loop librarystart_res = None Start residue for rebuilding. Normally use None.end_res = None Ending residue for rebuilding. Normally use None.worst_rebuild_percent = 10 If specified, this percentage of residues will be rebuilt. Otherwise, the entire model will be rebuilttarget_insert = None Length of insert to put from start_res to end_res Normally leave blank. Can be used to force the addition or subtraction of residues.target_insert_offset = None If non-zero, try to rebuild with this number of extra residues in each rebuilt segment. Normally leave blank. Can be used to force the addition or subtraction of residues.

• rebuild = True Rebuild model. Only for protein.

• rebuild_cycles = None Number of rebuilding cycles. Set to zero to skip rebuilding. Default is 1 (2 if thorough)

• trace_loops = True Rebuild model by iterative loop tracing. Only for protein.

• trace_loops_target_time = 60 Target time for completing a single trace_loops job. You can decrease it to speed it up (and miss some rebuilding) or increase it and possibly get some more successful rebuilding.

• loop_lib = False Rebuild model using loop library. Only for protein.

• rebuild_length = 8 Length of segments to be rebuilt

• rebuild_length_worst = 6 Length of segments to be rebuilt in rebuilding worst sections

• offset_length = 3 Offset of segments to be rebuilt

• rebuild_length_loop_lib = 3 Length of segments to be rebuilt with loop library (max=3)

• offset_length_loop_lib = 1 Offset of segments to be rebuilt with loop library

• start_res = None Start residue for rebuilding. Normally use None.

• end_res = None Ending residue for rebuilding. Normally use None.

• worst_rebuild_percent = 10 If specified, this percentage of residues will be rebuilt. Otherwise, the entire model will be rebuilt

• target_insert = None Length of insert to put from start_res to end_res Normally leave blank. Can be used to force the addition or subtraction of residues.

• target_insert_offset = None If non-zero, try to rebuild with this number of extra residues in each rebuilt segment. Normally leave blank. Can be used to force the addition or subtraction of residues.

• iterative_ss_refineiterative_ss = True Run iterative assignment of secondary structure and refinement with secondary structure restraintsss_refine_ncycle = None Overall cycles of iterative secondary_structure identification and refinement. Default is 10refine_cycles = None Cycles of refinement within iterative secondary structure and refinement. Default 2 (5 if thorough=True)parallel_runs = None Number of parallel runs of iterative secondary structure assignment and refinement. Best model from each group is taken each cycle. Default 1 (4 if thorough)combine_annotations = False Combine existing secondary_structure annotations each cycle with new annotationsreplace_side_chains = True Replace side chains during iterative optimizationregularize = True regularize secondary structure during iterative optimization

• iterative_ss = True Run iterative assignment of secondary structure and refinement with secondary structure restraints

• ss_refine_ncycle = None Overall cycles of iterative secondary_structure identification and refinement. Default is 10

• refine_cycles = None Cycles of refinement within iterative secondary structure and refinement. Default 2 (5 if thorough=True)

• parallel_runs = None Number of parallel runs of iterative secondary structure assignment and refinement. Best model from each group is taken each cycle. Default 1 (4 if thorough)

• combine_annotations = False Combine existing secondary_structure annotations each cycle with new annotations

• replace_side_chains = True Replace side chains during iterative optimization

• regularize = True regularize secondary structure during iterative optimization

• mr_rosettarun_mr_rosetta = False Run mr_rosetta to rebuild modelrosetta_models = 20 Number of rosetta models to buildrelax_models = 5 Number of relaxed rosetta models to buildrosetta_fixed_seed = None Fixed seed for Rosetta. Only use for regression tests.

• run_mr_rosetta = False Run mr_rosetta to rebuild model

• rosetta_models = 20 Number of rosetta models to build

• relax_models = 5 Number of relaxed rosetta models to build

• rosetta_fixed_seed = None Fixed seed for Rosetta. Only use for regression tests.

• controlmultiprocessing = *multiprocessing sge lsf pbs condor pbspro slurm Choices are multiprocessing (single machine) or queuing systemsqueue_run_command = None run command for queue jobs. For example qsub.nproc = 1 Number of processors to usemax_dirs = 1000 Maximum number of directories (map_to_model_xxxx)build_only = False Only build model (then stop)build_and_refine_only = False Only build model and refine (then stop)skip_combine_models = None Skip combine_models stepskip_final_refinement = None Skip final refinementstop_after_segment_and_split_map = False Only carry out through segment_and_split_map step (then stop)split_up_job = False Generate scripts and optionally run them (if run_split_jobs=True) to run very small parts of your overall job individually. This is used to speed up the overall process for big structures such as ribosomes. You can run this after you have set up the job and segmented_maps/segment_and_split_map_info.pkl is present. After these jobs are done, you run map_to_model once more with carry_on=True and it will combine everything and finish up.run_split_jobs = False Run the split jobs from split_up_job=True directly. Default is to write scripts to the split_jobs_directory. .short_caption = Run split jobssplit_jobs_commands_directory = commands Place where scripts to run split jobs will gosplit_jobs_logs_directory = logs Place where logs from split jobs will gosplit_jobs_source_command = None You can set the source command for split jobsremove_pdb_block = None Clear out any .pdb_block files written by previous runs of map_to_model. Normally only set to False when you are running jobs with split_up_job. Default is True unless do_only_one_thing is set.do_only_one_thing = None You can use this to not combine models. Just build and save. This is part of carry_on=True and is a way to build up pieces of your model.stop_after_combine_chain_types = False Stop after combining models from different chain typesmodel_start = None You can use this to specify skipping model-building of model numbers below this (used to start in the middle)model_end = None You can use this to specify skipping model-building of model numbers above his (used to end in the middle)resolve_size = None Size of resolve to use. coarse_grid = None Use a coarse grid in RESOLVE (saves on memory)random_seed = 77151 Random seed (allows duplicating calculations or getting different results each time.)verbose = False Verbose outputcomparison_model = None Comparison modelcarry_on = False You can try to carry on from where you left off. The info_pickle_file (default or whatever you specify) will be read and any files listed in partial_model_type that are present will be used. If they are not present they will be generated. With this keyword you can progressively build up your model from pieces. You can also run many jobs in parallel building just part of the chain using the keywords do_only_one_thing input_map_id_start input_map_id_end model_start model_end and specifying chain_type and turning on and off include_helices_strands_only include_phase_and_build and build_in_regions. After they all run, you can run from the beginning again with carry_on=True and all the intermediate files will be combined to yield a full model.memory_check = None Map-to-model checks to make sure you have enough memory on your machine to run. You can disable this by setting this keyword to False. The estimates are approximate so it is possible your job could run even if the check fails. Note the check does not take any other uses of the memory on your machine into account.em_side_density = None Use EM side chain density in assign_sequence (and trace_and_build if em_side_density_in_tnb is True). Default is True if scattering_table is electron.em_side_density_in_tnb = False EM side chain density in trace_and_build (if em_side_density is True)ignore_map_limitations = None Ignore limitations such as map cannot be sharpened

• multiprocessing = *multiprocessing sge lsf pbs condor pbspro slurm Choices are multiprocessing (single machine) or queuing systems

• queue_run_command = None run command for queue jobs. For example qsub.

• nproc = 1 Number of processors to use

• max_dirs = 1000 Maximum number of directories (map_to_model_xxxx)

• build_only = False Only build model (then stop)

• build_and_refine_only = False Only build model and refine (then stop)

• skip_combine_models = None Skip combine_models step

• skip_final_refinement = None Skip final refinement

• stop_after_segment_and_split_map = False Only carry out through segment_and_split_map step (then stop)

• split_up_job = False Generate scripts and optionally run them (if run_split_jobs=True) to run very small parts of your overall job individually. This is used to speed up the overall process for big structures such as ribosomes. You can run this after you have set up the job and segmented_maps/segment_and_split_map_info.pkl is present. After these jobs are done, you run map_to_model once more with carry_on=True and it will combine everything and finish up.

- run_split_jobs = False Run the split jobs from split_up_job=True directly. Default is to write scripts to the split_jobs_directory. .short_caption = Run split jobs
- split_jobs_commands_directory = commands Place where scripts to run split jobs will go
- split_jobs_logs_directory = logs Place where logs from split jobs will go
- split_jobs_source_command = None You can set the source command for split jobs
- remove_pdb_block = None Clear out any .pdb_block files written by previous runs of map_to_model. Normally only set to False when you are running jobs with split_up_job. Default is True unless do_only_one_thing is set.
- do_only_one_thing = None You can use this to not combine models. Just build and save. This is part of carry_on=True and is a way to build up pieces of your model.
- stop_after_combine_chain_types = False Stop after combining models from different chain types
- model_start = None You can use this to specify skipping model-building of model numbers below this (used to start in the middle)
- model_end = None You can use this to specify skipping model-building of model numbers above his (used to end in the middle)
- resolve_size = None Size of resolve to use.
- coarse_grid = None Use a coarse grid in RESOLVE (saves on memory)
- random_seed = 77151 Random seed (allows duplicating calculations or getting different results each time.)
- verbose = False Verbose output
- comparison_model = None Comparison model
- carry_on = False You can try to carry on from where you left off. The info_pickle_file (default or whatever you specify) will be read and any files listed in partial_model_type that are present will be used. If they are not present they will be generated. With this keyword you can progressively build up your model from pieces. You can also run many jobs in parallel building just part of the chain using the keywords do_only_one_thing input_map_id_start input_map_id_end model_start model_end and specifying chain_type and turning on and off include_helices_strands_only include_phase_and_build and build_in_regions. After they all run, you can run from the beginning again with carry_on=True and all the intermediate files will be combined to yield a full model.
- memory_check = None Map-to-model checks to make sure you have enough memory on your machine to run. You can disable this by setting this keyword to False. The estimates are approximate so it is possible your job could run even if the check fails. Note the check does not take any other uses of the memory on your machine into account.
- em_side_density = None Use EM side chain density in assign_sequence (and trace_and_build if em_side_density_in_tnb is True). Default is True if scattering_table is electron.
- em_side_density_in_tnb = False EM side chain density in trace_and_build (if em_side_density is True)
- ignore_map_limitations = None Ignore limitations such as map cannot be sharpened
- guiGUI-specific parameter required for output directoryoutput_dir = None
- output_dir = None

Evaluating map quality with mtriage

mtriage.html

Author(s)

- mtriage: Pavel Afonine, Billy Poon (GUI)

Purpose

The routine mtriage will evaluate the resolution in a map, compare half-maps, and provide some basic statistics about the map

Usage

How mtriage works:

Possible inputs to mtriage are:

one full map; one full map and two half-maps; one full map and a file with atomic model; one full map, a file with atomic model and two half-maps.

- one full map;
- one full map and two half-maps;
- one full map and a file with atomic model;
- one full map, a file with atomic model and two half-maps.

If only one map is provided, mtriage will calculte some basic map stitistics (min, max, mean, standard deviation), show histogram of map values and calculate resolution estimate d99.

If two half-maps are given in addition to the full map, then FSC curve between the two half-maps will be calculated as well as resolution estimate d_FSC(0.143).

If an atomic model is provided then two sets of statistics will be calculated: using original and masked maps. The soft mask is calculated using the atomic model as described in citation below. Also, some model-based resolution estimates will be calculated, such as d_model and d_FSC_model as well as map-model FSC curve.

Model file can be in PDB or mmCIF formats. Map file is expected to be CCP4, mrc or other related format.

Guide to resolution estimates:

d_FSC - highest resolution at which the experimental data are confident. Obtained from FSC curve calculated using two half-maps and taken at FSC=0.143. d99 - resolution cutoff beyond which Fourier map coefficients are negligeably small. Calculated form the map. d_model - resolution cutoff at which the model map is the most similar to the target (experimental) map. Requires map and model. For d_model to be meaningful, model is expected to fit the map as good as possible. d_FSC_model - resolution cutoff up to which the model and map Fourier coefficients are similar. Refer to Table 2 in article below for more details about interpretation of various resolution estimates.

- d_FSC - highest resolution at which the experimental data are confident. Obtained from FSC curve calculated using two half-maps and taken at FSC=0.143.

- d99 - resolution cutoff beyond which Fourier map coefficients are negligeably small. Calculated form the map.
- d_model - resolution cutoff at which the model map is the most similar to the target (experimental) map. Requires map and model. For d_model to be meaningful, model is expected to fit the map as good as possible.
- d_FSC_model - resolution cutoff up to which the model and map Fourier coefficients are similar.

Refer to Table 2 in article below for more details about interpretation of various resolution estimates.

Examples

Standard run of mtriage:

Running mtriage is easy. From the command-line you can type:

```
phenix.mtriage map.mrc
phenix.mtriage map.mrc half_map_1.map half_map_2.map
phenix.mtriage map.mrc model.pdb
phenix.mtriage map.mrc model.pdb half_map_1.map half_map_2.map
```

Possible Problems

All inputs maps are expected to be on the same grid, the program will stop otherwise. Box information (aka unit cell CRYST1 in PDB files, for example) must match across all inputs.

Specific limitations and problems:

Reference

New tools for the analysis and validation of cryo-EM maps and atomic models. P.V. Afonine, B.P. Klaholz, N.W. Moriarty, B.K. Poon, O.V. Sobolev, T.C. Terwilliger, P.D. Adams, and A. Urzhumtsev. Acta Cryst. D74, 814-840 (2018).

- New tools for the analysis and validation of cryo-EM maps and atomic models. P.V. Afonine, B.P. Klaholz, N.W. Moriarty, B.K. Poon, O.V. Sobolev, T.C. Terwilliger, P.D. Adams, and A. Urzhumtsev. Acta Cryst. D74, 814-840 (2018).

Additional information

List of all available keywords

scattering_table = wk1995 it1992 n_gaussian neutron *electron Scattering table (X-ray, neutron or electron)resolution = None Map resolution (d_FSC)mask_maps = None Mask out region outside moleculeauto_mask_if_no_model = True If mask_maps is set and no model is present, mask based on densityauto_mask_if_model = False If mask_maps is set and model is present, mask based on densityradius_smooth = None Mask smoothing radiusradius_smooth_ratio = 2 If mask smoothing radius is not specified it will be radius_smooth_ratio times the resolutionn_bins = None Number of bins for FSC curves. Suggested number is 5000. Alternative is default (None) which gives bins of width 100.nproc = 1 Number of processors to useshow_time = False Show individual run times for each stepinclude_curves = True Keep FSC curvesinclude_mask = True Keep maskwrapping = None You can ignore the wrapping information from map files and set itfsc_model_plot_file_name_prefix = fsc_model fsc_half_maps_file_name_prefix =

fsc_half_maps write_mask_file = True mask_file_name = mask.ccp4 soft_mask = True Use soft mask when running box_map_before_analysis. Reduces artifacts from boxing.box_map_before_analysis = False You can box the map (cut out box a little bigger than supplied model) before any analysis if you want. This will make the analysis faster, but it can introduce masking artifacts. Artifacts reduced with soft_mask = True. Requires a model. If model is similar in size or bigger than the map this will extend the map around the model using data from the opposite part of the map.ignore_symmetry_conflicts = False You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with box_map for example.job_title = None Job title in PHENIX GUI, not used on command linemap_modelfull_map = None Input full map filehalf_map = None Input half map filesmodel = None Input model filecomputemap_counts = True Compute map countsfsc_curve_model = True Compute model-map CC in reciprocal space: FSC(model map, data map)d_fsc_half_map_05 = False Compute d_half_map (FSC=0.5)d_fsc_model_05 = True Compute d_fsc_model (FSC=0.5)d_fsc_model_0 = True Compute d_fsc_model (FSC=0)d_fsc_model_0143 = True Compute d_fsc_model (FSC=0.143)d_model = True Resolution estimate using model and mapd_model_b0 = True Resolution estimate using model and map, assumin all atoms B=0d99 = True Resolution estimate d99outputwrite_fsc_as_xml = True Exports FSC curve (half maps) in XML format for EMDBguiGUI-specific parameter required for output directoryoutput_dir = None

- scattering_table = wk1995 it1992 n_gaussian neutron *electron Scattering table (X-ray, neutron or electron)
- resolution = None Map resolution (d_FSC)
- mask_maps = None Mask out region outside molecule
- auto_mask_if_no_model = True If mask_maps is set and no model is present, mask based on density
- auto_mask_if_model = False If mask_maps is set and model is present, mask based on density
- radius_smooth = None Mask smoothing radius
- radius_smooth_ratio = 2 If mask smoothing radius is not specified it will be radius_smooth_ratio times the resolution
- n_bins = None Number of bins for FSC curves. Suggested number is 5000. Alternative is default (None) which gives bins of width 100.
- nproc = 1 Number of processors to use
- show_time = False Show individual run times for each step
- include_curves = True Keep FSC curves
- include_mask = True Keep mask
- wrapping = None You can ignore the wrapping information from map files and set it
- fsc_model_plot_file_name_prefix = fsc_model
- fsc_half_maps_file_name_prefix = fsc_half_maps
- write_mask_file = True
- mask_file_name = mask.ccp4
- soft_mask = True Use soft mask when running box_map_before_analysis. Reduces artifacts from boxing.
- box_map_before_analysis = False You can box the map (cut out box a little bigger than supplied model) before any analysis if you want. This will make the analysis faster, but it can introduce masking artifacts. Artifacts reduced with soft_mask = True. Requires a model. If model is similar in size or bigger than the map this will extend the map around the model using data from the opposite part of the map.

- ignore_symmetry_conflicts = False You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with box_map for example.
- job_title = None Job title in PHENIX GUI, not used on command line
- map_modelfull_map = None Input full map filehalf_map = None Input half map filesmodel = None Input model file
- full_map = None Input full map file
- half_map = None Input half map files
- model = None Input model file
- computemap_counts = True Compute map countsfsc_curve_model = True Compute model-map CC in reciprocal space: FSC(model map, data map)d_fsc_half_map_05 = False Compute d_half_map (FSC=0.5)d_fsc_model_05 = True Compute d_fsc_model (FSC=0.5)d_fsc_model_0 = True Compute d_fsc_model (FSC=0)d_fsc_model_0143 = True Compute d_fsc_model (FSC=0.143)d_model = True Resolution estimate using model and mapd_model_b0 = True Resolution estimate using model and map, assumin all atoms B=0d99 = True Resolution estimate d99
- map_counts = True Compute map counts
- fsc_curve_model = True Compute model-map CC in reciprocal space: FSC(model map, data map)
- d_fsc_half_map_05 = False Compute d_half_map (FSC=0.5)
- d_fsc_model_05 = True Compute d_fsc_model (FSC=0.5)
- d_fsc_model_0 = True Compute d_fsc_model (FSC=0)
- d_fsc_model_0143 = True Compute d_fsc_model (FSC=0.143)
- d_model = True Resolution estimate using model and map
- d_model_b0 = True Resolution estimate using model and map, assumin all atoms B=0
- d99 = True Resolution estimate d99
- outputwrite_fsc_as_xml = True Exports FSC curve (half maps) in XML format for EMDB
- write_fsc_as_xml = True Exports FSC curve (half maps) in XML format for EMDB
- guiGUI-specific parameter required for output directoryoutput_dir = None
- output_dir = None

phenix.real_space_refine: a tool for refinement a model against a map

real_space_refine.html

Contents

- Description
- Contact author
- Features
- Video Tutorial
- GUI
- Usage examples
- Notes
- Half maps vs full map
- References
- Full set of parameters

Description

The program refines a model into a map. The map can be derived from X-ray or neutron crystallography, or Electron Microscopy, and its quality can vary from good to poor. The program aims at obtaining a model that fits the map as well as possible while possessing a meaningful geometry (no validation outliers, such as Ramachandran plot or rotamer outliers).

Contact author

For questions, bug reports, feature requests: Pavel Afonine (PAfonine@lbl.gov)

IMPORTANT: any bug or problem report should be sent after trying the latest nightly build of Phenix:

```
https://www.phenix-online.org/download/nightly_builds.cgi
```

Bug or problem reports should include at least 1) all input files used to run, 2) list of parameters that were customized, 3) error message. Please use file sharing tools such as Dropbox to send large files.

Features

- Graphical User Interface (GUI) is available

Graphical User Interface (GUI) is available

- Fast gradient-driven minimization of combined map and restraints target: T = Tmap + weight * Trestraints

Fast gradient-driven minimization of combined map and restraints target:

```
T = Tmap + weight * Trestraints
```

- Local grid search based fit to fix rotamer outliers or poor map-to-model fit

Local grid search based fit to fix rotamer outliers or poor map-to-model fit

- Morphing (map guided rigid body shifts of small continuous model fragments)

Morphing (map guided rigid body shifts of small continuous model fragments)

- Simulated Annealing refinement

Simulated Annealing refinement

- Rigid-body refinement Rigid groups need to be defined

Rigid-body refinement

Rigid groups need to be defined

- Rigid groups need to be defined
- Restraints: Standard: bond, angle, planarity, chirality, dihedral, nonbonded repulsion. Extra restraints: Ramachandran plot (two options: "oldfield" and "emsley"), C-beta deviations, rotamer, secondary structure, reference model, starting position

Restraints:

Standard: bond, angle, planarity, chirality, dihedral, nonbonded repulsion. Extra restraints: Ramachandran plot (two options: "oldfield" and "emsley"), C-beta deviations, rotamer, secondary structure, reference model, starting position

- Standard: bond, angle, planarity, chirality, dihedral, nonbonded repulsion.
- Extra restraints: Ramachandran plot (two options: "oldfield" and "emsley"), C-beta deviations, rotamer, secondary structure, reference model, starting position
- Fast and fully automated restraints/map weight optimization is always performed

Fast and fully automated restraints/map weight optimization is always performed

- Support for maps having non-zero origin

Support for maps having non-zero origin

- NCS (non-crystallographic symmetry) constraints only: NCS groups are determined automatically or can be provided manually NCS operators are refined

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- Input map format: CCP4 map file or MTZ file with Fourier map coefficients In case of CCP4 map resolution must be specified using "resolution" keyword

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In case of CCP4 map resolution must be specified using "resolution" keyword

- In case of CCP4 map resolution must be specified using "resolution" keyword
- ADP (B-factors) refinement against the map Enabled by default Currently done using reciprocal space Performed at the last macro-cycle only Runtime can benefit from multiple CPU

ADP (B-factors) refinement against the map

Enabled by default Currently done using reciprocal space Performed at the last macro-cycle only Runtime can benefit from multiple CPU

- Enabled by default
- Currently done using reciprocal space
- Performed at the last macro-cycle only
- Runtime can benefit from multiple CPU
- Custom restraints for selected atoms

Custom restraints for selected atoms

- Non-standard ligand support (ligand CIF needs to be provided)

Non-standard ligand support (ligand CIF needs to be provided)

- Input and Output model can be in PDB or mmCIF formats

Input and Output model can be in PDB or mmCIF formats

Video Tutorial

The tutorial video is available on the Phenix YouTube channel and covers the following topics:

- basic overview
- how to run phenix.real_space_refine via the GUI
- example of refinement with default parameters

GUI

Graphical User Interface is available.

Usage examples

Running with default settings: phenix.real_space_refine model.pdb map.ccp4 resolution=4.2 phenix.real_space_refine model.pdb map_coefficients.mtz This will do 5 macro-cycles of global real-space refinement with rotamer, Ramachandran plot and C-beta deviations restraints enabled. If NCS is present it will be used as constraints with NCS groups found by the program automatically. Note: if map is used then resolution needs to be provided. Rotamer, c-beta and Ramachandran plot restraints as well as using NCS constraints are enabled by default. To disable these restraints: phenix.real_space_refine model.pdb map.mtz ncs_constraints=False \ rotamer_restraints=False ramachandran_restraints=False \ c_beta_restraints=False Request output model to have bond and angle rmsd from ideal not greater than certain values: phenix.real_space_refine model.pdb map.mtz target_bonds_rmsd=0.01 \ target_angles_rmsd=1.0 Specify Fourier map coefficients to use: phenix.real_space_refine model.pdb map.mtz label='2FOFCWT,PH2FOFCWT' Run refinement using global minimization (default), local rotamer fitting, morphing and simulated annealing: phenix.real_space_refine model.pdb map.mtz \ run=minimization_global+local_grid_search+morphing+simulated_annealing If PDB file contains unknown to Phenix ligand a ligand CIF file needs to be provided. Ligand CIF file can be obtained using one of available tools in Phenix (see documentation for more details). Once CIF file is available it can be used as following: phenix.real_space_refine model.pdb map.mtz ligands.cif Group ADP (B-factor) refinement. Currently only one option available: restrained group ADP refinement with one isotropic B-factor per residue. This is enabled by default. The command below will do B factor refinement only: phenix.real_space_refine model.pdb map.mtz run=adp To make use of multiple CPU for B factor refinement use nproc=: phenix.real_space_refine model.pdb map.ccp4 resolution=3.2 nproc=12 Note: multiple CPUs will only be used for B-factor refinement (which is done by default). Running rigid-body refinement. Rigid body refinement can be run alone or in combination with other refinement strategies. This will run rigid-body refinement only: phenix.real_space_refine model.pdb map.mtz run=rigid_body rigid.eff where rigid.eff contains definitions of rigid body groups: refinement.rigid_body { group = chain A group = chain B or chain C group = chain Z } This will run rigid-body and individual coordinate refinement only: phenix.real_space_refine model.pdb map.mtz run=rigid_body+minimization_global In case NCS is present the program will attempt to determine NCS groups automatically and use to define NCS constraints. Automatically found definitions for NCS groups will be printed to the log. It is possible to provide NCS group definitions manually: phenix.real_space_refine model.pdb map.mtz ncs.eff where ncs.eff contains definitions of NCS groups: pdb_interpretation { ncs_group { reference = chain A selection = chain B } ncs_group { reference = chain C or chain D selection = chain X selection = chain Y selection = chain Z } } Note that user-supplied NCS groups will be filtered at all times. More on this here. NCS group definitions can be obtained using GUI or command line tool: phenix.simple_ncs_from_pdb model.pdb NCS operators are refined if NCS constraints are used. To disable refinement of NCS operators use refine_ncs_operators=False. Use if secondary structure (SS) restraints is controlled by secondary_structure.enabled keyword: phenix.real_space_refine model.pdb map.mtz \ secondary_structure.enabled=True If HELIX/SHEET records are present in PDB file they will be used to construct restraints. Otherwise one of Phenix tools will be run internally to derive secondary structure information from the input model. DNA/RNA specific restraints will be generated internally. Not all provided SS will be used by default: some H-bonds that exceed a certain threshold wi...

- Running with default settings: phenix.real_space_refine model.pdb map.ccp4 resolution=4.2 phenix.real_space_refine model.pdb map_coefficients.mtz This will do 5 macro-cycles of global real-space refinement with rotamer, Ramachandran plot and C-beta deviations restraints enabled. If NCS is present it will be used as constraints with NCS groups found by the program automatically. Note: if map is used then resolution needs to be provided.

Running with default settings:

```
phenix.real_space_refine model.pdb map.ccp4 resolution=4.2

phenix.real_space_refine model.pdb map_coefficients.mtz
```

This will do 5 macro-cycles of global real-space refinement with rotamer, Ramachandran plot and C-beta deviations restraints enabled. If NCS is present it will be used as constraints with NCS groups found by the program automatically. Note: if map is used then resolution needs to be provided.

- Rotamer, c-beta and Ramachandran plot restraints as well as using NCS constraints are enabled by default. To disable these restraints: phenix.real_space_refine model.pdb map.mtz ncs_constraints=False \ rotamer_restraints=False ramachandran_restraints=False \ c_beta_restraints=False

Rotamer, c-beta and Ramachandran plot restraints as well as using NCS constraints are enabled by default. To disable these restraints:

```
phenix.real_space_refine model.pdb map.mtz ncs_constraints=False \
rotamer_restraints=False ramachandran_restraints=False \
c_beta_restraints=False
```

- Request output model to have bond and angle rmsd from ideal not greater than certain values: phenix.real_space_refine model.pdb map.mtz target_bonds_rmsd=0.01 \ target_angles_rmsd=1.0

Request output model to have bond and angle rmsd from ideal not greater than certain values:

```
phenix.real_space_refine model.pdb map.mtz target_bonds_rmsd=0.01 \
target_angles_rmsd=1.0
```

- Specify Fourier map coefficients to use: phenix.real_space_refine model.pdb map.mtz label='2FOFCWT,PH2FOFCWT'

Specify Fourier map coefficients to use:

```
phenix.real_space_refine model.pdb map.mtz label='2FOFCWT,PH2FOFCWT'
```

- Run refinement using global minimization (default), local rotamer fitting, morphing and simulated annealing: phenix.real_space_refine model.pdb map.mtz \ run=minimization_global+local_grid_search+morphing+simulated_annealing

Run refinement using global minimization (default), local rotamer fitting, morphing and simulated annealing:

```
phenix.real_space_refine model.pdb map.mtz \
run=minimization_global+local_grid_search+morphing+simulated_annealing
```

- If PDB file contains unknown to Phenix ligand a ligand CIF file needs to be provided. Ligand CIF file can be obtained using one of available tools in Phenix (see documentation for more details). Once CIF file is available it can be used as following: phenix.real_space_refine model.pdb map.mtz ligands.cif

If PDB file contains unknown to Phenix ligand a ligand CIF file needs to be provided. Ligand CIF file can be obtained using one of available tools in Phenix (see documentation for more details). Once CIF file is available it can be used as following:

```
phenix.real_space_refine model.pdb map.mtz ligands.cif
```

- Group ADP (B-factor) refinement. Currently only one option available: restrained group ADP refinement with one isotropic B-factor per residue. This is enabled by default. The command below will do B factor refinement only: phenix.real_space_refine model.pdb map.mtz run=adp To make use of multiple CPU for B factor refinement use nproc=: phenix.real_space_refine model.pdb map.ccp4 resolution=3.2 nproc=12 Note: multiple CPUs will only be used for B-factor refinement (which is done by default).

Group ADP (B-factor) refinement. Currently only one option available: restrained group ADP refinement with one isotropic B-factor per residue. This is enabled by default. The command below will do B factor refinement only:

```
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```

To make use of multiple CPU for B factor refinement use nproc=:

```
phenix.real_space_refine model.pdb map.ccp4 resolution=3.2 nproc=12
```

Note: multiple CPUs will only be used for B-factor refinement (which is done by default).

- Running rigid-body refinement. Rigid body refinement can be run alone or in combination with other refinement strategies. This will run rigid-body refinement only: phenix.real_space_refine model.pdb map.mtz run=rigid_body rigid.eff where rigid.eff contains definitions of rigid body groups: refinement.rigid_body { group = chain A group = chain B or chain C group = chain Z } This will run rigid-body and individual coordinate refinement only: phenix.real_space_refine model.pdb map.mtz run=rigid_body+minimization_global

Running rigid-body refinement. Rigid body refinement can be run alone or in combination with other refinement strategies. This will run rigid-body refinement only:

```
phenix.real_space_refine model.pdb map.mtz run=rigid_body rigid.eff
```

where rigid.eff contains definitions of rigid body groups:

```
refinement.rigid_body {
 group = chain A
 group = chain B or chain C
 group = chain Z
}
```

This will run rigid-body and individual coordinate refinement only:

```
phenix.real_space_refine model.pdb map.mtz run=rigid_body+minimization_global
```

- In case NCS is present the program will attempt to determine NCS groups automatically and use to define NCS constraints. Automatically found definitions for NCS groups will be printed to the log. It is possible to provide NCS group definitions manually: phenix.real_space_refine model.pdb map.mtz ncs.eff where ncs.eff contains definitions of NCS groups: pdb_interpretation { ncs_group { reference = chain A selection = chain B } ncs_group { reference = chain C or chain D selection = chain X selection = chain Y selection = chain Z } } Note that user-supplied NCS groups will be filtered at all times. More on this here. NCS group definitions can be obtained using GUI or command line tool: phenix.simple_ncs_from_pdb model.pdb NCS operators are refined if NCS constraints are used. To disable refinement of NCS operators use refine_ncs_operators=False.

In case NCS is present the program will attempt to determine NCS groups automatically and use to define NCS constraints. Automatically found definitions for NCS groups will be printed to the log. It is possible to provide NCS group definitions manually:

```
phenix.real_space_refine model.pdb map.mtz ncs.eff
```

where ncs.eff contains definitions of NCS groups:

```
pdb_interpretation {
 ncs_group {
 reference = chain A
 selection = chain B
 }
 ncs_group {
 reference = chain C or chain D
 selection = chain X
 selection = chain Y
 selection = chain Z
 }
}
```

Note that user-supplied NCS groups will be filtered at all times. More on this here.

NCS group definitions can be obtained using GUI or command line tool:

```
phenix.simple_ncs_from_pdb model.pdb
```

NCS operators are refined if NCS constraints are used. To disable refinement of NCS operators use refine_ncs_operators=False.

- Use if secondary structure (SS) restraints is controlled by secondary_structure.enabled keyword: phenix.real_space_refine model.pdb map.mtz \ secondary_structure.enabled=True If HELIX/SHEET records are present in PDB file they will be used to construct restraints. Otherwise one of Phenix tools will be run internally to derive secondary structure information from the input model. DNA/RNA specific restraints will be generated internally. Not all provided SS will be used by default: some H-bonds that exceed a certain threshold will be filtered out. To force using all SS annotations use remove_outliers=False keyword. phenix.secondary_structure_restraints can create SS annotations and restraints given a model file. This will output HELIX/SHEET records: phenix.secondary_structure_restraints model.pdb format=pdb and this will create a parameter file that defines SS restraints ready to use for refinement: phenix.secondary_structure_restraints model.pdb format=phenix None SS annotation programs can determine SS 100% reliably in all cases! SS annotation given to refinement will be enforced on the refined model. This means that any errors in HELIX/SHEET records or in SS restraints file will be propagated onto refined model. Therefore it is critical to make sure that provided SS annotations are as correct as possible. It is recommended to use one of available programs (such as phenix.secondary_structure_restraints) to obtain initial guess at SS annotation and then verify it manually to remove incorrect and add missing annotations.

Use if secondary structure (SS) restraints is controlled by secondary_structure.enabled keyword:

```
phenix.real_space_refine model.pdb map.mtz \
secondary_structure.enabled=True
```

If HELIX/SHEET records are present in PDB file they will be used to construct restraints. Otherwise one of Phenix tools will be run internally to derive secondary structure information from the input model. DNA/RNA specific restraints will be generated internally. Not all provided SS will be used by default: some H-bonds that exceed a certain threshold will be filtered out. To force using all SS annotations use remove_outliers=False keyword.

phenix.secondary_structure_restraints can create SS annotations and restraints given a model file. This will output HELIX/SHEET records:

```
phenix.secondary_structure_restraints model.pdb format=pdb
```

and this will create a parameter file that defines SS restraints ready to use for refinement:

```
phenix.secondary_structure_restraints model.pdb format=phenix
```

None SS annotation programs can determine SS 100% reliably in all cases! SS annotation given to refinement will be enforced on the refined model. This means that any errors in HELIX/SHEET records or in SS restraints file will be propagated onto refined model. Therefore it is critical to make sure that provided SS annotations are as correct as possible. It is recommended to use one of available programs (such as phenix.secondary_structure_restraints) to obtain initial guess at SS annotation and then verify it manually to remove incorrect and add missing annotations.

- Weight between data and restraints is determined automatically to achieve specified rms deviations for covalent bonds and angles, which default to target_bonds_rmsd=0.01 and target_angles_rmsd=1.0. Alternatively, the value of the weight can be specified manually: phenix.real_space_refine model.pdb map.ccp4 resolution=4.2 weight=123.

Weight between data and restraints is determined automatically to achieve specified rms deviations for covalent bonds and angles, which default to target_bonds_rmsd=0.01 and target_angles_rmsd=1.0. Alternatively, the value of the weight can be specified manually:

```
phenix.real_space_refine model.pdb map.ccp4 resolution=4.2 weight=123.
```

- Number of refinement macro-cycles defaults to 5, which is sufficient in most cases. If model geometry or/and model-to-map fit is poor then using more macro-cycles may be helpful: phenix.real_space_refine model.pdb map.mtz macro_cycles=12

Number of refinement macro-cycles defaults to 5, which is sufficient in most cases. If model geometry or/and model-to-map fit is poor then using more macro-cycles may be helpful:

```
phenix.real_space_refine model.pdb map.mtz macro_cycles=12
```

- It is possible to define a bond between any pair of atoms, any number of bonds. Likewise, an angle restraint can be defined between any triplet of atoms, any number of such restraints. Similarly, dihedral, planarity and parallelity restraints can be added. This example adds two custom bonds (one of which is between symmetry related atoms) and one angle restraints: geometry_restraints { edits { bond { atom_selection_1 = chain A and resseq 123 and name O atom_selection_2 = chain Z and resname LIG and name ND1 symmetry_operation = -x,y,z distance_ideal = 1.4 sigma = 0.025 } bond { atom_selection_1 = chain B and resseq 321 and name O1 atom_selection_2 = chain Z and resseq 234 and name O2 distance_ideal = 1.3 sigma = 0.03 } angle { action = *add delete change atom_selection_1 = chain C and resseq 2 and name CA atom_selection_2 = chain X and resseq 4 and name HG atom_selection_3 = chain Z and resseq 8 and name O angle_ideal = 120. sigma = 5. } } }

It is possible to define a bond between any pair of atoms, any number of bonds. Likewise, an angle restraint can be defined between any triplet of atoms, any number of such restraints. Similarly, dihedral, planarity and parallelity restraints can be added. This example adds two custom bonds (one of which is between symmetry related atoms) and one angle restraints:

```
geometry_restraints {
 edits {
 bond {
 atom_selection_1 = chain A and resseq 123 and name O
 atom_selection_2 = chain Z and resname LIG and name ND1
 symmetry_operation = -x,y,z
 distance_ideal = 1.4
 sigma = 0.025
 }
 bond {
 atom_selection_1 = chain B and resseq 321 and name O1
 atom_selection_2 = chain Z and resseq 234 and name O2
 distance_ideal = 1.3
 sigma = 0.03
 }
 angle {
 action = *add delete change
 atom_selection_1 = chain C and resseq 2 and name CA
 atom_selection_2 = chain X and resseq 4 and name HG
 atom_selection_3 = chain Z and resseq 8 and name O
 angle_ideal = 120.
 sigma = 5.
 }
 }
}
```

The lines above need to be saved into a file, say edits.eff, and then given to refinement along with all other inputs: phenix.real_space_refine model.pdb map.mtz edits.eff To verify that these restraints (along with others) were actually used in refinement inspect .geo file that lists all restraints for all atoms. The GUI provides a grafical way of defining custom restraints.

The lines above need to be saved into a file, say edits.eff, and then given to refinement along with all other inputs:

```
phenix.real_space_refine model.pdb map.mtz edits.eff
```

To verify that these restraints (along with others) were actually used in refinement inspect .geo file that lists all restraints for all atoms.

The GUI provides a grafical way of defining custom restraints.

- Reference model restraints come in two flavors. One allows to restrain model to its initial coordinates: phenix.real_space_refine model.pdb map.mtz \ reference_coordinate_restraints.enabled=true \ reference_coordinate_restraints.selection="chain A and resseq 12:21" \ reference_coordinate_restraints.sigma=0.05 Here only residues from 12 to 21 in chain A will be restrained with the weight 1/0.05**2. Another allows to provide a reference model in one or several files. This may be a higher quility (resolution) model and it does not have to be idential to refining model (similarity 85% or better, which is a parameter). In this example two reference models are used: chain Z from reference_model_1.pdb file and chain C from reference_model_2.pdb file. Chain Z from the first file serves as a reference for chain A in refining model, and chain C from the second file serves as a reference for residued 123-321 in chain B of refining model: reference_model { enabled = True reference_group { reference = chain Z selection = chain A file_name = reference_model_1.pdb } reference_group { reference = chain C selection = chain B and resseq 123:321 file_name = reference_model_2.pdb } }

Reference model restraints come in two flavors. One allows to restrain model to its initial coordinates:

```
phenix.real_space_refine model.pdb map.mtz \
reference_coordinate_restraints.enabled=true \
reference_coordinate_restraints.selection="chain A and resseq 12:21" \
reference_coordinate_restraints.sigma=0.05
```

Here only residues from 12 to 21 in chain A will be restrained with the weight 1/0.05**2.

Another allows to provide a reference model in one or several files. This may be a higher quility (resolution) model and it does not have to be idential to refining model (similarity 85% or better, which is a parameter). In this example two reference models are used: chain Z from reference_model_1.pdb file and chain C from reference_model_2.pdb file. Chain Z from the first file serves as a reference for chain A in refining model, and chain C from the second file serves as a reference for residued 123-321 in chain B of refining model:

```
reference_model {
 enabled = True
 reference_group {
 reference = chain Z
 selection = chain A
 file_name = reference_model_1.pdb
 }
 reference_group {
 reference = chain C
 selection = chain B and resseq 123:321
 file_name = reference_model_2.pdb
 }
}
```

To use reference model restraints a parameter file (for example, called reference.eff) with lines like above needs to be created and provided as input: phenix.real_space_refine model.pdb map.mtz reference.eff

To use reference model restraints a parameter file (for example, called reference.eff) with lines like above needs to be created and provided as input:

```
phenix.real_space_refine model.pdb map.mtz reference.eff
```

- To see all parameters: phenix.real_space_refine -h

To see all parameters:

```
phenix.real_space_refine -h
```

Notes

Generally, refinement with all defaults (example #1 above) is sufficient. Secondary structure (SS) restraints strongly rely on correct SS annotation. If SHEET and HELIX records are available in PDB file header then they will be used to enforce SS as defined. Incorrect SHEET and HELIX records are very likely to result in incorrectly refined structure. If SHEET and HELIX records are not present in PDB file header and SS restraints are enabled then the program will annotate SS and use it as restraints. The SS annotation strongly depends on input model quality: for a model with errors SS annotation may be inaccurate resulting in poorly refine model. Please refer to SS documentation for more details. Including local fitting, morphing, or simulated annealing (local_grid_search+morphing+simulated_annealing) into refinement may significantly increase runtime. It is possible to extract a box with map and model in it and do refinement inside of that box. To extract box with map and model use phenix.map_box command (see Phenix documentation for more details). It is important that information provided in CRYST1 record in PDB file (if provided) matches box information of CCP4 formatted map or crystal symmetry information in MTZ file (whatever used). ADP (B-factor) values do not affect the refinement of coordinates. It is best to do ADP refinement once the model fits the map well. Normally this would be next to final step. Part of the model can be refined against entire map.

- Generally, refinement with all defaults (example #1 above) is sufficient.
- Secondary structure (SS) restraints strongly rely on correct SS annotation. If SHEET and HELIX records are available in PDB file header then they will be used to enforce SS as defined. Incorrect SHEET and HELIX records are very likely to result in incorrectly refined structure. If SHEET and HELIX records are not present in PDB file header and SS restraints are enabled then the program will annotate SS and use it as restraints. The SS annotation strongly depends on input model quality: for a model with errors SS annotation may be inaccurate resulting in poorly refine model. Please refer to SS documentation for more details.
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- ADP (B-factor) values do not affect the refinement of coordinates.
- It is best to do ADP refinement once the model fits the map well. Normally this would be next to final step.
- Part of the model can be refined against entire map.

Half maps vs full map

Real space refinement by phenix.real_space_refine uses a full map, not half-maps. You can create a full (averaged) map from two half-maps by simple averaging, or by density modification with ResolveCryoEM.

References

P.V. Afonine, J.J. Headd, T.C. Terwilliger & P.D. Adams. Computational Crystallography Newsletter (2013). Volume 4, Part 2, 43-44. http://phenix-online.org/newsletter/CCN_2013_07.pdf

- P.V. Afonine, J.J. Headd, T.C. Terwilliger & P.D. Adams. Computational Crystallography Newsletter (2013). Volume 4, Part 2, 43-44. http://phenix-online.org/newsletter/CCN_2013_07.pdf

Full set of parameters

debug = False Excessive output for debuggingmap_coefficients_label = None resolution = None write_initial_geo_file = True write_final_geo_file = False write_all_states = False write_pkl_stats = False model_format = *pdb *mmcif mask_and_he_map = False resolution_factor = 0.25 ncs_constraints = Auto refine_ncs_operators = Auto variable_rama_potential = False Switch between oldfield and emsley potentials during refinementweight = None real_space_target_exclude_selection = "element H or element D" Exclude selected atoms from real-space target calculationshow_statistics = True show_per_residue = True dry_run = False random_seed = 0 nproc = 1 gradients_method = fd linear quadratic *tricubic scattering_table = n_gaussian wk1995 it1992 *electron neutron Choices of scattering table for structure factors calculationsignore_symmetry_conflicts = False Ignore conflicting symmetry (normally use symmetry from map file)wrapping = Auto Wrap map outside of boundaries. Default is True for mtz file input, False otherwiseskip_map_model_overlap_check = False Skip map-model overlap checkabsolute_length_tolerance = 0.01 Length tolerance for comparing symmetriesabsolute_angle_tolerance = 0.01 Angle tolerance for comparing symmetriesjob_title = None Job title in PHENIX GUI, not used on command linerotamersfit = all *outliers_or_poormap outliers_and_poormap outliers poormap Fit side-chains regimestuneup = outliers outliers_and_poormap Switch remaining (after map fit) outliers to the nearest rotamerrestraintsenabled = True Use rotamer restraintssigma = Auto The smaller the value, the stronger the restraintstarget = max_distant min_distant exact_match fix_outliers Choice for the reference conformation to restrain torefinementrun = *minimization_global rigid_body *local_grid_search morphing simulated_annealing *adp *occupancy *nqh_flips morphing = every_macro_cycle once *first simulated_annealing = every_macro_cycle *once local_grid_search = every_macro_cycle once *last max_iterations = 100 macro_cycles = 5 target_bonds_rmsd = 0.01 target_angles_rmsd = 1 backbone_sample = None use_adp_restraints = True atom_radius = 3 do_ss_ideal = False do_ccd = False adpindividualisotropic = Auto Individual isotropic ADP refinementrestraints_weight = 1 Setting to 0 disables restraintsgroupmode = *one_per_residue two_per_residue one_per_chain selection = None occupanciesScope of atom selections for occupancy refinementindividual = None Selection(s) for individual atoms. None is default which is to refine the individual occupancies for atoms in alternative conformations or for atoms with partial occupancies only.remove_selection = None Occupancies of selected atoms will not be refined (even though they might satisfy the default criteria for occupancy refinement).constrained_groupSelections to define constrained occupancies. If only one selection is provided then one occupancy factor per selected atoms will be refined and it will be constrained between predefined max and min values.selection = None Atom selection string.rigid_bodygroup = None amberParameters for using Amber in refinement.use_amber = False Use Amber for all the gradients in refinementtopology_file_name = None A topology file needed by Amber. Can be generated using phenix.AmberPrep.coordinate_file_name = None A coordinate file needed by Amber. Can be generated using phenix.AmberPrep.order_file_name = None A file that maps amber atom numbers to phenix atom numbers.wxc_factor = 1 restraint_wt = 0 restraintmask = "" reference_file_name = "" bellymask = "" If given, turn on belly in sanderqmmask = "" If given, turn on QM/MM with the given maskqmcharge = 0 Charge of the QM/MM regionnetcdf_trajectory_file_name = "" If given, turn on writing netcdf trajectoryprint_amber_energies = False Print details of Amber energies during refinementqiQMworking_directory = None qm_restraintsselection = None selection for core of atoms to calculate new restraints via a QM geometry minimisationrun_in_macro_cycles = *first_only first...

- debug = False Excessive output for debugging
- map_coefficients_label = None
- resolution = None
- write_initial_geo_file = True
- write_final_geo_file = False
- write_all_states = False
- write_pkl_stats = False
- model_format = *pdb *mmcif
- mask_and_he_map = False
- resolution_factor = 0.25
- ncs_constraints = Auto
- refine_ncs_operators = Auto
- variable_rama_potential = False Switch between oldfield and emsley potentials during refinement
- weight = None
- real_space_target_exclude_selection = "element H or element D" Exclude selected atoms from real-space target calculation

- show_statistics = True
- show_per_residue = True
- dry_run = False
- random_seed = 0
- nproc = 1
- gradients_method = fd linear quadratic *tricubic
- scattering_table = n_gaussian wk1995 it1992 *electron neutron Choices of scattering table for structure factors calculations
- ignore_symmetry_conflicts = False Ignore conflicting symmetry (normally use symmetry from map file)
- wrapping = Auto Wrap map outside of boundaries. Default is True for mtz file input, False otherwise
- skip_map_model_overlap_check = False Skip map-model overlap check
- absolute_length_tolerance = 0.01 Length tolerance for comparing symmetries
- absolute_angle_tolerance = 0.01 Angle tolerance for comparing symmetries
- job_title = None Job title in PHENIX GUI, not used on command line
- rotamersfit = all *outliers_or_poormap outliers_and_poormap outliers poormap Fit side-chains regimestuneup = outliers outliers_and_poormap Switch remaining (after map fit) outliers to the nearest rotamerrestraintsenabled = True Use rotamer restraintssigma = Auto The smaller the value, the stronger the restraintstarget = max_distant min_distant exact_match fix_outliers Choice for the reference conformation to restrain to
- fit = all *outliers_or_poormap outliers_and_poormap outliers poormap Fit side-chains regimes
- tuneup = outliers outliers_and_poormap Switch remaining (after map fit) outliers to the nearest rotamer
- restraintsenabled = True Use rotamer restraintssigma = Auto The smaller the value, the stronger the restraintstarget = max_distant min_distant exact_match fix_outliers Choice for the reference conformation to restrain to
- enabled = True Use rotamer restraints
- sigma = Auto The smaller the value, the stronger the restraints
- target = max_distant min_distant exact_match fix_outliers Choice for the reference conformation to restrain to
- refinementrun = *minimization_global rigid_body *local_grid_search morphing simulated_annealing *adp *occupancy *nqh_flips morphing = every_macro_cycle once *first simulated_annealing = every_macro_cycle *once local_grid_search = every_macro_cycle once *last max_iterations = 100 macro_cycles = 5 target_bonds_rmsd = 0.01 target_angles_rmsd = 1 backbone_sample = None use_adp_restraints = True atom_radius = 3 do_ss_ideal = False do_ccd = False adpindividualisotropic = Auto Individual isotropic ADP refinementrestraints_weight = 1 Setting to 0 disables restraintsgroupmode = *one_per_residue two_per_residue one_per_chain selection = None occupanciesScope of atom selections for occupancy refinementindividual = None Selection(s) for individual atoms. None is default which is to refine the individual occupancies for atoms in alternative conformations or for atoms with partial occupancies only.remove_selection = None Occupancies of selected atoms will not be refined (even though they might satisfy the default criteria for occupancy refinement).constrained_groupSelections to define constrained occupancies. If only one selection is provided then one occupancy factor per selected atoms will be refined and it will be constrained between predefined max and min values.selection = None Atom selection string.rigid_bodygroup = None

- run = *minimization_global rigid_body *local_grid_search morphing simulated_annealing *adp *occupancy *nqh_flips
- morphing = every_macro_cycle once *first
- simulated_annealing = every_macro_cycle *once
- local_grid_search = every_macro_cycle once *last
- max_iterations = 100
- macro_cycles = 5
- target_bonds_rmsd = 0.01
- target_angles_rmsd = 1
- backbone_sample = None
- use_adp_restraints = True
- atom_radius = 3
- do_ss_ideal = False
- do_ccd = False
- adpindividualisotropic = Auto Individual isotropic ADP refinementrestraints_weight = 1 Setting to 0 disables restraintsgroupmode = *one_per_residue two_per_residue one_per_chain selection = None
- individualisotropic = Auto Individual isotropic ADP refinementrestraints_weight = 1 Setting to 0 disables restraints
- isotropic = Auto Individual isotropic ADP refinement
- restraints_weight = 1 Setting to 0 disables restraints
- groupmode = *one_per_residue two_per_residue one_per_chain selection = None
- mode = *one_per_residue two_per_residue one_per_chain
- selection = None
- occupanciesScope of atom selections for occupancy refinementindividual = None Selection(s) for individual atoms. None is default which is to refine the individual occupancies for atoms in alternative conformations or for atoms with partial occupancies only.remove_selection = None Occupancies of selected atoms will not be refined (even though they might satisfy the default criteria for occupancy refinement).constrained_groupSelections to define constrained occupancies. If only one selection is provided then one occupancy factor per selected atoms will be refined and it will be constrained between predefined max and min values.selection = None Atom selection string.
- individual = None Selection(s) for individual atoms. None is default which is to refine the individual occupancies for atoms in alternative conformations or for atoms with partial occupancies only.
- remove_selection = None Occupancies of selected atoms will not be refined (even though they might satisfy the default criteria for occupancy refinement).
- constrained_groupSelections to define constrained occupancies. If only one selection is provided then one occupancy factor per selected atoms will be refined and it will be constrained between predefined max and min values.selection = None Atom selection string.
- selection = None Atom selection string.
- rigid_bodygroup = None
- group = None

- amberParameters for using Amber in refinement.use_amber = False Use Amber for all the gradients in refinementtopology_file_name = None A topology file needed by Amber. Can be generated using phenix.AmberPrep.coordinate_file_name = None A coordinate file needed by Amber. Can be generated using phenix.AmberPrep.order_file_name = None A file that maps amber atom numbers to phenix atom numbers.wxc_factor = 1 restraint_wt = 0 restraintmask = "" reference_file_name = "" bellymask = "" If given, turn on belly in sanderqmmask = "" If given, turn on QM/MM with the given maskqmcharge = 0 Charge of the QM/MM regionnetcdf_trajectory_file_name = "" If given, turn on writing netcdf trajectoryprint_amber_energies = False Print details of Amber energies during refinement
- use_amber = False Use Amber for all the gradients in refinement
- topology_file_name = None A topology file needed by Amber. Can be generated using phenix.AmberPrep.
- coordinate_file_name = None A coordinate file needed by Amber. Can be generated using phenix.AmberPrep.
- order_file_name = None A file that maps amber atom numbers to phenix atom numbers.
- wxc_factor = 1
- restraint_wt = 0
- restraintmask = ""
- reference_file_name = ""
- bellymask = "" If given, turn on belly in sander
- qmmask = "" If given, turn on QM/MM with the given mask
- qmcharge = 0 Charge of the QM/MM region
- netcdf_trajectory_file_name = "" If given, turn on writing netcdf trajectory
- print_amber_energies = False Print details of Amber energies during refinement
- qiQMworking_directory = None qm_restraintsselection = None selection for core of atoms to calculate new restraints via a QM geometry minimisationrun_in_macro_cycles = *first_only first_and_last all last_only test the steps of the refinement that the restraints generation is runbuffer = 3.5 distance to include entire residues into the enviroment of the coreignore_lack_of_h_on_ligand = False skip check on protonation of ligand for entities such as MgF3capping_groups = True cleanup = all *most None calculate = *in_situ_opt starting_energy final_energy starting_strain final_strain starting_bound final_bound starting_binding final_binding gradients Choose QM calculations to runwrite_files = *restraints pdb_core pdb_buffer pdb_final_core pdb_final_buffer which ligand or cluster files to writeprotein_optimisation_freeze = *all None main_chain main_chain_to_beta main_chain_to_delta torsions the parts of protein residues that are frozen when an amino acid is the main selectionremove_water = False restraints_filename = Auto restraints filename is based on model name if not specifiedignore_x_h_distance_protein = False skip check on transfer of proton during QM optimisationinclude_nearest_neighbours_in_optimisation = False include the side chains of protein in the QM optimisationinclude_inter_residue_restraints = False do_not_update_restraints = False For testing and maybe getting strain energy of standard restraintsbuffer_selection = None use this instead of distance from selectionspecific_atom_chargesatom_selection = None charge = None specific_atom_multiplicitiesatom_selection = None multiplicity = None freeze_specific_atomsatom_selection = None packageprogram = *mopac test charge = Auto multiplicity = Auto method = Auto basis_set = Auto solvent_model = None nproc = 1 read_output_to_skip_opt_if_available = False ignore_input_differences = False view_output = None qm_gradientsselection = None selection for core of atoms to calculate new restraints via a QM geometry

minimisationrun_in_macro_cycles = first_only first_and_last *all last_only test the steps of the refinement that the restraints generation is runbuffer = 3.5 distance to include entire residues into the enviroment of the coreignore_lack_of_h_on_ligand = False skip check on protonation of ligand for entities such as MgF3capping_groups = True cleanup = all *most None specific_atom_chargesatom_selection = None charge = None specific_atom_multiplicitiesatom_selection = None multiplicity = None packageprogram = *mopac test charge = Auto multiplicity = Auto method = Auto basis_set = Auto solvent_model = None nproc = 1 read_output_to_skip_opt_if_available = False ignore_input_differences = False view_output = None

- working_directory = None
- qm_restraintsselection = None selection for core of atoms to calculate new restraints via a QM geometry minimisationrun_in_macro_cycles = *first_only first_and_last all last_only test the steps of the refinement that the restraints generation is runbuffer = 3.5 distance to include entire residues into the enviroment of the coreignore_lack_of_h_on_ligand = False skip check on protonation of ligand for entities such as MgF3capping_groups = True cleanup = all *most None calculate = *in_situ_opt starting_energy final_energy starting_strain final_strain starting_bound final_bound starting_binding final_binding gradients Choose QM calculations to runwrite_files = *restraints pdb_core pdb_buffer pdb_final_core pdb_final_buffer which ligand or cluster files to writeprotein_optimisation_freeze = *all None main_chain main_chain_to_beta main_chain_to_delta torsions the parts of protein residues that are frozen when an amino acid is the main selectionremove_water = False restraints_filename = Auto restraints filename is based on model name if not specifiedignore_x_h_distance_protein = False skip check on transfer of proton during QM optimisationinclude_nearest_neighbours_in_optimisation = False include the side chains of protein in the QM optimisationinclude_inter_residue_restraints = False do_not_update_restraints = False For testing and maybe getting strain energy of standard restraintsbuffer_selection = None use this instead of distance from selectionspecific_atom_chargesatom_selection = None charge = None specific_atom_multiplicitiesatom_selection = None multiplicity = None freeze_specific_atomsatom_selection = None packageprogram = *mopac test charge = Auto multiplicity = Auto method = Auto basis_set = Auto solvent_model = None nproc = 1 read_output_to_skip_opt_if_available = False ignore_input_differences = False view_output = None
- selection = None selection for core of atoms to calculate new restraints via a QM geometry minimisation
- run_in_macro_cycles = *first_only first_and_last all last_only test the steps of the refinement that the restraints generation is run
- buffer = 3.5 distance to include entire residues into the enviroment of the core
- ignore_lack_of_h_on_ligand = False skip check on protonation of ligand for entities such as MgF3
- capping_groups = True
- cleanup = all *most None
- calculate = *in_situ_opt starting_energy final_energy starting_strain final_strain starting_bound final_bound starting_binding final_binding gradients Choose QM calculations to run
- write_files = *restraints pdb_core pdb_buffer pdb_final_core pdb_final_buffer which ligand or cluster files to write
- protein_optimisation_freeze = *all None main_chain main_chain_to_beta main_chain_to_delta torsions the parts of protein residues that are frozen when an amino acid is the main selection
- remove_water = False
- restraints_filename = Auto restraints filename is based on model name if not specified

- ignore_x_h_distance_protein = False skip check on transfer of proton during QM optimisation
- include_nearest_neighbours_in_optimisation = False include the side chains of protein in the QM optimisation
- include_inter_residue_restraints = False
- do_not_update_restraints = False For testing and maybe getting strain energy of standard restraints
- buffer_selection = None use this instead of distance from selection
- specific_atom_chargesatom_selection = None charge = None
- atom_selection = None
- charge = None
- specific_atom_multiplicitiesatom_selection = None multiplicity = None
- atom_selection = None
- multiplicity = None
- freeze_specific_atomsatom_selection = None
- atom_selection = None
- packageprogram = *mopac test charge = Auto multiplicity = Auto method = Auto basis_set = Auto solvent_model = None nproc = 1 read_output_to_skip_opt_if_available = False ignore_input_differences = False view_output = None
- program = *mopac test
- charge = Auto
- multiplicity = Auto
- method = Auto
- basis_set = Auto
- solvent_model = None
- nproc = 1
- read_output_to_skip_opt_if_available = False
- ignore_input_differences = False
- view_output = None
- qm_gradientsselection = None selection for core of atoms to calculate new restraints via a QM geometry minimisationrun_in_macro_cycles = first_only first_and_last *all last_only test the steps of the refinement that the restraints generation is runbuffer = 3.5 distance to include entire residues into the enviroment of the coreignore_lack_of_h_on_ligand = False skip check on protonation of ligand for entities such as MgF3capping_groups = True cleanup = all *most None specific_atom_chargesatom_selection = None charge = None specific_atom_multiplicitiesatom_selection = None multiplicity = None packageprogram = *mopac test charge = Auto multiplicity = Auto method = Auto basis_set = Auto solvent_model = None nproc = 1 read_output_to_skip_opt_if_available = False ignore_input_differences = False view_output = None
- selection = None selection for core of atoms to calculate new restraints via a QM geometry minimisation
- run_in_macro_cycles = first_only first_and_last *all last_only test the steps of the refinement that the restraints generation is run
- buffer = 3.5 distance to include entire residues into the enviroment of the core

• ignore_lack_of_h_on_ligand = False skip check on protonation of ligand for entities such as MgF3

• capping_groups = True

• cleanup = all *most None

• specific_atom_chargesatom_selection = None charge = None

• atom_selection = None

• charge = None

• specific_atom_multiplicitiesatom_selection = None multiplicity = None

• atom_selection = None

• multiplicity = None

• packageprogram = *mopac test charge = Auto multiplicity = Auto method = Auto basis_set = Auto solvent_model = None nproc = 1 read_output_to_skip_opt_if_available = False ignore_input_differences = False view_output = None

• program = *mopac test

• charge = Auto

• multiplicity = Auto

• method = Auto

• basis_set = Auto

• solvent_model = None

• nproc = 1

• read_output_to_skip_opt_if_available = False

• ignore_input_differences = False

• view_output = None

• reference_modelenabled = False Restrains the dihedral angles to a high-resolution reference structure to reduce overfitting at low resolution. You will need to specify a reference PDB file (in the input list in the main window) to use this option.file = None use_starting_model_as_reference = False sigma = 1 limit = 15 hydrogens = False Include dihedrals with hydrogen atomsmain_chain = True Include dihedrals formed by main chain atomsside_chain = True Include dihedrals formed by side chain atomsfix_outliers = True Try to fix rotamer outliers in refined modelstrict_rotamer_matching = False Make sure that rotamers in refinement model matches those in reference model even when they are not outliersauto_shutoff_for_ncs = False Do not apply to parts of structure covered by NCS restraintssecondary_structure_only = False Only apply reference model restraints to secondary structure elements (helices and sheets)reference_groupreference = None selection = None file_name = None this is to used internally to disambiguate cases where multiple reference models contain the same chain ID. This normally does not need to be set by the usersearch_optionsSet of parameters for NCS search procedure. Some of them also used for filtering user-supplied ncs_group.exclude_selection = element H or element D or water Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS.chain_similarity_threshold = 0.85 Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residueschain_max_rmsd = 100 limit of rms difference between chains to be considered as copiesresidue_match_radius = 1000 Maximum allowed distance difference between pairs of matching atoms of two residuestry_shortcuts = False Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.minimum_number_of_atoms_in_copy = 3 Do not create ncs groups where master and copies would contain less than specified amount of atomsvalidate_user_supplied_groups = True Enable validation of user-supplied ncs_group. Need to exercise a lot of caution turning this off. This option is for developers only.

• enabled = False Restrains the dihedral angles to a high-resolution reference structure to reduce overfitting at low resolution. You will need to specify a reference PDB file (in the input list in the main window) to use this option.

• file = None

• use_starting_model_as_reference = False

• sigma = 1

• limit = 15

• hydrogens = False Include dihedrals with hydrogen atoms

• main_chain = True Include dihedrals formed by main chain atoms

• side_chain = True Include dihedrals formed by side chain atoms

• fix_outliers = True Try to fix rotamer outliers in refined model

• strict_rotamer_matching = False Make sure that rotamers in refinement model matches those in reference model even when they are not outliers

• auto_shutoff_for_ncs = False Do not apply to parts of structure covered by NCS restraints

• secondary_structure_only = False Only apply reference model restraints to secondary structure elements (helices and sheets)

• reference_groupreference = None selection = None file_name = None this is to used internally to disambiguate cases where multiple reference models contain the same chain ID. This normally does not need to be set by the user

• reference = None

• selection = None

• file_name = None this is to used internally to disambiguate cases where multiple reference models contain the same chain ID. This normally does not need to be set by the user

• search_optionsSet of parameters for NCS search procedure. Some of them also used for filtering user-supplied ncs_group.exclude_selection = element H or element D or water Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS.chain_similarity_threshold = 0.85 Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residueschain_max_rmsd = 100 limit of rms difference between chains to be considered as copiesresidue_match_radius = 1000 Maximum allowed distance difference between pairs of matching atoms of two residuestry_shortcuts = False Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.minimum_number_of_atoms_in_copy = 3 Do not create ncs groups where master and copies would contain less than specified amount of atomsvalidate_user_supplied_groups = True Enable validation of user-supplied ncs_group. Need to exercise a lot of caution turning this off. This option is for developers only.

• exclude_selection = element H or element D or water Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS.

• chain_similarity_threshold = 0.85 Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residues

• chain_max_rmsd = 100 limit of rms difference between chains to be considered as copies

• residue_match_radius = 1000 Maximum allowed distance difference between pairs of matching atoms of two residues

• try_shortcuts = False Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.

• minimum_number_of_atoms_in_copy = 3 Do not create ncs groups where master and copies would contain less than specified amount of atoms

• validate_user_supplied_groups = True Enable validation of user-supplied ncs_group. Need to exercise a lot of caution turning this off. This option is for developers only.

• pdb_interpretationsort_atoms = True use_ncs_to_build_restraints = True show_restraints_histograms = True flip_symmetric_amino_acids = True superpose_ideal_ligand = *None all SF4 F3S DVT disable_uc_volume_vs_n_atoms_check = False allow_polymer_cross_special_position = False correct_hydrogens = True c_beta_restraints = True use_neutron_distances = False Use neutron X-H distances (which are longer than X-ray ones)disulfide_bond_exclusions_selection_string = None exclusion_distance_cutoff = 3 If SG of CYS forming SS bond is closer than this distance to an atom that it may coordinate then this SG is excluded from SS bond.link_distance_cutoff = 3 Length of link between the linked residuesdisulfide_distance_cutoff = 3 add_angle_and_dihedral_restraints_for_disulfides = True dihedral_function_type = *determined_by_sign_of_periodicity all_sinusoidal all_harmonic chir_volume_esd = 0.2 max_reasonable_bond_distance = 50 nonbonded_distance_cutoff = None default_vdw_distance = 1 min_vdw_distance = 1 nonbonded_buffer = 1 **EXPERIMENTAL, developers only**nonbonded_weight = 100 Weighting of nonbonded restraints term. By default, this will be set to 16 if explicit hydrogens are used (this was the default in earlier versions of Phenix), or 100 if hydrogens are missing.const_shrink_donor_acceptor = 0 **EXPERIMENTAL, developers only**vdw_1_4_factor = 0.8 min_distance_sym_equiv = 0.5 custom_nonbonded_symmetry_exclusions = None translate_cns_dna_rna_residue_names = None proceed_with_excessive_length_bonds = False restraints_librarycdl = True Use Conformation Dependent Library (CDL) for geometry restraintsinclude_modified_amino_acid_in_cdl = False mcl = True Use Metal Coordination Library (MCL) for tetrahedral Zn++ and iron-sulfur clusters SF4, FES, F3S, ...cis_pro_eh99 = True omega_cdl = False Use Omega Conformation Dependent Library (omega-CDL) for geometry restraintscdl_svl = False rdl = False rdl_selection = *all TRP hpdl = False cdl_nucleotidesenable = False Use RestraintsLib for DNA and RNA for geometry restraintsesd = *phenix csd factor = 2 user_suppliedpath = None Root directory of user supplied restraintsaction = *pre post Choice to look here before GeoStd, the default, or aftersecondary_structuress_by_chain = True Only applies if search_method = from_ca. Find secondary structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much slower with ss_by_chain=False. If your model is complete use ss_by_chain=True. If your model is many fragments, use ss_by_chain=False.from_ca_conservative = False various parameters changed to make from_ca method more conservative, hopefully to closer resemble ksdssp.max_rmsd = 1 Only applies if search_method = from_ca. Maximum rmsd to consider two chains with identical sequences as the same for ss identificationuse_representative_chains = True Only applies if search_method = from_ca. Use a representative of all chains with the same sequence. Alternative is to examine each chain individually. Can be much slower with use_representative_of_chain=False if there are many symmetry copies. Ignored unless ss_by_chain is True.max_representative_chains = 100 Only applies if search_method = from_ca. Maximum number of representative chainsenabled = True Turn on secondary structure restraints (main switch)proteinenabled = True Turn on secondary structure restraints for proteinsearch_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.distance_ideal_n_o = 2.9 Target length for N-O hydrogen bonddistance_cut_n_o = 3.5 Hydrogen bond with length exceeding this value will not be establishedremove_outliers = True If true, h-bonds exceeding distance_cut_n_o length will not be establishedrestrain_hbond_angles = True helixserial_number = None helix_identifier = None enabled = True Restrain this particular helixselection = None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraint...

- sort_atoms = True
- use_ncs_to_build_restraints = True
- show_restraints_histograms = True
- flip_symmetric_amino_acids = True
- superpose_ideal_ligand = *None all SF4 F3S DVT
- disable_uc_volume_vs_n_atoms_check = False
- allow_polymer_cross_special_position = False
- correct_hydrogens = True
- c_beta_restraints = True
- use_neutron_distances = False Use neutron X-H distances (which are longer than X-ray ones)
- disulfide_bond_exclusions_selection_string = None
- exclusion_distance_cutoff = 3 If SG of CYS forming SS bond is closer than this distance to an atom that it may coordinate then this SG is excluded from SS bond.
- link_distance_cutoff = 3 Length of link between the linked residues
- disulfide_distance_cutoff = 3
- add_angle_and_dihedral_restraints_for_disulfides = True
- dihedral_function_type = *determined_by_sign_of_periodicity all_sinusoidal all_harmonic
- chir_volume_esd = 0.2
- max_reasonable_bond_distance = 50
- nonbonded_distance_cutoff = None
- default_vdw_distance = 1
- min_vdw_distance = 1
- nonbonded_buffer = 1 **EXPERIMENTAL, developers only**
- nonbonded_weight = 100 Weighting of nonbonded restraints term. By default, this will be set to 16 if explicit hydrogens are used (this was the default in earlier versions of Phenix), or 100 if hydrogens are missing.
- const_shrink_donor_acceptor = 0 **EXPERIMENTAL, developers only**
- vdw_1_4_factor = 0.8
- min_distance_sym_equiv = 0.5
- custom_nonbonded_symmetry_exclusions = None
- translate_cns_dna_rna_residue_names = None
- proceed_with_excessive_length_bonds = False

- restraints_librarycdl = True Use Conformation Dependent Library (CDL) for geometry restraintsinclude_modified_amino_acid_in_cdl = False mcl = True Use Metal Coordination Library (MCL) for tetrahedral Zn++ and iron-sulfur clusters SF4, FES, F3S, ...cis_pro_eh99 = True omega_cdl = False Use Omega Conformation Dependent Library (omega-CDL) for geometry restraintscdl_svl = False rdl = False rdl_selection = *all TRP hpdl = False cdl_nucleotidesenable = False Use RestraintsLib for DNA and RNA for geometry restraintsesd = *phenix csd factor = 2 user_suppliedpath = None Root directory of user supplied restraintsaction = *pre post Choice to look here before GeoStd, the default, or after
- cdl = True Use Conformation Dependent Library (CDL) for geometry restraints
- include_modified_amino_acid_in_cdl = False
- mcl = True Use Metal Coordination Library (MCL) for tetrahedral Zn++ and iron-sulfur clusters SF4, FES, F3S, ...
- cis_pro_eh99 = True
- omega_cdl = False Use Omega Conformation Dependent Library (omega-CDL) for geometry restraints
- cdl_svl = False
- rdl = False
- rdl_selection = *all TRP
- hpdl = False
- cdl_nucleotidesenable = False Use RestraintsLib for DNA and RNA for geometry restraintsesd = *phenix csd factor = 2
- enable = False Use RestraintsLib for DNA and RNA for geometry restraints
- esd = *phenix csd
- factor = 2
- user_suppliedpath = None Root directory of user supplied restraintsaction = *pre post Choice to look here before GeoStd, the default, or after
- path = None Root directory of user supplied restraints
- action = *pre post Choice to look here before GeoStd, the default, or after
- secondary_structuress_by_chain = True Only applies if search_method = from_ca. Find secondary structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much slower with ss_by_chain=False. If your model is complete use ss_by_chain=True. If your model is many fragments, use ss_by_chain=False.from_ca_conservative = False various parameters changed to make from_ca method more conservative, hopefully to closer resemble ksdssp.max_rmsd = 1 Only applies if search_method = from_ca. Maximum rmsd to consider two chains with identical sequences as the same for ss identificationuse_representative_chains = True Only applies if search_method = from_ca. Use a representative of all chains with the same sequence. Alternative is to examine each chain individually. Can be much slower with use_representative_of_chain=False if there are many symmetry copies. Ignored unless ss_by_chain is True.max_representative_chains = 100 Only applies if search_method = from_ca. Maximum number of representative chainsenabled = True Turn on secondary structure restraints (main switch)proteinenabled = True Turn on secondary structure restraints for proteinsearch_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.distance_ideal_n_o = 2.9 Target length for N-O hydrogen bonddistance_cut_n_o = 3.5 Hydrogen bond with length exceeding this value will not be establishedremove_outliers = True If true, h-bonds exceeding distance_cut_n_o length will not be establishedrestrain_hbond_angles = True helixserial_number = None helix_identifier = None enabled = True Restrain this particular helixselection =

None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuehbonddonor = None acceptor = None sheetenabled = True Restrain this particular sheetfirst_strand = None sheet_id = None sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuestrandselection = None sense = parallel antiparallel *unknown bond_start_current = None bond_start_previous = None hbonddonor = None acceptor = None nucleic_acidenabled = True Turn on secondary structure restraints for nucleic acidshbond_distance_cutoff = 3.4 Hydrogen bonds with length exceeding this limit will not be establishedscale_bonds_sigma = 1 All sigmas for h-bond length will be multiplied by this number. The smaller number is tighter restraints.base_pairenabled = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residuesaenger_class = 0 Type of base-pairingrestrain_planarity = False Apply planarity restraint to this base-pairplanarity_sigma = 0.176 restrain_hbonds = True Restrain hydrogen bonds length for this base-pairrestrain_hb_angles = True Restrain angles around hydrogen bonds for this base-pairrestrain_parallelity = True Apply parallelity restraint to this base-pairparallelity_target = 0 parallelity_sigma = 0.0335 stacking_pairenabled = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residueangle = 0 sigma = 0.027

- ss_by_chain = True Only applies if search_method = from_ca. Find secondary structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much slower with ss_by_chain=False. If your model is complete use ss_by_chain=True. If your model is many fragments, use ss_by_chain=False.
- from_ca_conservative = False various parameters changed to make from_ca method more conservative, hopefully to closer resemble ksdssp.
- max_rmsd = 1 Only applies if search_method = from_ca. Maximum rmsd to consider two chains with identical sequences as the same for ss identification
- use_representative_chains = True Only applies if search_method = from_ca. Use a representative of all chains with the same sequence. Alternative is to examine each chain individually. Can be much slower with use_representative_of_chain=False if there are many symmetry copies. Ignored unless ss_by_chain is True.
- max_representative_chains = 100 Only applies if search_method = from_ca. Maximum number of representative chains
- enabled = True Turn on secondary structure restraints (main switch)
- proteinenabled = True Turn on secondary structure restraints for proteinsearch_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.distance_ideal_n_o = 2.9 Target length for N-O hydrogen bonddistance_cut_n_o = 3.5 Hydrogen bond with length exceeding this value will not be establishedremove_outliers = True If true, h-bonds exceeding distance_cut_n_o length will not be establishedrestrain_hbond_angles = True helixserial_number = None helix_identifier = None enabled = True Restrain this particular helixselection = None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles

to a particular valuehbonddonor = None acceptor = None sheetenabled = True Restrain this particular sheetfirst_strand = None sheet_id = None sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuestrandselection = None sense = parallel antiparallel *unknown bond_start_current = None bond_start_previous = None hbonddonor = None acceptor = None

- enabled = True Turn on secondary structure restraints for protein
- search_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.
- distance_ideal_n_o = 2.9 Target length for N-O hydrogen bond
- distance_cut_n_o = 3.5 Hydrogen bond with length exceeding this value will not be established
- remove_outliers = True If true, h-bonds exceeding distance_cut_n_o length will not be established
- restrain_hbond_angles = True
- helixserial_number = None helix_identifier = None enabled = True Restrain this particular helixselection = None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuehbonddonor = None acceptor = None
- serial_number = None
- helix_identifier = None
- enabled = True Restrain this particular helix
- selection = None
- helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.
- sigma = 0.05
- slack = 0
- top_out = False
- angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.
- angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular value
- hbonddonor = None acceptor = None
- donor = None
- acceptor = None
- sheetenabled = True Restrain this particular sheetfirst_strand = None sheet_id = None sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuestrandselection = None sense = parallel antiparallel *unknown bond_start_current = None bond_start_previous = None hbonddonor = None acceptor = None
- enabled = True Restrain this particular sheet
- first_strand = None
- sheet_id = None

- sigma = 0.05
- slack = 0
- top_out = False
- angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.
- angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular value
- strandselection = None sense = parallel antiparallel *unknown bond_start_current = None bond_start_previous = None
- selection = None
- sense = parallel antiparallel *unknown
- bond_start_current = None
- bond_start_previous = None
- hbonddonor = None acceptor = None
- donor = None
- acceptor = None
- nucleic_acidenabled = True Turn on secondary structure restraints for nucleic acidshbond_distance_cutoff = 3.4 Hydrogen bonds with length exceeding this limit will not be establishedscale_bonds_sigma = 1 All sigmas for h-bond length will be multiplied by this number. The smaller number is tighter restraints.base_pairenabled = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residuesaenger_class = 0 Type of base-pairingrestrain_planarity = False Apply planarity restraint to this base-pairplanarity_sigma = 0.176 restrain_hbonds = True Restrain hydrogen bonds length for this base-pairrestrain_hb_angles = True Restrain angles around hydrogen bonds for this base-pairrestrain_parallelity = True Apply parallelity restraint to this base-pairparallelity_target = 0 parallelity_sigma = 0.0335 stacking_pairenabled = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residueangle = 0 sigma = 0.027
- enabled = True Turn on secondary structure restraints for nucleic acids
- hbond_distance_cutoff = 3.4 Hydrogen bonds with length exceeding this limit will not be established
- scale_bonds_sigma = 1 All sigmas for h-bond length will be multiplied by this number. The smaller number is tighter restraints.
- base_pairenabled = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residuesaenger_class = 0 Type of base-pairingrestrain_planarity = False Apply planarity restraint to this base-pairplanarity_sigma = 0.176 restrain_hbonds = True Restrain hydrogen bonds length for this base-pairrestrain_hb_angles = True Restrain angles around hydrogen bonds for this base-pairrestrain_parallelity = True Apply parallelity restraint to this base-pairparallelity_target = 0 parallelity_sigma = 0.0335
- enabled = True Restraint this particular base-pair
- base1 = None Selection string selecting at least one atom in the desired residue
- base2 = None Selection string selecting at least one atom in the desired residue

- saenger_class = 0 Type of base-pairing
- restrain_planarity = False Apply planarity restraint to this base-pair
- planarity_sigma = 0.176
- restrain_hbonds = True Restrain hydrogen bonds length for this base-pair
- restrain_hb_angles = True Restrain angles around hydrogen bonds for this base-pair
- restrain_parallelity = True Apply parallelity restraint to this base-pair
- parallelity_target = 0
- parallelity_sigma = 0.0335
- stacking_pairenabled = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residueangle = 0 sigma = 0.027
- enabled = True Restraint this particular base-pair
- base1 = None Selection string selecting at least one atom in the desired residue
- base2 = None Selection string selecting at least one atom in the desired residue
- angle = 0
- sigma = 0.027
- reference_coordinate_restraintsRestrains coordinates in Cartesian space to stay near their starting positions. This is intended for use in generating simulated annealing omit maps, to prevent refined atoms from collapsing in on the region missing atoms. For conserving geometry quality at low resolution, the more flexible reference model restraints should be used instead.enabled = False exclude_outliers = True selection = all sigma = 0.2 limit = 1 top_out = False reference_is_average_alt_confs = False Sets the reference coordinate to the average of two alt. confs. Ignores selection option.
- enabled = False
- exclude_outliers = True
- selection = all
- sigma = 0.2
- limit = 1
- top_out = False
- reference_is_average_alt_confs = False Sets the reference coordinate to the average of two alt. confs. Ignores selection option.
- automatic_restraintsside_chain_parallelity = False
- side_chain_parallelity = False
- automatic_linkinglink_all = False If True, bond restraints will be generated for any appropriate ligand-protein or ligand-nucleic acid covalent bonds. This includes sugars, amino acid modifications, and other prosthetic groups.link_none = False link_metals = Auto link_residues = True link_amino_acid_rna_dna = False link_carbohydrates = True link_ligands = True link_small_molecules = False metal_coordination_cutoff = 3 amino_acid_bond_cutoff = 1.9 inter_residue_bond_cutoff = 2.2 buffer_for_second_row_elements = 0.5 carbohydrate_bond_cutoff = 1.99 ligand_bond_cutoff = 1.99 small_molecule_bond_cutoff = 1.98 exclude_hydrogens_from_bonding_decisions = False
- link_all = False If True, bond restraints will be generated for any appropriate ligand-protein or ligand-nucleic acid covalent bonds. This includes sugars, amino acid modifications, and other prosthetic

groups.

- link_none = False
- link_metals = Auto
- link_residues = True
- link_amino_acid_rna_dna = False
- link_carbohydrates = True
- link_ligands = True
- link_small_molecules = False
- metal_coordination_cutoff = 3
- amino_acid_bond_cutoff = 1.9
- inter_residue_bond_cutoff = 2.2
- buffer_for_second_row_elements = 0.5
- carbohydrate_bond_cutoff = 1.99
- ligand_bond_cutoff = 1.99
- small_molecule_bond_cutoff = 1.98
- exclude_hydrogens_from_bonding_decisions = False
- include_in_automatic_linkingselection_1 = None selection_2 = None bond_cutoff = 4.5
- selection_1 = None
- selection_2 = None
- bond_cutoff = 4.5
- exclude_from_automatic_linkingselection_1 = None selection_2 = None
- selection_1 = None
- selection_2 = None
- apply_cis_trans_specificationcis_trans_mod = cis *trans residue_selection = None Residues containing C-alpha atom of omega dihedral
- cis_trans_mod = cis *trans
- residue_selection = None Residues containing C-alpha atom of omega dihedral
- apply_cif_restraintsrestraints_file_name = None residue_selection = None
- restraints_file_name = None
- residue_selection = None
- apply_cif_modificationdata_mod = None residue_selection = None
- data_mod = None
- residue_selection = None
- apply_cif_linkdata_link = None residue_selection_1 = None residue_selection_2 = None
- data_link = None
- residue_selection_1 = None
- residue_selection_2 = None

- peptide_linkramachandran_restraints = False !!! OBSOLETED. Kept for backward compatibility only !!! Restrains peptide backbone to fall within allowed regions of Ramachandran plot. Although it does not eliminate outliers, it can significantly improve the percent favored and percent outliers at low resolution. Probably not useful (and maybe even harmful) at resolutions much higher than 3.5A.cis_threshold = 45 apply_all_trans = False discard_omega = False discard_psi_phi = True apply_peptide_plane = False omega_esd_override_value = None rama_weight = 1 scale_allowed = 1 rama_potential = *oldfield emsley rama_selection = None Selection of part of the model for which Ramachandran restraints will be set up.restrain_rama_outliers = True Apply restraints to Ramachandran outliersrestrain_rama_allowed = True Apply restraints to residues in allowed region on Ramachandran plotrestrain_allowed_outliers_with_emsley = False In case of restrain_rama_outliers=True and/or restrain_rama_allowed=True still restrain these residues with emsley. Make sense only in case of using oldfield potential.oldfieldesd = 10 weight_scale = 1 dist_weight_max = 10 weight = None plot_cutoff = 0.027
- ramachandran_restraints = False !!! OBSOLETED. Kept for backward compatibility only !!! Restrains peptide backbone to fall within allowed regions of Ramachandran plot. Although it does not eliminate outliers, it can significantly improve the percent favored and percent outliers at low resolution. Probably not useful (and maybe even harmful) at resolutions much higher than 3.5A.
- cis_threshold = 45
- apply_all_trans = False
- discard_omega = False
- discard_psi_phi = True
- apply_peptide_plane = False
- omega_esd_override_value = None
- rama_weight = 1
- scale_allowed = 1
- rama_potential = *oldfield emsley
- rama_selection = None Selection of part of the model for which Ramachandran restraints will be set up.
- restrain_rama_outliers = True Apply restraints to Ramachandran outliers
- restrain_rama_allowed = True Apply restraints to residues in allowed region on Ramachandran plot
- restrain_allowed_outliers_with_emsley = False In case of restrain_rama_outliers=True and/or restrain_rama_allowed=True still restrain these residues with emsley. Make sense only in case of using oldfield potential.
- oldfieldesd = 10 weight_scale = 1 dist_weight_max = 10 weight = None plot_cutoff = 0.027
- esd = 10
- weight_scale = 1
- dist_weight_max = 10
- weight = None
- plot_cutoff = 0.027
- ramachandran_plot_restraintsenabled = True favored = *oldfield emsley emsley8k phi_psi_2 allowed = *oldfield emsley emsley8k phi_psi_2 outlier = *oldfield emsley emsley8k phi_psi_2 selection = None Selection of part of the model for which Ramachandran restraints will be set up.inject_emsley8k_into_oldfield_favored = True Backdoor to disable temporary dirty hack to use

botholdfieldweight = 0 Direct weight value. If 0 the weight will be calculated as following: (w, op.esd, op.dist_weight_max, 2.0, op.weight_scale) 1 / esd^2 * max(2.0, min(current_distance_to_allowed, dist_weight_max)) * weight_scale max(2.0, current_distance_to_allowed) 1 / esd^2 * weight_scale * max(distance_to_allowed_cutoff, current_distance_to_allowed) weight_scale(=0.01) * max(distance_weight_min(=2.), min(distance_weight_max(=10.), current_distance_to_allowed))weight_scale = 0.01 distance_weight_min = 2 minimum coefficient when scaling depending on how far the residue is from allowed region.distance_weight_max = 10 maximum coefficient when scaling depending on how far the residue is from allowed region.plot_cutoff = 0.027 emsleyweight = 1 scale_allowed = 1 emsley8kweight_favored = 5 weight_allowed = 10 weight_outlier = 10 phi_psi_2favored_strategy = *closest highest_probability random weighted_random allowed_strategy = *closest highest_probability random weighted_random outlier_strategy = *closest highest_probability random weighted_random

- enabled = True
- favored = *oldfield emsley emsley8k phi_psi_2
- allowed = *oldfield emsley emsley8k phi_psi_2
- outlier = *oldfield emsley emsley8k phi_psi_2
- selection = None Selection of part of the model for which Ramachandran restraints will be set up.
- inject_emsley8k_into_oldfield_favored = True Backdoor to disable temporary dirty hack to use both
- oldfieldweight = 0 Direct weight value. If 0 the weight will be calculated as following: (w, op.esd, op.dist_weight_max, 2.0, op.weight_scale) 1 / esd^2 * max(2.0, min(current_distance_to_allowed, dist_weight_max)) * weight_scale max(2.0, current_distance_to_allowed) 1 / esd^2 * weight_scale * max(distance_to_allowed_cutoff, current_distance_to_allowed) weight_scale(=0.01) * max(distance_weight_min(=2.), min(distance_weight_max(=10.), current_distance_to_allowed))weight_scale = 0.01 distance_weight_min = 2 minimum coefficient when scaling depending on how far the residue is from allowed region.distance_weight_max = 10 maximum coefficient when scaling depending on how far the residue is from allowed region.plot_cutoff = 0.027
- weight = 0 Direct weight value. If 0 the weight will be calculated as following: (w, op.esd, op.dist_weight_max, 2.0, op.weight_scale) 1 / esd^2 * max(2.0, min(current_distance_to_allowed, dist_weight_max)) * weight_scale max(2.0, current_distance_to_allowed) 1 / esd^2 * weight_scale * max(distance_to_allowed_cutoff, current_distance_to_allowed) weight_scale(=0.01) * max(distance_weight_min(=2.), min(distance_weight_max(=10.), current_distance_to_allowed))
- weight_scale = 0.01
- distance_weight_min = 2 minimum coefficient when scaling depending on how far the residue is from allowed region.
- distance_weight_max = 10 maximum coefficient when scaling depending on how far the residue is from allowed region.
- plot_cutoff = 0.027
- emsleyweight = 1 scale_allowed = 1
- weight = 1
- scale_allowed = 1
- emsley8kweight_favored = 5 weight_allowed = 10 weight_outlier = 10
- weight_favored = 5
- weight_allowed = 10

- weight_outlier = 10
- phi_psi_2favored_strategy = *closest highest_probability random weighted_random allowed_strategy = *closest highest_probability random weighted_random outlier_strategy = *closest highest_probability random weighted_random
- favored_strategy = *closest highest_probability random weighted_random
- allowed_strategy = *closest highest_probability random weighted_random
- outlier_strategy = *closest highest_probability random weighted_random
- rna_sugar_pucker_analysisbond_min_distance = 1.2 bond_max_distance = 1.8 epsilon_range_min = 155 epsilon_range_max = 310 delta_range_2p_min = 129 delta_range_2p_max = 162 delta_range_3p_min = 65 delta_range_3p_max = 104 p_distance_c1p_outbound_line_2p_max = 2.9 o3p_distance_c1p_outbound_line_2p_max = 2.4 bond_detection_distance_tolerance = 0.5 enable = True
- bond_min_distance = 1.2
- bond_max_distance = 1.8
- epsilon_range_min = 155
- epsilon_range_max = 310
- delta_range_2p_min = 129
- delta_range_2p_max = 162
- delta_range_3p_min = 65
- delta_range_3p_max = 104
- p_distance_c1p_outbound_line_2p_max = 2.9
- o3p_distance_c1p_outbound_line_2p_max = 2.4
- bond_detection_distance_tolerance = 0.5
- enable = True
- show_histogram_slotsbond_lengths = 5 nonbonded_interaction_distances = 5 bond_angle_deviations_from_ideal = 5 dihedral_angle_deviations_from_ideal = 5 chiral_volume_deviations_from_ideal = 5
- bond_lengths = 5
- nonbonded_interaction_distances = 5
- bond_angle_deviations_from_ideal = 5
- dihedral_angle_deviations_from_ideal = 5
- chiral_volume_deviations_from_ideal = 5
- show_max_itemsnot_linked = 5 bond_restraints_sorted_by_residual = 5 nonbonded_interactions_sorted_by_model_distance = 5 bond_angle_restraints_sorted_by_residual = 5 dihedral_angle_restraints_sorted_by_residual = 3 chirality_restraints_sorted_by_residual = 3 planarity_restraints_sorted_by_residual = 3 residues_with_excluded_nonbonded_symmetry_interactions = 12 fatal_problem_max_lines = 10
- not_linked = 5
- bond_restraints_sorted_by_residual = 5
- nonbonded_interactions_sorted_by_model_distance = 5

- bond_angle_restraints_sorted_by_residual = 5
- dihedral_angle_restraints_sorted_by_residual = 3
- chirality_restraints_sorted_by_residual = 3
- planarity_restraints_sorted_by_residual = 3
- residues_with_excluded_nonbonded_symmetry_interactions = 12
- fatal_problem_max_lines = 10
- ncs_groupThe definition of one NCS group. Note, that almost always in refinement programs they will be checked and filtered if needed.reference = None Residue selection string for the complete master NCS copyselection = None Residue selection string for each NCS copy location in ASU
- reference = None Residue selection string for the complete master NCS copy
- selection = None Residue selection string for each NCS copy location in ASU
- ncs_searchSet of parameters for NCS search procedure. Some of them also used for filtering user-supplied ncs_group.enabled = True Enable NCS restraints or constraints in refinement (in some cases may be switched on inside refinement program).exclude_selection = water Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS.chain_similarity_threshold = 0.85 Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residueschain_max_rmsd = 999 limit of rms difference between chains to be considered as copiesresidue_match_radius = 1000 Maximum allowed distance difference between pairs of matching atoms of two residuestry_shortcuts = True Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.minimum_number_of_atoms_in_copy = 3 Do not create ncs groups where master and copies would contain less than specified amount of atomsvalidate_user_supplied_groups = True Enable validation of user-supplied ncs_group. Need to exercise a lot of caution turning this off. This option is for developers only.
- enabled = True Enable NCS restraints or constraints in refinement (in some cases may be switched on inside refinement program).
- exclude_selection = water Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS.
- chain_similarity_threshold = 0.85 Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residues
- chain_max_rmsd = 999 limit of rms difference between chains to be considered as copies
- residue_match_radius = 1000 Maximum allowed distance difference between pairs of matching atoms of two residues
- try_shortcuts = True Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.
- minimum_number_of_atoms_in_copy = 3 Do not create ncs groups where master and copies would contain less than specified amount of atoms
- validate_user_supplied_groups = True Enable validation of user-supplied ncs_group. Need to exercise a lot of caution turning this off. This option is for developers only.
- clash_guardnonbonded_distance_threshold = 0.5 max_number_of_distances_below_threshold = 100 max_fraction_of_distances_below_threshold = 0.1
- nonbonded_distance_threshold = 0.5

- max_number_of_distances_below_threshold = 100
- max_fraction_of_distances_below_threshold = 0.1
- geometry_restraintseditsexcessive_bond_distance_limit = 10 bondaction = *add delete change atom_selection_1 = None atom_selection_2 = None symmetry_operation = None The bond is between atom_1 and symmetry_operation * atom_2, with atom_1 and atom_2 given in fractional coordinates. Example: symmetry_operation = -x-1,-y,zdistance_ideal = None sigma = None slack = None limit = -1.0 top_out = False angleaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None angle_ideal = None sigma = None dihedralaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None atom_selection_4 = None angle_ideal = None alt_angle_ideals = None sigma = None periodicity = 1 planarityaction = *add delete change atom_selection = None sigma = None parallelityaction = *add delete change atom_selection_1 = None atom_selection_2 = None sigma = 0.027 target_angle_deg = 0 removeangles = None dihedrals = None chiralities = None planarities = None parallelities = None
- editsexcessive_bond_distance_limit = 10 bondaction = *add delete change atom_selection_1 = None atom_selection_2 = None symmetry_operation = None The bond is between atom_1 and symmetry_operation * atom_2, with atom_1 and atom_2 given in fractional coordinates. Example: symmetry_operation = -x-1,-y,zdistance_ideal = None sigma = None slack = None limit = -1.0 top_out = False angleaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None angle_ideal = None sigma = None dihedralaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None atom_selection_4 = None angle_ideal = None alt_angle_ideals = None sigma = None periodicity = 1 planarityaction = *add delete change atom_selection = None sigma = None parallelityaction = *add delete change atom_selection_1 = None atom_selection_2 = None sigma = 0.027 target_angle_deg = 0
- excessive_bond_distance_limit = 10
- bondaction = *add delete change atom_selection_1 = None atom_selection_2 = None symmetry_operation = None The bond is between atom_1 and symmetry_operation * atom_2, with atom_1 and atom_2 given in fractional coordinates. Example: symmetry_operation = -x-1,-y,zdistance_ideal = None sigma = None slack = None limit = -1.0 top_out = False
- action = *add delete change
- atom_selection_1 = None
- atom_selection_2 = None
- symmetry_operation = None The bond is between atom_1 and symmetry_operation * atom_2, with atom_1 and atom_2 given in fractional coordinates. Example: symmetry_operation = -x-1,-y,z
- distance_ideal = None
- sigma = None
- slack = None
- limit = -1.0
- top_out = False
- angleaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None angle_ideal = None sigma = None
- action = *add delete change
- atom_selection_1 = None
- atom_selection_2 = None

- atom_selection_3 = None
- angle_ideal = None
- sigma = None
- dihedralaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None atom_selection_4 = None angle_ideal = None alt_angle_ideals = None sigma = None periodicity = 1
- action = *add delete change
- atom_selection_1 = None
- atom_selection_2 = None
- atom_selection_3 = None
- atom_selection_4 = None
- angle_ideal = None
- alt_angle_ideals = None
- sigma = None
- periodicity = 1
- planarityaction = *add delete change atom_selection = None sigma = None
- action = *add delete change
- atom_selection = None
- sigma = None
- parallelityaction = *add delete change atom_selection_1 = None atom_selection_2 = None sigma = 0.027 target_angle_deg = 0
- action = *add delete change
- atom_selection_1 = None
- atom_selection_2 = None
- sigma = 0.027
- target_angle_deg = 0
- removeangles = None dihedrals = None chiralities = None planarities = None parallelities = None
- angles = None
- dihedrals = None
- chiralities = None
- planarities = None
- parallelities = None
- simulated_annealingstart_temperature = 5000 final_temperature = 300 cool_rate = 100 number_of_steps = 50 time_step = 0.0005 initial_velocities_zero_fraction = 0 interleave_minimization = False verbose = -1 n_print = 100 update_grads_shift = 0.3 random_seed = None
- start_temperature = 5000
- final_temperature = 300
- cool_rate = 100
- number_of_steps = 50

- time_step = 0.0005
- initial_velocities_zero_fraction = 0
- interleave_minimization = False
- verbose = -1
- n_print = 100
- update_grads_shift = 0.3
- random_seed = None
- outputsuffix = "_real_space_refined" Suffix string added to automatically generated output filenames
- suffix = "_real_space_refined" Suffix string added to automatically generated output filenames
- guiGUI-specific parameter required for output directoryoutput_dir = None run_validation = True
- output_dir = None
- run_validation = True

Reducing resolution of cryo-EM maps by adding random noise

reduce_cryoem_resolution.html

Author

- Randy J Read

Purpose

The reduce_cryoem_resolution tool constructs a realistic set of lower-resolution maps by adding random noise to the half-maps. Noise is added as a function of resolution of the Fourier terms so that the resulting half-map FSC curve looks typical of an experimental map at that resolution.

How reduce_cryoem_resolution works

A study of hundreds of FSC curves from maps in the EMDB at different resolutions showed that these FSC curves could be approximated very well with a simple functional form that depends only on the nominal resolution of the map. To reduce the resolution to match the generic curve requires evaluating the signal to noise ratio from the input half-maps, than adding resolution-dependent random noise to produce the desired FSC curve. To avoid including any information beyond the target resolution, the Fourier terms are filtered at the target resolution, where the half-map FSC is 0.143. The two half-maps and the corresponding averaged full map are produced, in addition to an optional plot showing half-map FSC curves for the original and reduced-resolution maps.

Running reduce_cryoem_resolution

This can be controlled by an input script using the phil syntax, either on the command line or in a phil file. The command to take two half-maps and limit a starting resolution of 3A to 4A, while producing the FSC plot, looks like this:

```
phenix.reduce_cryoEM_resolution hm1.mrc hm2.mrc d_min=3.0 target_d_min=4.0 plot=True
```

Note that the starting d_min is optional, but is useful if you want to compare the maps over their full resolution ranges.

It is recommended to start with half-maps when they are available, but it is also possible to reduce the resolution of a single full map. Here it is assumed that the map is close to perfect at the new resolution limit.

```
phenix.reduce_cryoEM_resolution full.mrc target_d_min=4.0
```

List of all available keywords

d_min = None Nominal resolution of input map(s) (Angstroms). Optional to see initial FSCtarget_d_min = None Desired high-resolution limit (Angstroms)mute = False Flag to mute standard outputmap_modelfull_map = None Input full map filehalf_map = None Input half map filesoutput_filesoutput_dir = "." Directory where output files will be writtenfile_root = "reduced_resolution" Filename root for resulting output map file(s)plot = False Flag to make and save FSC plot

- d_min = None Nominal resolution of input map(s) (Angstroms). Optional to see initial FSC
- target_d_min = None Desired high-resolution limit (Angstroms)
- mute = False Flag to mute standard output

- map_modelfull_map = None Input full map filehalf_map = None Input half map files
- full_map = None Input full map file
- half_map = None Input half map files
- output_filesoutput_dir = "." Directory where output files will be writtenfile_root = "reduced_resolution" Filename root for resulting output map file(s)plot = False Flag to make and save FSC plot
- output_dir = "." Directory where output files will be written
- file_root = "reduced_resolution" Filename root for resulting output map file(s)
- plot = False Flag to make and save FSC plot

Density modification of cryo EM maps with resolve_cryo_em

resolve_cryo_em.html

Author(s)

- resolve_cryo_em: Tom Terwilliger

Purpose

The routine resolve_cryo_em is a tool for carrying out density modification of cryo-EM maps

Usage

Density modification with resolve_cryo_em is carried out using two half-maps, along with the FSC-based resolution and a sequence file specifying the contents of the map.

Density modification can also be carried out by using the initial density-modified map or a single supplied model as a basis for generating multiple models, then using model-based density as part of the density modification target to yield a model-based density modified map.

Video Tutorial

The tutorial video is available on the Phenix YouTube channel and covers the following topics:

- basic overview
- how to run resolve_cryo_em via the GUI

How resolve_cryo_em works:

Density modification with resolve_cryo_em is based on two ideas. One is that the errors in Fourier coefficients representing a cryo-EM map are (to some extent) uncorrelated. This means that one Fourier coefficient does not know about the errors in another one. (Note that this is not including errors that are correlated simply because the molecule is small and is placed in a large map. Correlated errors in this context are those where one Fourier coefficient has been adjusted to compensate for errors in another one.)

The other is that some features in a map are known in advance. This could include features such as the flatness of the solvent region, distributions of map values in the solvent and macromolecule region, similarities of symmetry-related regions.

Then the way density modification works is that Fourier coefficients for the map are adjusted to agree both with the original map and with the expected features. This improves the Fourier coefficients, and the key result is that the map improves everywhere, not just where the information about expected features was available.

Unique features of density modification for cryo-EM are that two half-maps with independent errors are available in cryo-EM (allowing estimation of errors), and that the errors in Fourier coefficients are (more or less) distributed as two-dimensional Gaussians (i.e both phase and amplitude errors). This leads to many differences in implementation density modification in crystallography though core elements are identical.

Using resolve_cryo_em:

Normally you will access the functionality of resolve_cryo_em by running the ResolveCryoEM tool in the Phenix GUI. You can also run it from the command line. You may wish to run it from the command line in background with multiprocessing if you are running with denmod_with_models=True as this can take a long time (perhaps 1 day x 16 processors if you have 250 residues in the unique part of your model).

Half-maps: Supply two unmasked half maps. Half maps are maps created using half of the data. They are full-size, but just contain less accurate information. Normally half-maps are named something like my_map_1.ccp4 and my_map_2.ccp4, for half maps 1 and 2. The half-maps can be sharpened but it does not make much of a difference. They must be unmasked and they must still have noise in the solvent region. They can be boxed with a soft boundary (if they are boxed specify box_before_analysis=False so it isn't done again)

Sequence file: Supply a sequence file with the sequence of the molecule. Be sure to put in all copies of the molecule (i.e. a 24-mer needs 24 chains). (You can also supply one copy of the molecule and specify copies=24 if you want.)

Model file: You can supply a model; if you do then the model will be randomized and refined against the half-maps to yield 8 models for each half map. These will be used in model-based density modification.

Automatically-generated models: You can specify density_modify_with_model=True and not supply any model. In this case 8 models will be automatically generated for each half-map and used in model-based density modification.

Procedure used by resolve_cryo_em

The basic inputs to resolve_cryo_em are:

```
Two unmasked half-maps
sequence file or molecular mass or solvent fraction
```

Optional inputs are:

```
Full map (Recombined with density-modified half maps)
Model (Used to generate multi-model representation for density modification)
Symmetry file (Supply with model or if automatic model generation is
 used to limit the modeling to the unique part of the map. Alternatively,
 used in density modification in cases where the symmetry is not part
 of the reconstruction process).
Mask file (Used to identify the region where the macromolecule is present)
```

The procedure used by resolve_cryo_em has several steps:

```
Boxing of maps: If the supplied maps are much larger than the molecule,
the maps are trimmed down to about 5 A bigger than the largest dimension
of the molecule (estimated from a low-res mask and the molecular
volume based on sequence or as specified) in each direction, using
a soft mask at the edges of the box.

Resolution estimate and half-map sharpening of maps: The half-maps are
compared as a function of resolution and the resolution (FSC=0.143)
is estimated and the maps are sharpened based on the estimated map quality
of the full (averaged) map. A full map is calculated.

Generation of map-value (density) histograms: The full map is analyzed
to identify the distribution of map values in the solvent and
macromolecule region. These histograms are optionally used in density
modification.

Map-phasing of half-maps: Each half map is used in a process of
map-based estimation of new Fourier coefficients using a
maximum-likelihood procedure. In essence, one Fourier coefficient is
```

```
removed from the map at a time. Then a new value of that Fourier coefficient
is found that maximizes the likelihood of the map given all the other
Fourier coefficients. The likelihood is calculated from the agreement of
the histograms of the (new) map with expected histograms. For example,
if the solvent region is expected to be flat, then the new map has a
high likelihood if it is flat in the solvent region. By varying the one
Fourier coefficient the flatness of the solvent region will change. Similarly
the histograms of density in the region of the macromolecule will change
depending on the value of the one Fourier coefficient. The best value for
each Fourier coefficient is then used to calculated a map-phasing map.

Estimation of errors: Fourier coefficients for the two starting
half-maps and the two density-modified maps are compared to give FSC
values as a function of resolution. These FSC values are used to estimate
correlated and uncorrelated errors in the four maps and to identify
optimal weighting between original and density-modified maps.

Recombination of original and map-phasing half-maps: Based on the
estimated errors in original and map-phasing half-maps, all four maps
are recombined to create a new density-modified map. Additionally,
each half-map and associated map-phasing half map are recombined to
create two new density-modified half-maps.

Optional real-space and sigma weighting: The smoothed local rms differences
between original half maps and between density-modified half maps are
used (optionally) to identify location-specific weighting for the
original and density-modified maps. The variance of Fourier coefficients
among the four maps are used (optionally) to weight individual final
Fourier coefficients.

Optional starting with unsharpened maps. The input half maps are used in the
density modification step without adjusting the resolution dependence
of the maps with half-map sharpening. This can be useful
for high-resolution maps. Note that whether or not this option is
selected, normally an overall B-factor is still applied to the input maps
based on the nominal resolution.  The difference therefore is in the
details of the resolution dependence which is adjusted in the default
case based on a comparison of half-maps, but not with density
modify unsharpened maps. The overall B-factor
applied before density modification is controlled by the remove_aniso
and b_iso keywords.  By default remove_aniso=True and b_iso is
target_b_ratio*resolution (target_b_ratio=10), so before density
modification whatever resolution dependence and anisotropy is present in
the each map is adjusted to yield an overall B of b_iso and no anisotropy.

Optional alternative final sharpening of maps. The final sharpening normally
consists of two parts. One is scaling the map in shells of resolution
based on the estimated correlation of the map coefficients at each
resolution with the true ones. This is controlled by the
keyword final_scale_with_fsc_ref. The se...
```

Procedure used by resolve_cryo_em for density modification with model-building

Density modification with model-building adds additional cycles to the density modification procedure in which multiple models are built using map_to_model and the averaged density and uncertainty in the average density is used to combine the model density with the initial density-modified map.

The procedure includes:

```
Create initial density-modified half-maps and full map

Create N (typically 16) variants of the full map by changing the resolution
cutoff, spectral_scaling, and blurring of the map.

Build a model into each modified full map. If symmetry file supplied,
 build only the unique part of the model.
```

```
Refine some of the models against half-map 1 and some against half-map 2

Create one composite model based on all models

Create model density for each half-map based on the models refined againt
that map. This model density will have a mean value and variance for each
point in the map near to at least 3 models.

Create composite density for each half-maps by combining the model density
with the density-modified half-map, weighting the model density according to
its consistency among models.

Density-modify each composite half-map, and create a new set of density
modified half-maps and full map, as in the procedure for standard
density modification.

Sharpen the resulting maps using model-based sharpening with the composite
model.

Note that this procedure takes a lot of computation.
```

Note on R-values in density modification

The R-value for density modification is quite different from a refinement R value or Rfree value. Basically the statistical (maximum-likelihood) density modification procedure allows calculation of each structure factor based on the values of all the other structure factors and the expectations about the map (i.e, flat solvent, distribution of density values in protein, ncs, etc.) These map-phasing structure factors are (at least in the first cycle) independent of the starting structure factors and can be compared with the measured data and an R-value obtained. The values of these R-values range from about 0.25 to 0.5 in most cases, and when the map is very good usually the R values are smaller. These R values do not involve any atomic model so they are quite unlike a refinement R.

For discussion of how map-likelihood structure factors are calculated in Resolve, see the Map-likelihood phasing reference below.

Examples

Standard run of resolve_cryo_em:

You can use resolve_cryo_em to density-modify a cryo-EM map:

```
phenix.resolve_cryo_em half_map_A.mrc half_map_B.mrc seq_file=seq.dat
```

What to expect with density modification

If density modification is working well, you should see:

```
An improvement in reported resolution. Typically this is small..from 0.05 A
to 0.5 A. If it is zero...it probably did not work.

More fine detail. Density modification mostly changes
the high-resolution part of your map. (The low-resolution part of your map
is generally already very good, and usually if it is not then density
modification can't fix it. Occasionally low and high-resolution aspects
of a map can improve.) Compare your original (auto-sharpened) map with
your density-modified map and see if side chains and carbonyl oxygens are more
clear and if the connectivity of the chain improves.

If you supply or generate models, you should see improved high-resolution
detail in the region of the model. You are not likely to see improvement
```

```
away from the model as you would in crystallography. The multiple models
generated are basically identifying what locations in your map are
compatible with being the location of an atom, based on the density already
there and the geometrical restrictions on a model. That information then
is included in density modification and improves density in the region of
the model but is not strong enough to improve density elsewhere.
```

What to try to improve your map

Usually you should first just run with all defaults, supplying two half-maps and a sequence file and nominal resolution. See how the map looks and use it as a baseline. Then some things to try to improve the map are:

```
Add a full map file to recombine it with density-modified half-maps. This
can be helpful if your full map has been processed in a way that makes it
much better than an average of your half-maps.

Add a mask file to exclude part of the map. If your map has some huge noise
peaks or just density that is not part of your molecule, you can try to
exclude it by supplying a mask file (a map file with 1 where your molecule is
located and 0 elsewhere).

If you have a model, you can supply the model and it will be used to generate
a multi-model representation of your molecule. This multi-model representation
will be used in density modification. Note that you should supply a
symmetry file if there is symmetry in your reconstruction so that only
the unique part needs to be analyzed.

If you don't have a model but your resolution is high (about 3.5 A or better),
you can improve your map by automatically generating a multi-model
representation of your structure. This is used in the same way as the
multi-model representation starting with a supplied model, and a symmetry
file is helpful to identify just the unique part of the molecule.

You can try changing
the resolution (overall) and the density-modification resolution. Normally
the resolution is the gold-standard resolution from comparing your half-maps.
However varying this may help. The density-modification resolution is
normally set automatically to be somewhat finer than the overall resolution,
but its exact value does have an effect so you could try different values.

You can try changing some of the defaults that occasionally help. One is
setting density_modify_unsharpened_maps=True (density modifies the original
half-maps without auto-sharpening them). Another is setting
final_scale_with_fsc_ref=False (applies final resolution-dependent scaling
that is part-way between standard half-map FSC-based sharpening and
constant amplitudes over all resolutions).
These usually are best for high_resolution maps (&lt; 3 A).

Real_space_weighting, sigma_weighting, and spectral_scaling may improve the
final map. Real-space weighting weights original and density-modified maps
based on their local uncertainties. Sigma-weighting weights them by
uncertainties in individual Fourier coefficients. Spectral scaling scales the
final map by resolution to match the resolution-dependence of a typical
protein.
```

How to tell how well density modification is working

Density modification changes the sharpening of your map, so just because the map looks better or worse doesn't necessarily mean that anything important has happened.

If you have two maps and you want to know which is the better one, here are some things you can do:

```
1. Create a matched version of your maps that have the same
resolution-dependence using phenix.map_sharpening. Run it like this:
```

```
phenix.map_sharpening n_bins=100 auto_sharpen_methods=external_map_sharpening
 external_map_file=target_map.map map_to_change.map resolution=4.1
 sharpened_map_file=map_to_change_matched.map

Now you can look at target_map.map and map_to_change_matched.map and
differences you see are due to intrinsic properties of the maps, not just
sharpening.

2. If you have a good model, refine the model against each map. Then use
phenix.mtriage to estimate the resolution where the FSC(map,model)==0.5
for each map. The map with a lower value of this resolution may be better.

3. As in #2, once you have a pair of maps and a model refined against
each map, you can run phenix.map_sharpening with each map and model and
it will print out the average FSC for each. The one with the higher
overall FSC may be better.
```

Output files from cryo EM density modification

The principal output files from ResolveCryoEM cryo-EM density modification are:

```
denmod_map.ccp4 -- a density-modified (improved) map
denmod_resolution_map.ccp4 -- a map showing the resolution of denmod_map.ccp4
```

You can view the density-modified map (denmod_map.ccp4), colored based on the resolution from denmod_resolution_map.ccp4, using ChimeraX.

Note that ResolveCryoEM may output a model called "merged_model.pdb" if you run it using a model. This merged_model.pdb is not necessarily an improved model. It is just a merged version of the input models. If you do not supply a model ResolveCryoEM will not output any models.

What to do next after cryo EM density modification

You can use your density-modified cryo-EM map anywhere that you need your clearest map (refinement, model-building, ligand fitting, validation). Do keep in mind that this is map can have some bias, or systematic errors. For example density that is in the solvent region can be flattened somewhat. If a model is used in density modification, be careful in interpreting any features that are not present at all without density modification.

Often the next step after resolve_cryo_em will be model-building with predict_and_build or map_to_model. Alternatively you might carry out docking with EmPlacement or DockInMap. You might also try map sharpening with the map_sharpening tool and compare that map with the density-modified one.

Possible Problems

If the half-maps have been masked the procedure may not work well.

If the solvent noise is very non-uniform the procedure may work poorly. By default a rectangular solid region enclosing the molecule is cut out and used in density modification. You can supply a boxed map and set the keyword box_before_analysis=False to avoid this.

If the maps have very prominent density away from the macromolecule this may interfere with density modification. You can get around this by supplying a mask (as a map, 1=inside the molecule).

If there is non-macromolecule but real density in the maps this may interfere with density modification (for example, lipid density).

Specific limitations and problems:

If your computer does not have a lot of memory and you use multiple processors (nproc bigger than 1) then resolve_cryo_em can crash. If this happens try nproc=1.

Density modification introduces some correlations between half-maps due to solvent flattening. This can have a small effect on the resolution estimates obtained with half-map FSC. The resolution estimates provided by the program are corrected for this effect.

If you use the real_space_weighting or sigma_weighting or sharpening_type=local_final_half_map options there may be some extra correlations between half-maps introduced. Calculating resolution using FSC between these density-modified maps can lead to overstating the resolution. The resolution estimates provided by the program are before applying these weighting schemes (unless you specify local_methods_final_cycle = False and run multiple cycles) so they are not normally affected by this.

The density modification procedure works best in the resolution range of about 4.5 A or better, but occasionally works very well at lower resolutions (not tested beyond 9 A). The procedure does not work in every case (we do not have a great metric for success but about half of cases appear to be improved according to the metrics described above). Varying the parameters can have a substantial effect on the outcome.

Model-based density modification necessarily biases the map towards the models that are built. By building multiple models, the effect of this bias is reduced but not eliminated. For example if the starting map has an error that causes models to be built with a side chain the wrong place, the new model- based density will show even more density in that location. That is, this process can accentuate errors in the original maps if they match plausible locations of atoms in a model. Balancing this, the procedure generally does not appear to yield density at positions where there is no density in the original map even if atoms are located there in a model. It is important that the original or non-model-based maps be consulted to evaluate any specific density in the map.

Literature

Improvement of cryo-EM maps by density modification. T.C. Terwilliger, S.J. Ludtke, R.J. Read, P.D. Adams, and Afonine. Nature Methods 17, 923-927 (2020). Map-likelihood phasing. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 57, 1763-75 (2001).

- Improvement of cryo-EM maps by density modification. T.C. Terwilliger, S.J. Ludtke, R.J. Read, P.D. Adams, and Afonine. Nature Methods 17, 923-927 (2020).
- Map-likelihood phasing. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 57, 1763-75 (2001).

Additional information

List of all available keywords

job_title = None Job title in PHENIX GUI, not used on command lineinput_fileshalf_map_file_name_1 = None Half map file namehalf_map_file_name_2 = None Half map file namemap_file_name = None Optional map file name. If you supply a full map it will be used in recombination with the density modified half maps. If you have a good full map this may be useful.target_map_file_name = None Input target map file namemask_file_name = None Input mask file name (as map file with 1 for macromolecule, 0 solvent, smoothly varying between 1 and 0). Used instead of automatic mask to define region where map correlations are calculated.model_1 = None Model to be randomized to generate multiple-model representation of structure in density modification. Normally eight models will be generated for each half-map. If multiple model_1 and model_2 files are supplied then they will be used directly and should be refined against half_map_1 or denmod_half_map_1 before supplying them. Normally supply at least minimum_model_files corresponding to half-map 1 and the same for half-map 2. Generated automatically if density_modify_with_model is set and no model is supplied. If supplied, at least minimum_model_files must be supplied for each half-map.model_2 = None Model corresponding to half-map 2 (unique part of model) to use in density modification. All model_2 models should be refined against half_map_2 or denmod_half_map_2 before supplying them. Normally supply at least minimum_model_files corresponding to half-map 2 and the same for half-map 1. Generated automatically if density_modify_with_model is set.minimum_model_files = 3 Minimum number of models representing half-maps 1 and 2. NOTE: must be at least 2.denmod_map_file_name = None Optional input density-modified full map. Normally generated automaticallydenmod_half_map_file_name_1 = None Optional input density-modified half map file 1. Normally generated automaticallydenmod_half_map_file_name_2 = None Optional input density-modified half map file 2. Normally generated automaticallytruncate_density_file_name = None Input model file containing atomic coordinates around which density is to be truncated at 2 sigma in density modification. This can be used to truncate high density for heavy atoms or high density that is away from the macromolecule so that they do not interfere with density modification. Note that this means that the density in these regions will typically be much lower after density modification. See also rad_mask, the radius around the coordinates to mask.rad_mask = 5 Radius around truncate_density atom positions to truncate density in density modificationhistograms_file_name = None Input histograms file name. Normally this file is created automaticallyguess_histograms_path = True If histograms path is too long, try to guess the path based on the resolve default path and the working directorysymmetry_file = None Symmetry file used to apply symmetry during density modification and model-building. Required if density_modify_with_model is set and ncs_copies is greater than 1. Normally not used as NCS is part of the map symmetryseq_file = None Sequence file used to estimate volume occupied by the molecule and to generate models (if density_modify_with_model is set) . Either supply all copies of each chain or supply the unique part of the sequence file and use ncs_copies to specify how many copies of this are present. If ncs_copies = 1 or None, then the seq_file must include all copies (i.e, if a dimer, must have two copies of the chain) and no NCS will be applied. (Fasta format or sequences separated by blank lines)output_filespdb_out = None Merged model (composite model obtained from model-building steps)initial_map_file_name = initial_map.ccp4 Output scaled original map file name. Only written out if apply_cc_star_to_initial_map is True. (Otherwise the initial map is highly sharpened and not suitable for viewing).model_density_map_file_name = model_density_12.ccp4 Output map containing densit...

- job_title = None Job title in PHENIX GUI, not used on command line
- input_fileshalf_map_file_name_1 = None Half map file namehalf_map_file_name_2 = None Half map file namemap_file_name = None Optional map file name. If you supply a full map it will be used in recombination with the density modified half maps. If you have a good full map this may be useful.target_map_file_name = None Input target map file namemask_file_name = None Input mask file name (as map file with 1 for macromolecule, 0 solvent, smoothly varying between 1 and 0). Used instead of automatic mask to define region where map correlations are calculated.model_1 = None Model to be randomized to generate multiple-model representation of structure in density modification. Normally eight models will be generated for each half-map. If multiple model_1 and model_2 files are supplied then they will be used directly and should be refined against half_map_1 or denmod_half_map_1 before supplying them. Normally supply at least minimum_model_files corresponding to half-map 1 and the same for half-map 2. Generated automatically if density_modify_with_model is set and no model is supplied. If supplied, at least minimum_model_files must be supplied for each half-map.model_2 = None Model corresponding to half-map 2 (unique part of model) to use in density modification. All model_2 models should be refined against half_map_2 or denmod_half_map_2 before supplying them. Normally supply at least minimum_model_files corresponding to half-map 2 and the same for half-map 1. Generated

automatically if density_modify_with_model is set.minimum_model_files = 3 Minimum number of models representing half-maps 1 and 2. NOTE: must be at least 2.denmod_map_file_name = None Optional input density-modified full map. Normally generated automaticallydenmod_half_map_file_name_1 = None Optional input density-modified half map file 1. Normally generated automaticallydenmod_half_map_file_name_2 = None Optional input density-modified half map file 2. Normally generated automaticallytruncate_density_file_name = None Input model file containing atomic coordinates around which density is to be truncated at 2 sigma in density modification. This can be used to truncate high density for heavy atoms or high density that is away from the macromolecule so that they do not interfere with density modification. Note that this means that the density in these regions will typically be much lower after density modification. See also rad_mask, the radius around the coordinates to mask.rad_mask = 5 Radius around truncate_density atom positions to truncate density in density modificationhistograms_file_name = None Input histograms file name. Normally this file is created automaticallyguess_histograms_path = True If histograms path is too long, try to guess the path based on the resolve default path and the working directorysymmetry_file = None Symmetry file used to apply symmetry during density modification and model-building. Required if density_modify_with_model is set and ncs_copies is greater than 1. Normally not used as NCS is part of the map symmetryseq_file = None Sequence file used to estimate volume occupied by the molecule and to generate models (if density_modify_with_model is set) . Either supply all copies of each chain or supply the unique part of the sequence file and use ncs_copies to specify how many copies of this are present. If ncs_copies = 1 or None, then the seq_file must include all copies (i.e, if a dimer, must have two copies of the chain) and no NCS will be applied. (Fasta format or sequences separated by blank lines)

- half_map_file_name_1 = None Half map file name
- half_map_file_name_2 = None Half map file name
- map_file_name = None Optional map file name. If you supply a full map it will be used in recombination with the density modified half maps. If you have a good full map this may be useful.
- target_map_file_name = None Input target map file name
- mask_file_name = None Input mask file name (as map file with 1 for macromolecule, 0 solvent, smoothly varying between 1 and 0). Used instead of automatic mask to define region where map correlations are calculated.
- model_1 = None Model to be randomized to generate multiple-model representation of structure in density modification. Normally eight models will be generated for each half-map. If multiple model_1 and model_2 files are supplied then they will be used directly and should be refined against half_map_1 or denmod_half_map_1 before supplying them. Normally supply at least minimum_model_files corresponding to half-map 1 and the same for half-map 2. Generated automatically if density_modify_with_model is set and no model is supplied. If supplied, at least minimum_model_files must be supplied for each half-map.
- model_2 = None Model corresponding to half-map 2 (unique part of model) to use in density modification. All model_2 models should be refined against half_map_2 or denmod_half_map_2 before supplying them. Normally supply at least minimum_model_files corresponding to half-map 2 and the same for half-map 1. Generated automatically if density_modify_with_model is set.
- minimum_model_files = 3 Minimum number of models representing half-maps 1 and 2. NOTE: must be at least 2.
- denmod_map_file_name = None Optional input density-modified full map. Normally generated automatically

• denmod_half_map_file_name_1 = None Optional input density-modified half map file 1. Normally generated automatically

• denmod_half_map_file_name_2 = None Optional input density-modified half map file 2. Normally generated automatically

• truncate_density_file_name = None Input model file containing atomic coordinates around which density is to be truncated at 2 sigma in density modification. This can be used to truncate high density for heavy atoms or high density that is away from the macromolecule so that they do not interfere with density modification. Note that this means that the density in these regions will typically be much lower after density modification. See also rad_mask, the radius around the coordinates to mask.

• rad_mask = 5 Radius around truncate_density atom positions to truncate density in density modification

• histograms_file_name = None Input histograms file name. Normally this file is created automatically

• guess_histograms_path = True If histograms path is too long, try to guess the path based on the resolve default path and the working directory

• symmetry_file = None Symmetry file used to apply symmetry during density modification and model-building. Required if density_modify_with_model is set and ncs_copies is greater than 1. Normally not used as NCS is part of the map symmetry

• seq_file = None Sequence file used to estimate volume occupied by the molecule and to generate models (if density_modify_with_model is set) . Either supply all copies of each chain or supply the unique part of the sequence file and use ncs_copies to specify how many copies of this are present. If ncs_copies = 1 or None, then the seq_file must include all copies (i.e, if a dimer, must have two copies of the chain) and no NCS will be applied. (Fasta format or sequences separated by blank lines)

• output_filespdb_out = None Merged model (composite model obtained from model-building steps)initial_map_file_name = initial_map.ccp4 Output scaled original map file name. Only written out if apply_cc_star_to_initial_map is True. (Otherwise the initial map is highly sharpened and not suitable for viewing).model_density_map_file_name = model_density_12.ccp4 Output map containing density used as model-based target in density modification.denmod_map_file_name = denmod_map.ccp4 Denmod map file namedenmod_blur_50_map_file_name = None Output scaled denmod map file name, blurred with B of 50denmod_half_map_1_file_name = denmod_half_map_1.ccp4 Output scaled denmod half map 1 file namedenmod_half_map_2_file_name = denmod_half_map_2.ccp4 Output scaled denmod half map 2 file namemap_phasing_half_map_1_file_name = None Map phasing half map 1 file namemap_phasing_half_map_2_file_name = None Map phasing half map 2 file nameoutput_mask_file_name = None Output mask filetemp_dir = None Temporary directory. Default is resolve_cryo_em_xx where xx creates new directory. If specified, temp_dir will be used or created and used, and the working directory will be resolve_cryo_em_xx inside the specified temp_dir. Does not apply if running from GUI.restore_full_size = False Restore full size of maps on output by padding with zeroes.resample_on_fine_grid = False Resample output maps on fine grid with scale of resampling_ratio to starting resolution of map. Requires input maps have origin at (0, 0, 0).resampling_ratio_to_resolution = 0.2 Ratio of gridding for output map to starting resolution of mapoutput_pixel_size = None You can specify the output pixel size. This will set resample_on_fine_grid to True and adjust the gridding to result in the specified pixel size.use_cubic_boxing = False When boxing map, make a cubic boxsoft_mask_at_end = False When boxing map, create a soft mask around the edges at the end. This is in addition to the soft_mask that is used in the analysis step. Does not apply if restore_full_size is used.local_resolution_smoothing_radius_ratio = 1 Ratio of smoothing radius to resolution (local resolution only)adjust_local_resolution_coloring = False Adjust local resolution coloring based on limits of resolution. Default is to use a standard coloring scheme.output_directory = None Location for output files

- pdb_out = None Merged model (composite model obtained from model-building steps)
- initial_map_file_name = initial_map.ccp4 Output scaled original map file name. Only written out if apply_cc_star_to_initial_map is True. (Otherwise the initial map is highly sharpened and not suitable for viewing).
- model_density_map_file_name = model_density_12.ccp4 Output map containing density used as model-based target in density modification.
- denmod_map_file_name = denmod_map.ccp4 Denmod map file name
- denmod_blur_50_map_file_name = None Output scaled denmod map file name, blurred with B of 50
- denmod_half_map_1_file_name = denmod_half_map_1.ccp4 Output scaled denmod half map 1 file name
- denmod_half_map_2_file_name = denmod_half_map_2.ccp4 Output scaled denmod half map 2 file name
- map_phasing_half_map_1_file_name = None Map phasing half map 1 file name
- map_phasing_half_map_2_file_name = None Map phasing half map 2 file name
- output_mask_file_name = None Output mask file
- temp_dir = None Temporary directory. Default is resolve_cryo_em_xx where xx creates new directory. If specified, temp_dir will be used or created and used, and the working directory will be resolve_cryo_em_xx inside the specified temp_dir. Does not apply if running from GUI.
- restore_full_size = False Restore full size of maps on output by padding with zeroes.
- resample_on_fine_grid = False Resample output maps on fine grid with scale of resampling_ratio to starting resolution of map. Requires input maps have origin at (0, 0, 0).
- resampling_ratio_to_resolution = 0.2 Ratio of gridding for output map to starting resolution of map
- output_pixel_size = None You can specify the output pixel size. This will set resample_on_fine_grid to True and adjust the gridding to result in the specified pixel size.
- use_cubic_boxing = False When boxing map, make a cubic box
- soft_mask_at_end = False When boxing map, create a soft mask around the edges at the end. This is in addition to the soft_mask that is used in the analysis step. Does not apply if restore_full_size is used.
- local_resolution_smoothing_radius_ratio = 1 Ratio of smoothing radius to resolution (local resolution only)
- adjust_local_resolution_coloring = False Adjust local resolution coloring based on limits of resolution. Default is to use a standard coloring scheme.
- output_directory = None Location for output files
- crystal_inforesolution = None Estimated resolution of full map data. Used to set dm_resolutionoriginal_resolution = None Original resolution (set automatically).minimum_resolution = None Minimum resolution (set automatically).n_xyz = None Gridding for resolve density modification calculation. Normally set automatically as same as gridding of input map after boxing.auto_gridding = None Automatically set gridding in resolve density modification based on the resolution (not same as input map)use_model_mask = False Use model mask if a model is supplied. Only applies if use_model_as_target is Falsesoft_mask = True Use soft mask when boxing mapsoft_mask_radius = None Radius for soft mask when boxing map. Default is twice the resolution of the mapncs_copies = 1 Number of copies in the entire structure of whatever is in seq_file or whatever is specified as the molecular_mass. You can leave ncs_copies = None or ncs_copies = 1 and specify entire molecule in

seq_file or molecular mass. Or you can set ncs_copies = xx and specify just the unique part of the sequence or molecular_mass.box_before_analysis = None Cut out and soft-mask a box of density around the molecule before density modification. If you have already boxed your map specify False, otherwise normally specify True.box_cushion = 5 Buffer around box of density around molecule to be cut out for density modificationdensity_select = False Use density_select option to cut out molecule in box_before_analysisbox_with_model_if_present = True Use model to box density before analysis (if present)close_index = None Index within this value of edge is considered close. If edge of proposed box is within close_index of end, take the whole box.min_close_index = 10 Smallest value for estimate of close_indexmin_close_index_ratio = 10 Close index will be larger of min_close_index and cell grid units divided by min_close_index_ratio.solvent_content = None Solvent fraction (content) of the cell. You can specify the fraction of the volume of this cell that is taken up by the macromolecule. Normally set automatically. Values go from 0 to 1. This number applies to the boxed map if boxing is done here.solvent_content_iterations = 3 Iterations of solvent fraction estimationmolecular_mass = None Molecular mass of molecule in Da. Used as alternative method of specifying solvent content. If ncs_copies is specified, total molecular mass = molecular_mass times ncs_copies.fraction_of_max_mask_threshold = None threshold of standard deviation map in low resolution mask identification of solvent content. Used if no solvent content and no molecular mass and no sequence file are specified. A good value is 0.05.wang_radius = None Local averaging radius for solvent identification in masking. Default is 1.5* resolutiondenmod_wang_radius = None Local averaging radius for solvent identification in density modification Default is 1.5* resolutionsmoothing_radius = None Radius for mask smoothing. Default is twice resolution.buffer_radius = None Radius for mask buffer in map_box. Default is smoothing radius value.minimum_smoothing_radius = 5 Minimum radius for mask smoothing.minimum_solvent_content = 0.5 Stop and ask for a bigger box if solvent content appears to be less than minimum_solvent_contentmaximum_solvent_content = 0.9999 Stop and ask for a smaller box if solvent content appears to be more than maximum_solvent_contentoverall_delta_b = -30 Overall delta B to apply to model map coefficients, relative to overall B of the map. (-30 means get the overall B of the map, subtract 30, set B of atoms in model to this value.)n_bins = 100 Number of resolution bins for sharpening. Default is 100. More bins may help slightly but will slow down density modification.set_resolution_to_dm_resolution = True Set resolution to dm_resolution once it is determineddm_resolution_offset_list = None List of resolution offsets to try for density modification. Applied if quick = Falsedm_resolution_list = None List of resolutions to try for de...

- resolution = None Estimated resolution of full map data. Used to set dm_resolution
- original_resolution = None Original resolution (set automatically).
- minimum_resolution = None Minimum resolution (set automatically).
- n_xyz = None Gridding for resolve density modification calculation. Normally set automatically as same as gridding of input map after boxing.
- auto_gridding = None Automatically set gridding in resolve density modification based on the resolution (not same as input map)
- use_model_mask = False Use model mask if a model is supplied. Only applies if use_model_as_target is False
- soft_mask = True Use soft mask when boxing map
- soft_mask_radius = None Radius for soft mask when boxing map. Default is twice the resolution of the map
- ncs_copies = 1 Number of copies in the entire structure of whatever is in seq_file or whatever is specified as the molecular_mass. You can leave ncs_copies = None or ncs_copies = 1 and specify entire

molecule in seq_file or molecular mass. Or you can set ncs_copies = xx and specify just the unique part of the sequence or molecular_mass.

• box_before_analysis = None Cut out and soft-mask a box of density around the molecule before density modification. If you have already boxed your map specify False, otherwise normally specify True.

• box_cushion = 5 Buffer around box of density around molecule to be cut out for density modification

• density_select = False Use density_select option to cut out molecule in box_before_analysis

• box_with_model_if_present = True Use model to box density before analysis (if present)

• close_index = None Index within this value of edge is considered close. If edge of proposed box is within close_index of end, take the whole box.

• min_close_index = 10 Smallest value for estimate of close_index

• min_close_index_ratio = 10 Close index will be larger of min_close_index and cell grid units divided by min_close_index_ratio.

• solvent_content = None Solvent fraction (content) of the cell. You can specify the fraction of the volume of this cell that is taken up by the macromolecule. Normally set automatically. Values go from 0 to 1. This number applies to the boxed map if boxing is done here.

• solvent_content_iterations = 3 Iterations of solvent fraction estimation

• molecular_mass = None Molecular mass of molecule in Da. Used as alternative method of specifying solvent content. If ncs_copies is specified, total molecular mass = molecular_mass times ncs_copies.

• fraction_of_max_mask_threshold = None threshold of standard deviation map in low resolution mask identification of solvent content. Used if no solvent content and no molecular mass and no sequence file are specified. A good value is 0.05.

• wang_radius = None Local averaging radius for solvent identification in masking. Default is 1.5* resolution

• denmod_wang_radius = None Local averaging radius for solvent identification in density modification Default is 1.5* resolution

• smoothing_radius = None Radius for mask smoothing. Default is twice resolution.

• buffer_radius = None Radius for mask buffer in map_box. Default is smoothing radius value.

• minimum_smoothing_radius = 5 Minimum radius for mask smoothing.

• minimum_solvent_content = 0.5 Stop and ask for a bigger box if solvent content appears to be less than minimum_solvent_content

• maximum_solvent_content = 0.9999 Stop and ask for a smaller box if solvent content appears to be more than maximum_solvent_content

• overall_delta_b = -30 Overall delta B to apply to model map coefficients, relative to overall B of the map. (-30 means get the overall B of the map, subtract 30, set B of atoms in model to this value.)

• n_bins = 100 Number of resolution bins for sharpening. Default is 100. More bins may help slightly but will slow down density modification.

• set_resolution_to_dm_resolution = True Set resolution to dm_resolution once it is determined

• dm_resolution_offset_list = None List of resolution offsets to try for density modification. Applied if quick = False

• dm_resolution_list = None List of resolutions to try for density modification. Normally set automatically

- dm_resolution = None High-resolution limit for density modification. Normally set automatically from input resolution.
- dm_res_a = 2.4 dm_resolution will be guessed from resolution with the formula: res_dm = a + b*(res-res_c) + d * (res-res_c)**2
- dm_res_b = 0.99 dm_resolution will be guessed from resolution with the formula: res_dm = a + b*(res-res_c) + d * (res-res_c)**2
- dm_res_c = 3.0 dm_resolution will be guessed from resolution with the formula: res_dm = a + b*(res-res_c) + d * (res-res_c)**2
- dm_res_d = -0.2 dm_resolution will be guessed from resolution with the formula: res_dm = a + b*(res-res_c) + d * (res-res_c)**2
- chain_type = *None PROTEIN RNA DNA Chain type. Determined automatically from sequence file if not given. Mixed chain types are fine (leave blank if so).
- sequence = None Optional sequence. Can be used instead of a seq_file
- scattering_table = n_gaussian wk1995 it1992 *electron neutron Choice of scattering table for structure factor calculations. Standard for X-ray is n_gaussian, for cryoEM is electron.
- strategycalculate_local_resolution = True Get local resolution mapgeometric_scale = True Use geometric mean of low-res scale factor and shell scale factor in scaling. Prior to FSCref scaling, the low-res scale factor would make Fourier coefficients in each shell of resolution match those in the lowest-resolution shell (on average). The shell scale factor would make them match the resolution dependence of the (scaled) half-maps. The geometric mean is partway between these. Note: not applied if final_scale_with_fsc_ref is True.geometric_scale_half_maps_only = False Use geometric mean only for half-maps, not map combinationgeometric_scale_not_half_maps = False Use geometric mean except for half-maps, only map combinationtarget_value = None Target value in scaling. None means scale to input map. You can use geometric_scale = False target_value = 1000 final_scale_with_fsc_ref = False to have a more-or-less resolution-independent final map (an E-map).final_scale_with_fsc_ref = True Adjust final resolution-dependent average Fourier coefficient amplitudes to match estimated correlation with true cofficients (FSCref). This uses fixed_target_value = 1000 in recombination and turns off geometric_scale in map recombinationfixed_target_value = 1000 Fixed target value to use if specified.include_all_in_target_norm = False Include all input data in calculating average values for normalization. Alternative is just half-map datarequire_significant_hm12 = None Require significant FSC for half-maps (within min_hm12ab_ratio of FSC for density-modified maps). Default is True except when models are used.half_map_fsc_to_keep_only = 0.9 If half-map fsc is this value or greater, skip density modification in this shell. Used to prevent density-modified information from making low-resolution terms worse than they started out.optimize_r_value = False Optimize choice of histograms/sharpening based on R-value for density modification. Alternative is based on estimated improvement in resolution.optimize_sharpening_method = None Try unsharpened and half-map sharpened map coefficients as starting point for density modification, then auto-set overall sharpening if remove_aniso = true, then density modify and choose based on minimum density modification R valuedensity_modify_density_target_start_with_denmod = False Start with density-modified maps in density_modify_density_targetdensity_modify_model_simple_merge = False Merge model_density with original without another density modification run. Normally choose either density_modify_model_simple_merge or density_modify_density_target for model-based density modification.density_modify_density_target = True Use density from unmerged models as target for density modificationuse_model_variance = True Use estimate of model variance to weight density target.mask_density_target = True Mask density target to use only region where model was presentuse_merged_models_in_denmod = False Use merged model in density modification (one for

each half map)use_tight_mask = False Use tight mask in average_mapsmodel_mask_radius = 3 Radius for masking density from modelno_denmod_after_model = False No density modification after recombining model information. Not implemented.use_correlated_fsc = False Assume errors in model-based FSC values are highly correlateduse_maximum_inside_mask = False Use map_1 density (original) if higher than map_2 (model-based) inside map_2 mask in average_mapssd_smoothing_radius_ratio = 0.75 Radius for smoothing density sd estimates as ratio to resolution. Used in average_mapssmooth_model_map = False Smooth model map in average_mapsweight_on_model = 1 Weight on model in average_mapssharpen_map_data_before_scaling = True Sharpen input map data before using it to scale half-map and model datause_half_datasets_for_sigmas_by_reflection = False Use differences between F1 and F2 for sigmas (times 0.707). Alternative is use these differences in shells only. Do not use this as it introduces correlations between h...

- calculate_local_resolution = True Get local resolution map
- geometric_scale = True Use geometric mean of low-res scale factor and shell scale factor in scaling. Prior to FSCref scaling, the low-res scale factor would make Fourier coefficients in each shell of resolution match those in the lowest-resolution shell (on average). The shell scale factor would make them match the resolution dependence of the (scaled) half-maps. The geometric mean is partway between these. Note: not applied if final_scale_with_fsc_ref is True.
- geometric_scale_half_maps_only = False Use geometric mean only for half-maps, not map combination
- geometric_scale_not_half_maps = False Use geometric mean except for half-maps, only map combination
- target_value = None Target value in scaling. None means scale to input map. You can use geometric_scale = False target_value = 1000 final_scale_with_fsc_ref = False to have a more-or-less resolution-independent final map (an E-map).
- final_scale_with_fsc_ref = True Adjust final resolution-dependent average Fourier coefficient amplitudes to match estimated correlation with true cofficients (FSCref). This uses fixed_target_value = 1000 in recombination and turns off geometric_scale in map recombination
- fixed_target_value = 1000 Fixed target value to use if specified.
- include_all_in_target_norm = False Include all input data in calculating average values for normalization. Alternative is just half-map data
- require_significant_hm12 = None Require significant FSC for half-maps (within min_hm12ab_ratio of FSC for density-modified maps). Default is True except when models are used.
- half_map_fsc_to_keep_only = 0.9 If half-map fsc is this value or greater, skip density modification in this shell. Used to prevent density-modified information from making low-resolution terms worse than they started out.
- optimize_r_value = False Optimize choice of histograms/sharpening based on R-value for density modification. Alternative is based on estimated improvement in resolution.
- optimize_sharpening_method = None Try unsharpened and half-map sharpened map coefficients as starting point for density modification, then auto-set overall sharpening if remove_aniso = true, then density modify and choose based on minimum density modification R value
- density_modify_density_target_start_with_denmod = False Start with density-modified maps in density_modify_density_target
- density_modify_model_simple_merge = False Merge model_density with original without another density modification run. Normally choose either density_modify_model_simple_merge or

density_modify_density_target for model-based density modification.

• density_modify_density_target = True Use density from unmerged models as target for density modification

• use_model_variance = True Use estimate of model variance to weight density target.

• mask_density_target = True Mask density target to use only region where model was present

• use_merged_models_in_denmod = False Use merged model in density modification (one for each half map)

• use_tight_mask = False Use tight mask in average_maps

• model_mask_radius = 3 Radius for masking density from model

• no_denmod_after_model = False No density modification after recombining model information. Not implemented.

• use_correlated_fsc = False Assume errors in model-based FSC values are highly correlated

• use_maximum_inside_mask = False Use map_1 density (original) if higher than map_2 (model-based) inside map_2 mask in average_maps

• sd_smoothing_radius_ratio = 0.75 Radius for smoothing density sd estimates as ratio to resolution. Used in average_maps

• smooth_model_map = False Smooth model map in average_maps

• weight_on_model = 1 Weight on model in average_maps

• sharpen_map_data_before_scaling = True Sharpen input map data before using it to scale half-map and model data

• use_half_datasets_for_sigmas_by_reflection = False Use differences between F1 and F2 for sigmas (times 0.707). Alternative is use these differences in shells only. Do not use this as it introduces correlations between half-datasets.

• fix_amplitudes = False Fix amplitudes, only change phases

• use_variance = True Use model variance in model-based density modification

• refine_symmetry_in_denmod = True Refine symmetry operators in density modification.

• use_symmetry_in_denmod = False Use symmetry in density modification. Normally not used as maps already have symmetry applied.

• symmetry_in_denmod_last_cycle_only = True Use symmetry in density modification only on last cycle (if use_symmetry_in_denmod = True).

• fraction_ncs = None Fraction of macromolecule expected to follow symmetry. Use None to identify automatically

• use_denmod_map_in_denmod_with_model = False Use density-modified map to start density modification with model

• refine = True Refine models

• refine_after_randomize = False Refine models after running randomize routine. Note that randomize (used in density_modify_with_single_model) already refines the multiple models but does not refine B. This refines them further including B values. It may be better to keep B values of zero for the multiple-models and skip this.

• sequence_models = None Get sequences of models. Default is True if models are generated automatically at a resolution of sequence_models_resolution or better (normally 3.5 A) and False otherwise. Note that if a very big model this can take a really long time.

• sequence_models_resolution = 3.5 Resolution at which sequence_models is normally applied if models are generated automatically.

• rerun_with_avail_seq = True Rerun sequence alignment with parts of the sequence already used removed. With a large structure this can take a long time.

• write_map_files = True Write out final map files

• write_boxed_map_files = False Write out boxed versions of input map files

• delta_resolution = 0.0 0.5 Resolution offsets to try in model-building

• macro_cycles = 3 Cycles of real-space refinement in model-building and randomization

• build_thoroughness = *None quick medium thorough Thoroughness of model-building

• build_methods = high_density_from_model no_high_density_from_model Build methods to try in model-building. Allowed methods: trace_and_build phase_and_build high_density_from_model no_high_density_from_model quick medium

• backup_build_methods = phase_and_build Build methods to try in model-building if initial methods do not yield enough models. Allowed methods: trace_and_build phase_and_build high_density_from_model no_high_density_from_model quick medium

• vary_blur_and_spectral_scaling_with_model = True In density_modify_with_model, generate maps with and without blur_by_resolution and spectral_scaling (4 maps)

• density_modify_with_model = None Run model-based density modification. If one model is supplied, randomize and generate multi-model representation for each half-map. Models are deleted unless verbose = True. If multiple models (at least minimum_model_files) are supplied for each half-map, use them. Otherwise generate models and refine against half-maps. Use model information in density modification.

• randomize_single_model = True Randomize single model if supplied.

• density_modify_with_single_model = None Density modify with single model by randomizing model before use Normally used internally only.

• model_cycles = 1 Number of cycles of model-based density modification. Normally use 1 if you supply a single model (if you use more it will generate new models on cycles 2-...). Use 1 or a few if you are generating new models.

• model_sharpen = True Carry out model-based sharpening based on final model if density modifying with model information

• use_model_as_target = True Use model as target in density modification. Alternative is to use density based on model as target for density modification.

• omit_with_model = False Use omit maps in density_modify_with_model. Only applies if use_model_as_target is True. Not implemented

• n_box_target = None Target number of boxes in omit maps

• scale_rmsd = 1 Scale on RMSD estimate for model-map agreement

• patterns = False Use resolve_patterns to identify patterns in map and use them in density modification. Requires box_before_analysis.

• update_starting_map = True Update starting half maps each cycle using half-maps from previous cycle.. Alternative is to restart with original (sharpened) half maps each cycle.

- update_histograms = True Update histograms each cycle using working map. Alternative is to use input histogram file or generate histograms once.
- database_list = None List of histogram database entries to try one at a time if optimizing histograms. Database 0 is local database for this run. Databases 1 2 3 4 5 are 1 - 3 A databases. Default is 5 for quick [0, 5] for non-quick
- database_number = None Use this database entry by default (0 is local database, 5 std)
- rad_smooth_ratio = 1 Ratio of rad_smooth to resolution
- spectral_scaling = False Scale average Fourier coefficient amplitude vs resolution to match spectrum expected for a protein in a box of solvent. This can restore resolution-dependent features of a map.
- blur_by_resolution = False Blur after half-map averaging by B = 10 times resolution. This can be used along with blur by resolution factor to apply a slight blurring that may make the final map easier to interpret.
- blur_by_resolution_factor = 10 Blur after half-map averaging by B = blur_by_resolution_factor times resolution
- sigma_weighting = None In recombination step, weight individual fourier coefficients based on estimated variances: 1/sqrt(normalized_variance). Can be combined with real_space_weighting. Can cause correlations between density-modified half maps so that half-map FSC values may overstate resolution.
- average_neighbors = False In sigma_weighting, average neighboring weights
- apply_cc_star_to_initial_map = None Apply resolution-dependent cc_star (FSCref) weighting to initial map before density modification. Default is True. If False, apply any other scaling but set cc_star = 1.
- local_methods_final_cycle = True Apply any local sharpening/averaging only on last cycle
- real_space_weighting = None In recombination step, weight maps in half-map averaging based on local half-map variance in real space (smoothed with smoothing_radius). Can be combined with sigma_weighting. Can cause correlations between density-modified half maps so that half-map FSC values may overstate resolution.
- sharpening_type = None half_map *final_half_map local_final_half_map Sharpening to apply at each stage. Half-map means scale data in each shell to maximum then multiply by estimated FSCref from input first half maps. Final_half_map means same except use estimate of FSCref for final maps. Final_local_half_map uses local FSCref estimates.
- cycles = 1 Cycles of density modification
- damping = None Damping of shifts on each cycle.
- minimum_r_value_improvement_per_cycle = 0.001 Minimum r_value improvement to keep cycling. Does not apply if dampling is used.
- force_cycles = True Keep results of last cycle even if not better
- minimum_improvement_per_cycle = -1 Minimum improvement to keep cycling. Does not apply if dampling is used.
- zero_half_dataset_correlation = False Assume correlation between density-modified half-datasets is zero.
- very_high_error = 100 If an error (asqr, bsqr, csqr, ssqr) is this big or bigger, require it to keep this value for all subsequent bins

- cc_cut = 0.2 Estimate of minimum highly reliable CC in half-map FSC. Used to decide at what CC value to smooth the remaining CC values. Also used to decide whether scale_using_last will apply
- correct_for_final_value = True Correct FSC values for correlated errors by assuming that the last smoothed value represents the correlated error
- max_cc_for_rescale = 0.1 Min reliable CC in half-maps. Used along with cc_cut and scale_using_last to correct for small errors in FSC estimation at high resolution. If the value of FSC near the high-resolution limit is above max_cc_for_rescale, assume these values are correct and do not correct them.
- scale_using_last = 3 If set, assume that the last scale_using_last bins in the FSC for half-map or model sharpening are about zero (corrects for errors int the half-map process). Only applies if these values are less than cc_cut
- mask_atoms_atom_radius = 5 Atom radius in model masking
- control_no_denmod = False Run dummy density modification just flattening solvent as estimated by probability mask
- zero_solvent = False Zero solvent region (for testing)
- randomize_solvent = False Randomize solvent region (for testing). Also set solvent_content and solvent_noise_ratio to run this.
- solvent_noise_ratio = 0.4 Solvent noise ratio for randomize_solvent
- use_bins_for_solvent_noise = True Use bins in solvent noise
- require_non_identical_half_maps = True Require half maps that are not identical
- density_modify_unsharpened_maps = None Use unsharpened (original) half maps in density modification. Normally using the sharpened map is better but it may be worth trying with unsharpened maps.
- default_fom = 0.9 Default FOM value if unknown
- anisotropyremove_aniso = True Remove anisotropy from data and optionally sharpen before density modificationb_iso = None Target overall B value for anisotropy correction. Ignored if remove_aniso = False. If None, default is target_b_ratio*resolutionmax_b_iso = None Default maximum overall B value for anisotropy correction. Ignored if remove_aniso = False. Recommended if b_iso is set. If used, default is minimum of (max_b_iso, B of dataset, target_b_ratio*resolution)target_b_ratio_list = 10. 20. Set of ratios of target B to resolution to try. Applied if quick = False and ignored if remove_aniso = False or b_iso is settarget_b_ratio = 10. Default ratio of target B value to resolution for anisotropy correction. Ignored if remove_aniso = False. Ignored if quick or b_iso is set. If used, default is minimum of (max_b_iso, B of dataset, target_b_ratio*resolution)
- remove_aniso = True Remove anisotropy from data and optionally sharpen before density modification
- b_iso = None Target overall B value for anisotropy correction. Ignored if remove_aniso = False. If None, default is target_b_ratio*resolution
- max_b_iso = None Default maximum overall B value for anisotropy correction. Ignored if remove_aniso = False. Recommended if b_iso is set. If used, default is minimum of (max_b_iso, B of dataset, target_b_ratio*resolution)
- target_b_ratio_list = 10. 20. Set of ratios of target B to resolution to try. Applied if quick = False and ignored if remove_aniso = False or b_iso is set
- target_b_ratio = 10. Default ratio of target B value to resolution for anisotropy correction. Ignored if remove_aniso = False. Ignored if quick or b_iso is set. If used, default is minimum of (max_b_iso, B of dataset, target_b_ratio*resolution)

- controlquick = True Quick run, do not optimize dm_resolution, target_b_iso or histogram database. Turns off real_space_weighting and sigma_weighting unless they are set.ignore_symmetry_conflicts = False Ignore symmetry conflicts in input fileswrapping = False Allow wrapping of map around boundariesignore_limitations = False Ignore restrictions on what can be runstop_after_mask = False Stop after getting maskverbose = False Verbose outputmax_dirs = 1000 Maximum number of directories (trace_and_build_xxx)resolve_size = None Size of resolve to use. resolve_command_file = None File with commands for resolvemultiprocessing = *multiprocessing sge lsf pbs condor pbspro slurm Choices are multiprocessing (single machine) or queuing systemsqueue_run_command = None run command for queue jobs. For example qsub.nproc = 1 Number of processors to use. NOTE: by default multiple processors will only be used in the map-to-model step (this is because multiprocessing requires writing out nproc sets of huge files and it can be very slow with distributed queues.). You can override this with force_nproc = True.force_nproc = False Use all processors in all steps. If not set then multiprocessing will not be used in average_mapsrandom_seed = 171731 Random seedmax_wait_time = 1 Max time to wait for files to appear. Increase if files seem to be missing during run.test_gui = None Run from command line but test GUI features
- quick = True Quick run, do not optimize dm_resolution, target_b_iso or histogram database. Turns off real_space_weighting and sigma_weighting unless they are set.
- ignore_symmetry_conflicts = False Ignore symmetry conflicts in input files
- wrapping = False Allow wrapping of map around boundaries
- ignore_limitations = False Ignore restrictions on what can be run
- stop_after_mask = False Stop after getting mask
- verbose = False Verbose output
- max_dirs = 1000 Maximum number of directories (trace_and_build_xxx)
- resolve_size = None Size of resolve to use.
- resolve_command_file = None File with commands for resolve
- multiprocessing = *multiprocessing sge lsf pbs condor pbspro slurm Choices are multiprocessing (single machine) or queuing systems
- queue_run_command = None run command for queue jobs. For example qsub.
- nproc = 1 Number of processors to use. NOTE: by default multiple processors will only be used in the map-to-model step (this is because multiprocessing requires writing out nproc sets of huge files and it can be very slow with distributed queues.). You can override this with force_nproc = True.
- force_nproc = False Use all processors in all steps. If not set then multiprocessing will not be used in average_maps
- random_seed = 171731 Random seed
- max_wait_time = 1 Max time to wait for files to appear. Increase if files seem to be missing during run.
- test_gui = None Run from command line but test GUI features
- guiGUI-specific parameter required for output directoryoutput_dir = None
- output_dir = None

Cryo-EM Validation tools in Phenix

validation_cryo_em.html

Contents

- Overview
- Model
- Model and Data Fit
- Data
- References

Overview

Phenix has multiple tools for validating cryo-EM maps and models. They are combined into a single tool called phenix.validation_cryoem. This can be run in the GUI by selecting "Comprehensive Validation (cryo-EM)" either from the "Validation" section or the "Cryo-EM" section.

Opening the program will show a window for supplying your input files and basic settings.

You will need at least a model and one map. If you provide half maps, be sure to indicate which map is the full map and which maps are the half maps. To do so, after adding the files, you can right-click on the row of a map file and select either "Full map" or "Half map."

Lastly, before running the program, enter the resolution of the map.

Important note: The default scattering table for cryo-EM tools is "electron." A consequence of this setting is that Phenix does not have electron scattering factors for ions. You may get error messages if your cryo-EM structures have ions.

Once the program finishes the analysis, you will get a page that summarizes all the results, as well as additional pages for more detailed information about each section.

Similar to the "Comprehensive Validation (X-ray/Neutron)" GUI, the model and map can be viewed in Coot and the pages with more detail information will allow you to zoom into the more problematic areas of your structure.

There is also a button, "Export Table 1," for exporting the summary statistics into plain text, comma separated text, or rich-text format.

Model

This page contains the model-related metrics with additional pages for each major metric.

Generally, clicking on a row of a table or clicking a point in the graph

will zoom in on that particular atom/residue in Coot.

The results are subdivided into several categories listed below.

Clashes: this shows areas of the structure where atoms may be too close to each other. In Coot, this is shown by red dots. CaBLAM: this is the output from phenix.cablam, which looks at several measures based on the geometry of the C-α atoms. Cβ: this looks for any deviations in the C-β atoms. Cis/Twisted: Cis peptides are generally rare (~5% of prolines and ~0.03% of other residues), so make sure that a cis conformation is really supported by the map. Twisted peptides are almost always incorrect. Rotamers: this checks that the sidechains are in the expected regions according to compiled distributions of the X-angles in the Top8000 database. Ramachandran: this checks that the dihedral angles of the protein residues are in the expected regions according to the Top8000 database. Geometry restraints: this checks that the bond lengths, bond angles, dihedral angles, and other geometric restraints do not deviate too much from the mean value.

- Clashes: this shows areas of the structure where atoms may be too close to each other. In Coot, this is shown by red dots.
- CaBLAM: this is the output from phenix.cablam, which looks at several measures based on the geometry of the C-α atoms.
- Cβ: this looks for any deviations in the C-β atoms.
- Cis/Twisted: Cis peptides are generally rare (~5% of prolines and ~0.03% of other residues), so make sure that a cis conformation is really supported by the map. Twisted peptides are almost always incorrect.
- Rotamers: this checks that the sidechains are in the expected regions according to compiled distributions of the X-angles in the Top8000 database.
- Ramachandran: this checks that the dihedral angles of the protein residues are in the expected regions according to the Top8000 database.
- Geometry restraints: this checks that the bond lengths, bond angles, dihedral angles, and other geometric restraints do not deviate too much from the mean value.

Model and Data Fit

This page shows how well the model fits into the map by calculating different correlation coefficients. Please check the first reference for a more detailed description of each correlation coefficient (CC). Table 3 from the reference is reproduced below.

Correlation coefficients per chain and per residue are also shown in plots.

The per residue plot is shown for a single chain. To change the chain, you can click on the per chain plot or use the drop down menu. Clicking on the per residue plot will center on the residue in Coot.

Data

This page shows the output from phenix.mtriage. The output is the same as if you were to run phenix.mtriage separately. More details can be found in the Mtriage documemtation

The results are subdivided into several categories listed below.

Summary: this shows basic statistics about the maps and summarizes the resolution estimates FSC (Half-maps): this shows a graph plotting the Fourier shell coefficient curve based on the half-maps (if available) with and without masking. The intersections of the curves with FSC=0.143 are shown. FSC (Model-map): this shows a graph plotting the Fourier shell coefficent curve based on the model map with and without masking. The intersections of the curves with FSC=0.5 are shown.

- Summary: this shows basic statistics about the maps and summarizes the resolution estimates
- FSC (Half-maps): this shows a graph plotting the Fourier shell coefficient curve based on the half-maps (if available) with and without masking. The intersections of the curves with FSC=0.143 are shown.
- FSC (Model-map): this shows a graph plotting the Fourier shell coefficent curve based on the model map with and without masking. The intersections of the curves with FSC=0.5 are shown.

In the pages with graphs, you can save the graph and export the data used to generate the graphs into a text file.

References

New tools for the analysis and validation of cryo-EM maps and atomic models. P.V. Afonine, B.P. Klaholz, N.W. Moriarty, B.K. Poon, O.V. Sobolev, T.C. Terwilliger, P.D. Adams, and A. Urzhumtsev. Acta Cryst. D74, 814-840 (2018).

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